



US 20250286400A1

(19) **United States**

(12) **Patent Application Publication**
Parness et al.

(10) **Pub. No.: US 2025/0286400 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **ENERGY STORAGE CONFIGURATION
MANAGEMENT BETWEEN A VEHICLE AND
A PROPERTY ASSOCIATED WITH THE
VEHICLE**

(71) Applicant: **TOYOTA MOTOR ENGINEERING
& MANUFACTURING NORTH
AMERICA, INC.**, Plano, TX (US)

(72) Inventors: **Maximilian Parness**, Takoma Park,
MD (US); **Norman Lu**, Fairview, TX
(US)

(73) Assignees: **TOYOTA MOTOR ENGINEERING
& MANUFACTURING NORTH
AMERICA, INC.**, Plano, TX (US);
**TOYOTA JIDOSHA KABUSHIKI
KAISHA**, AICHI-KEN (JP)

(21) Appl. No.: **18/599,152**

(22) Filed: **Mar. 7, 2024**

Publication Classification

(51) **Int. Cl.**
H02J 7/35 (2006.01)
B60L 53/51 (2019.01)
H02J 1/10 (2006.01)
H02J 7/00 (2006.01)
(52) **U.S. Cl.**
CPC **H02J 7/35** (2013.01); **B60L 53/51**
(2019.02); **H02J 1/109** (2020.01); **H02J**
7/00034 (2020.01)

(57) **ABSTRACT**

An example operation includes one or more of receiving, via a power management apparatus, mobility data comprising movements of a vehicle from one or more systems associated with the vehicle, receiving, via the power management apparatus, energy consumption data from one or more power consuming systems at a real property that is associated with the vehicle, determining a recommended battery for the vehicle and a recommended power generation system for the real property based on the mobility data and the energy consumption data, and displaying an identifier of the recommended battery and an identifier of the recommended power generation system via a user interface of the power management apparatus.

200

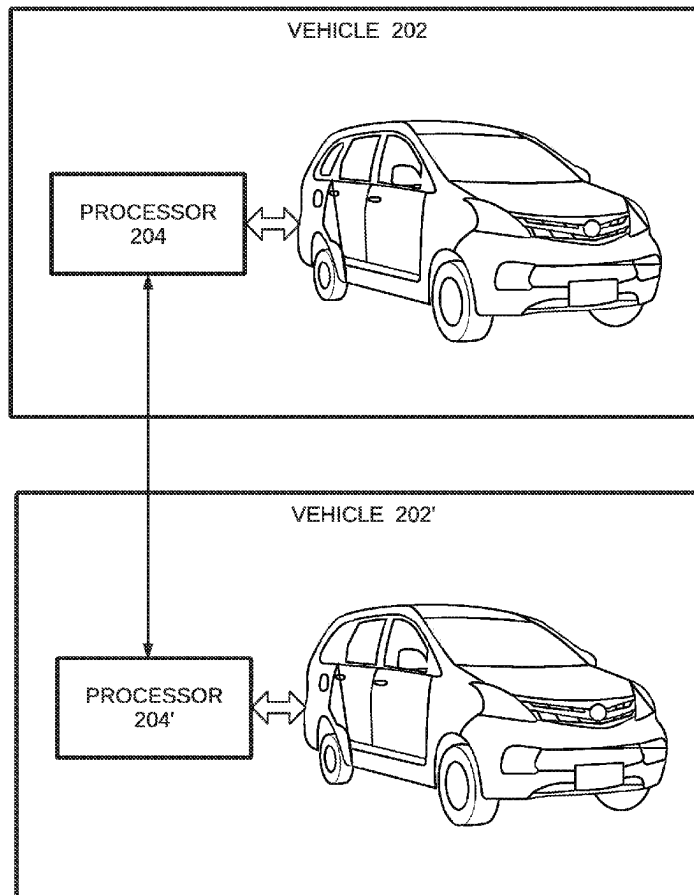


FIG. 1A

100A

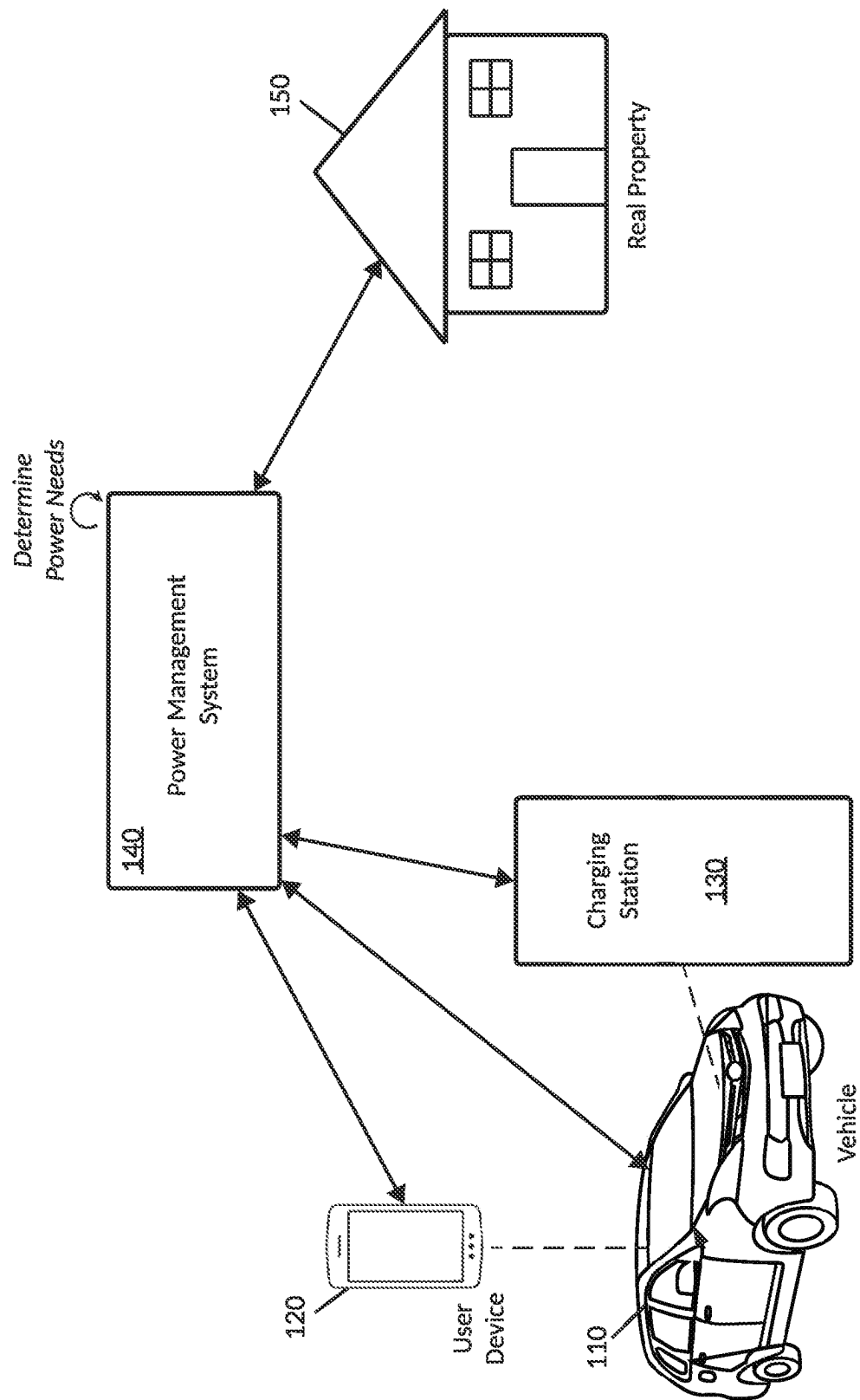


FIG. 1B

100B

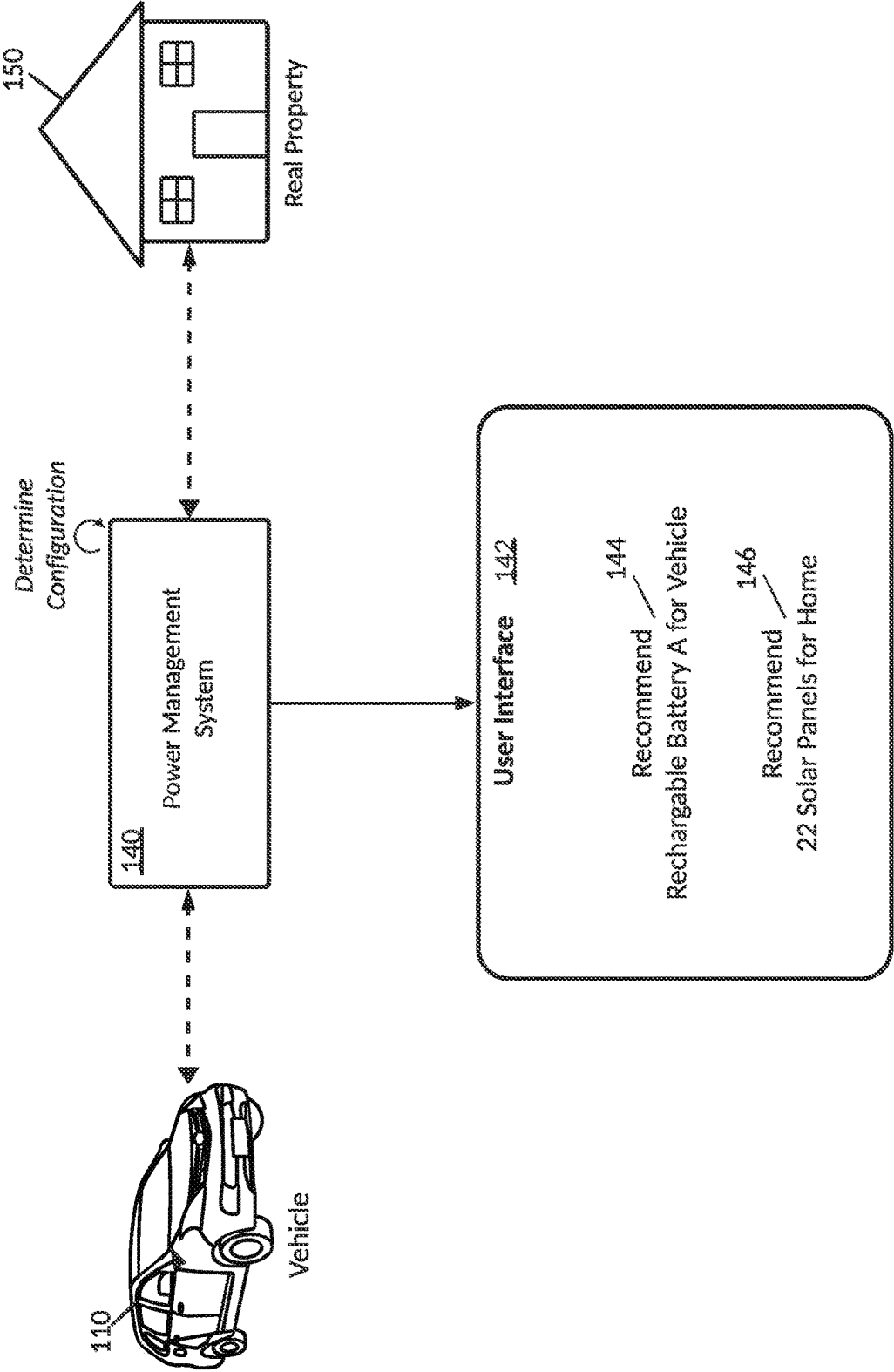


FIG. 1C

100C

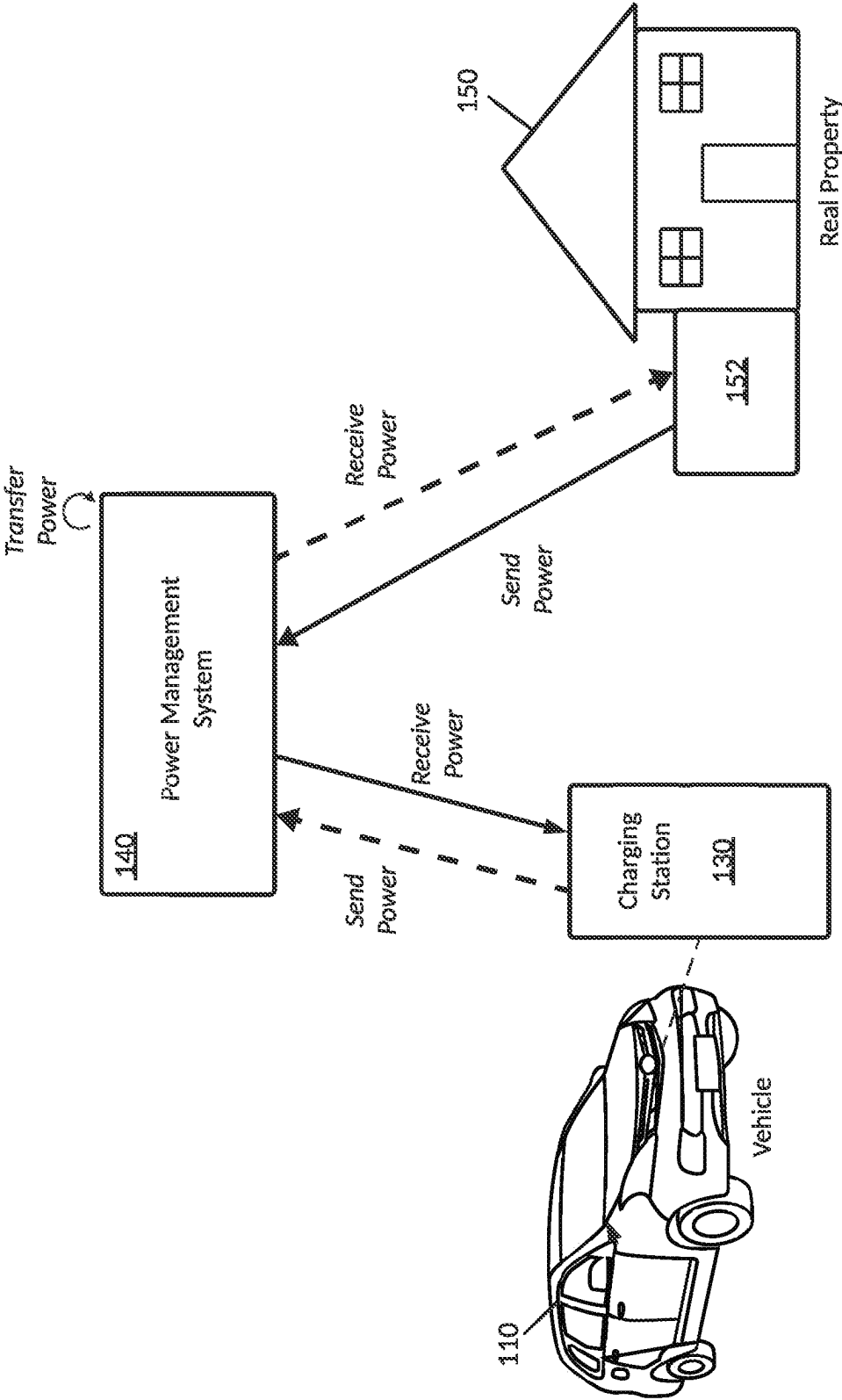
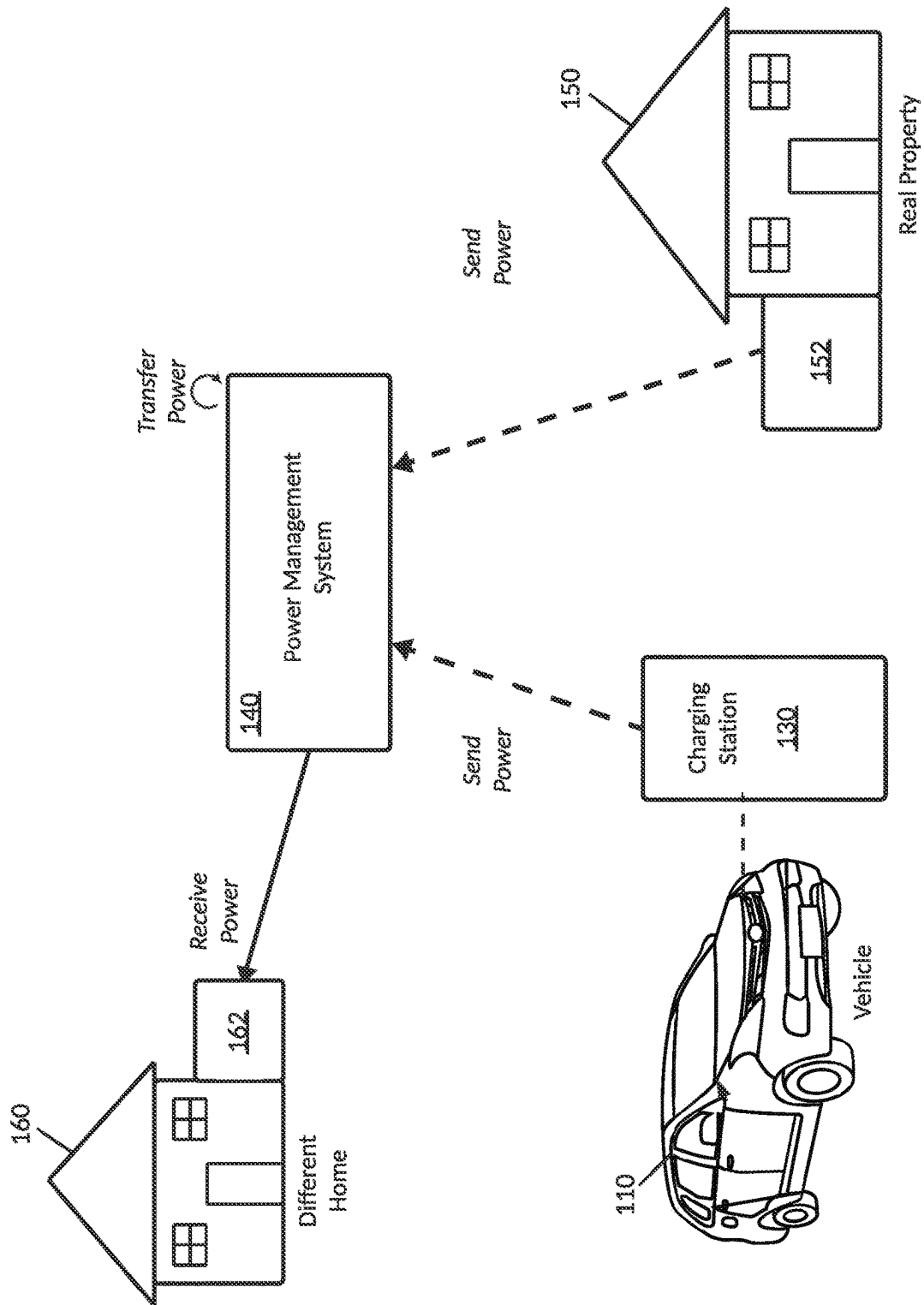


FIG. 1D

100D



200

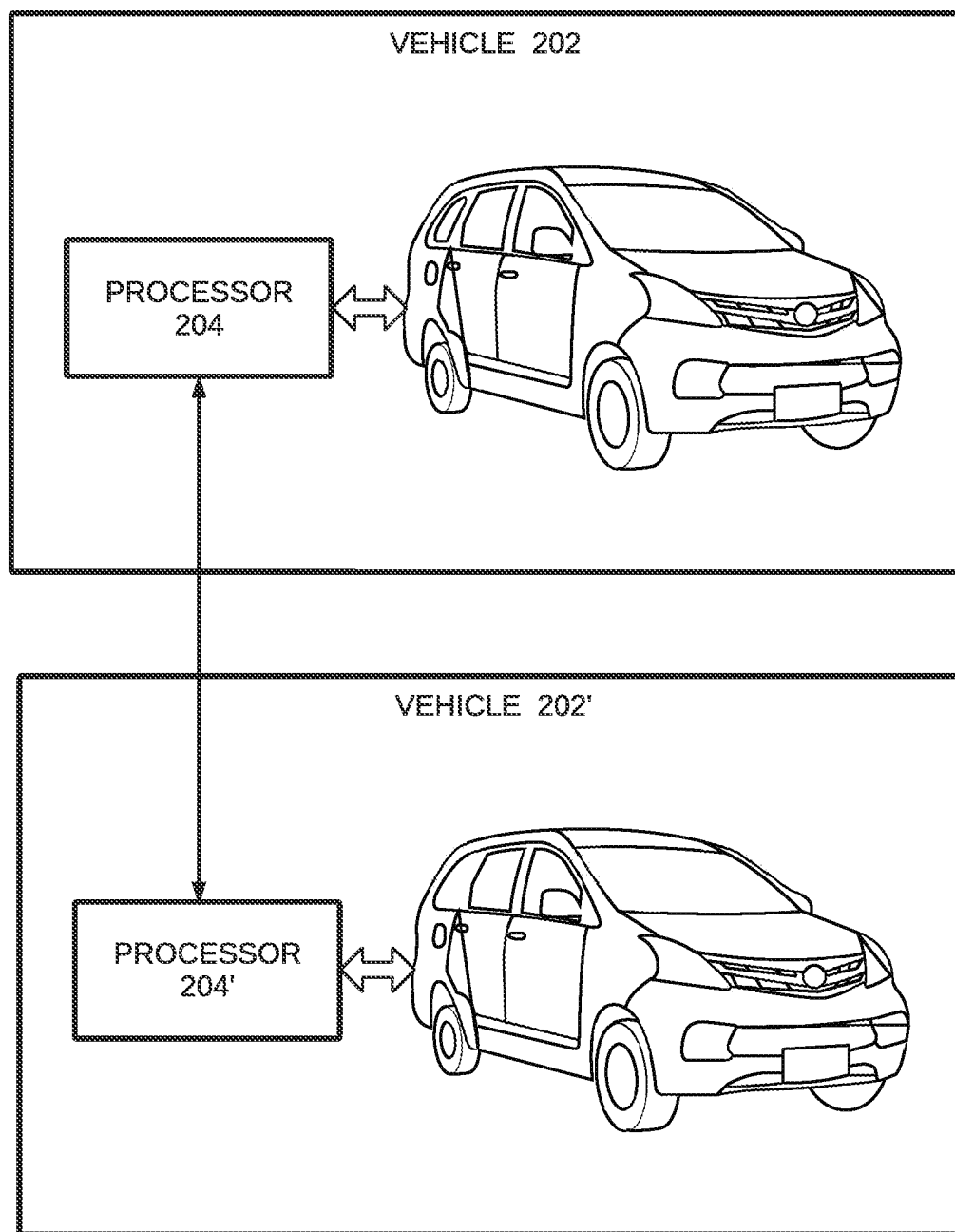


FIG. 2A

210

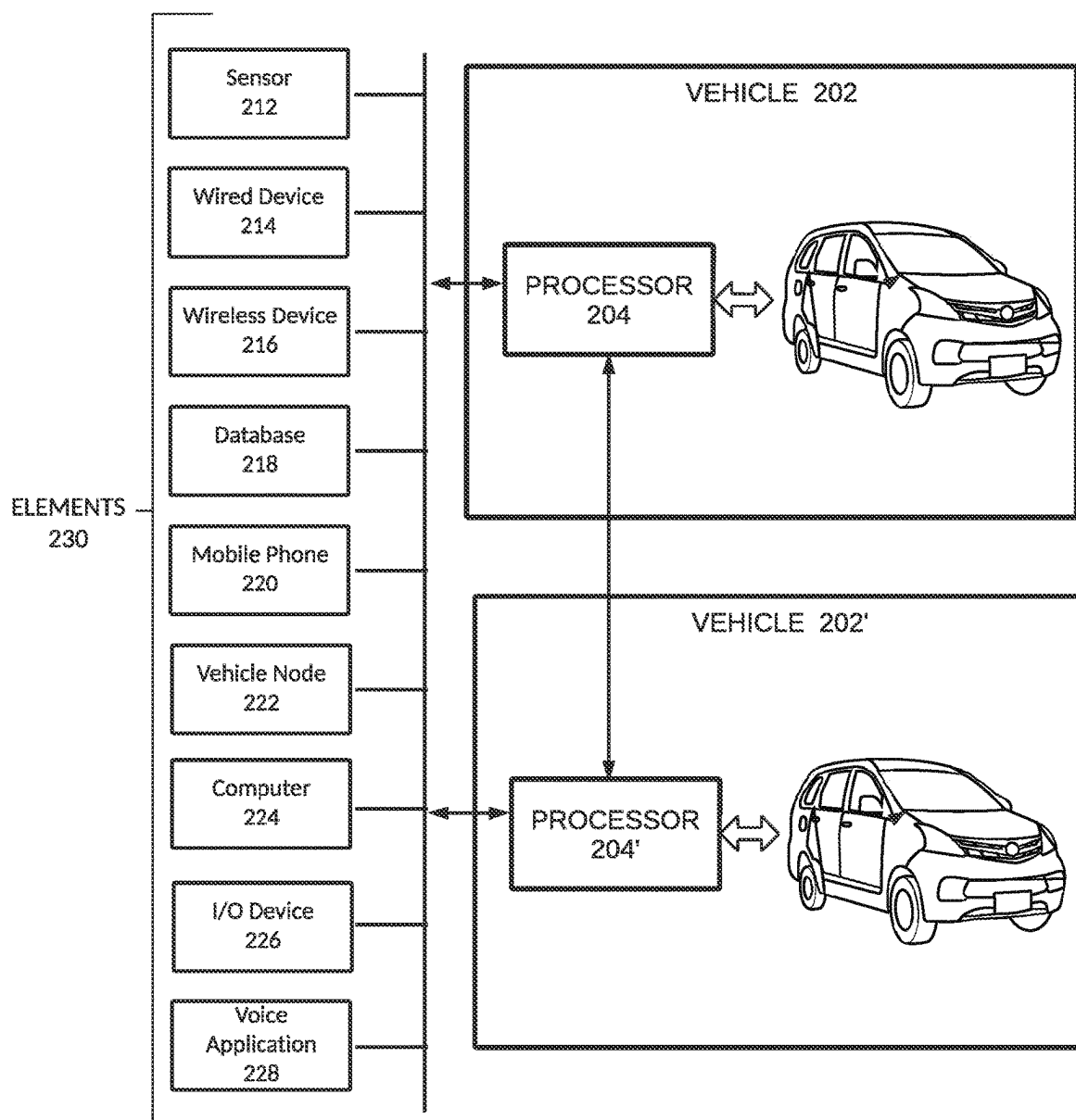


FIG. 2B

240

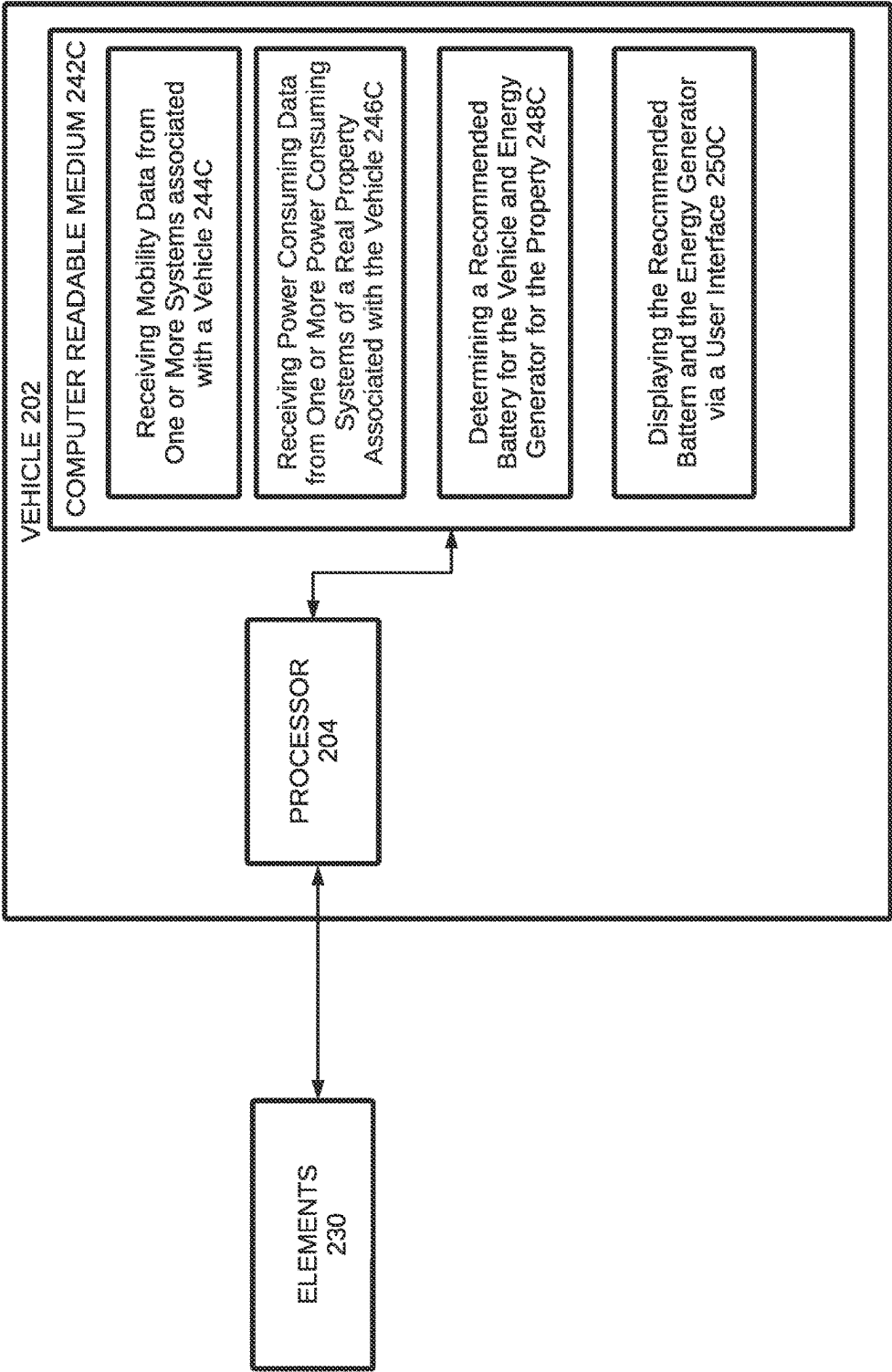


FIG. 2C

250

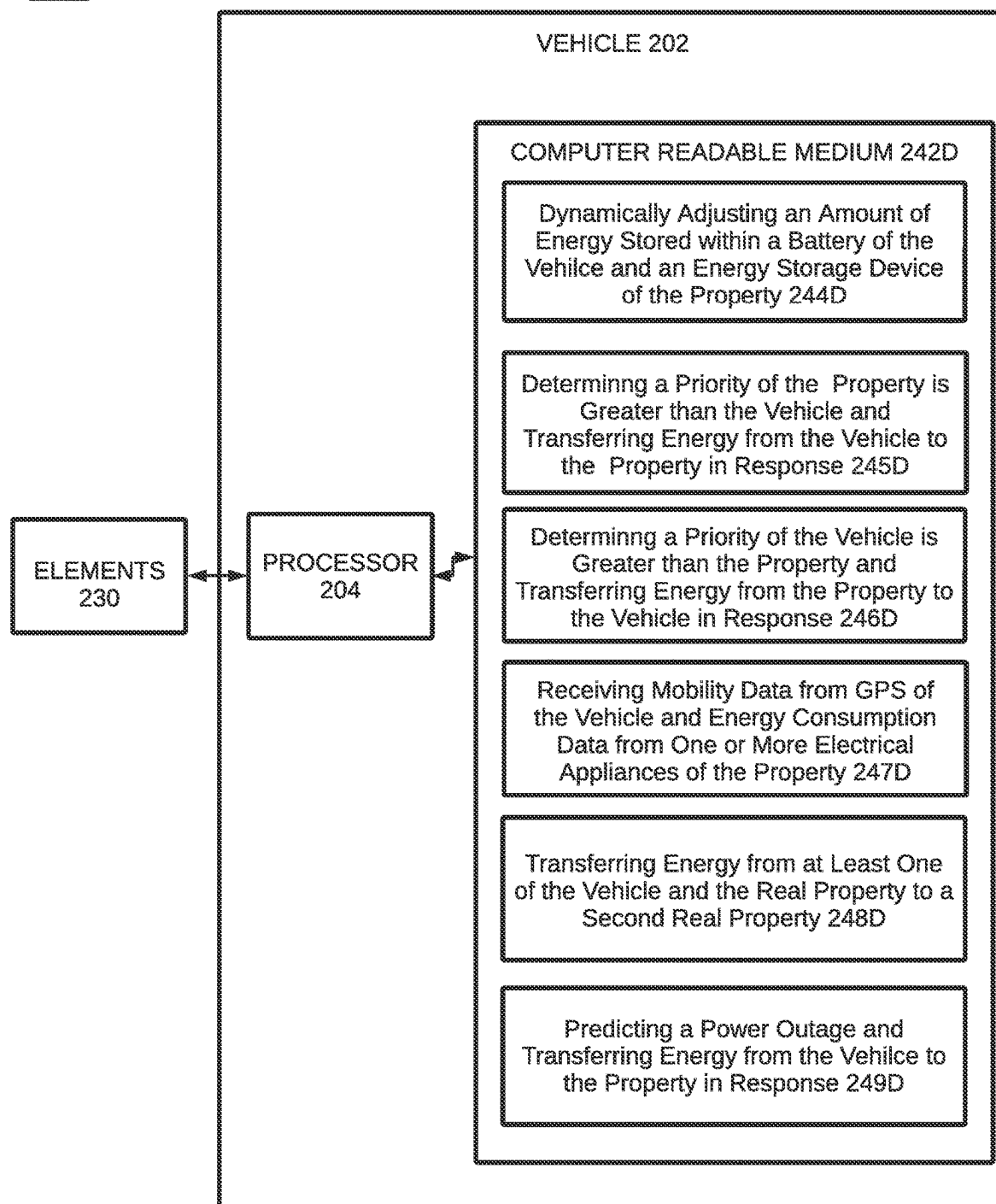


FIG. 2D

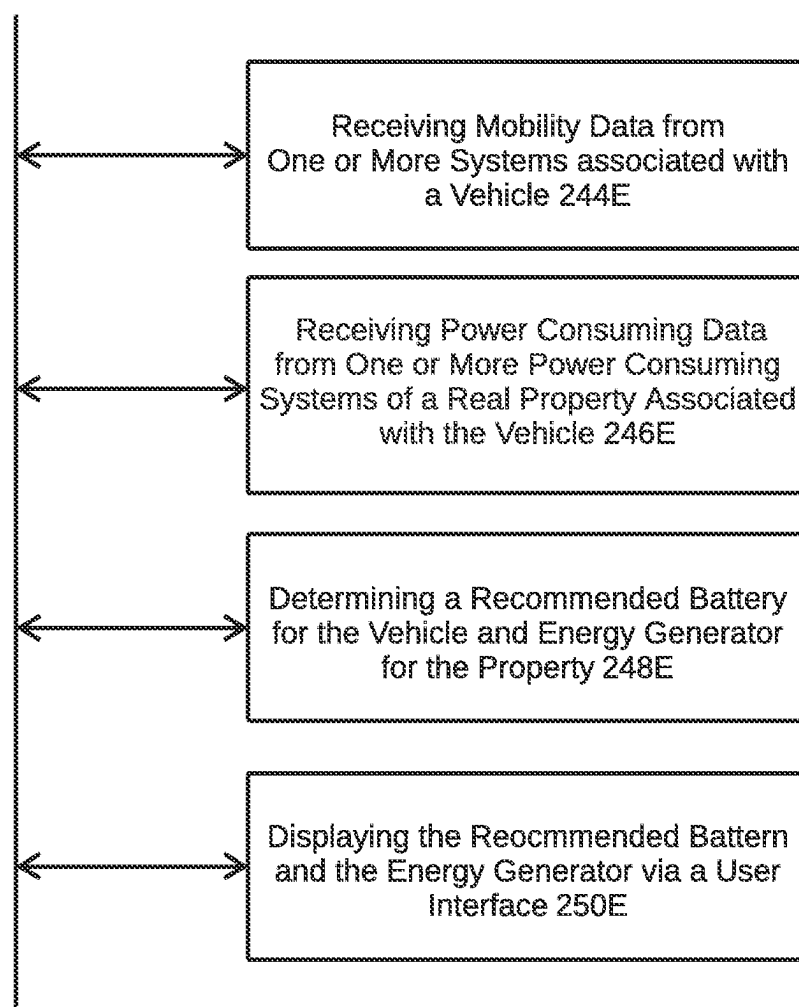
260

FIG. 2E

270

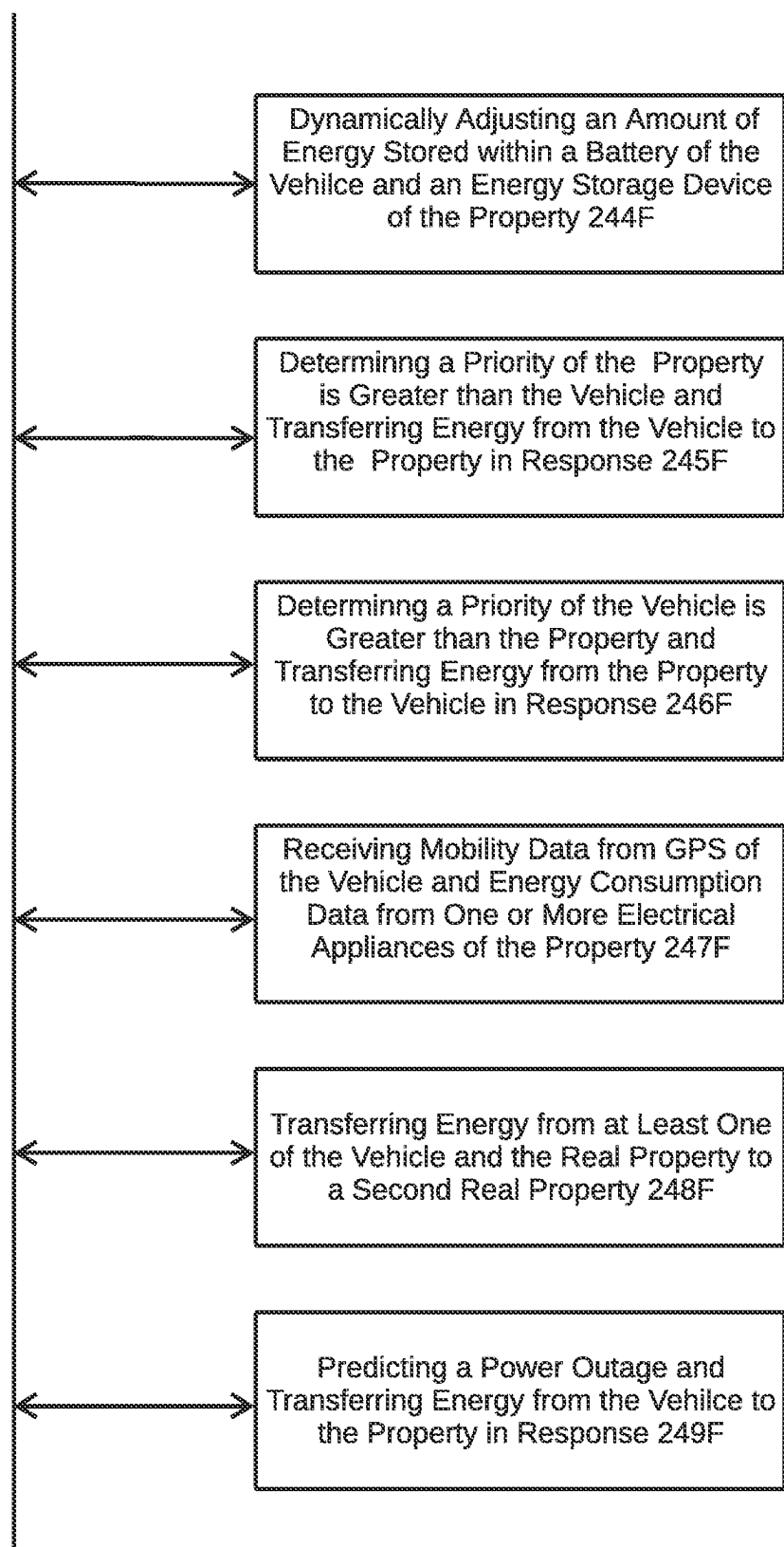


FIG. 2F

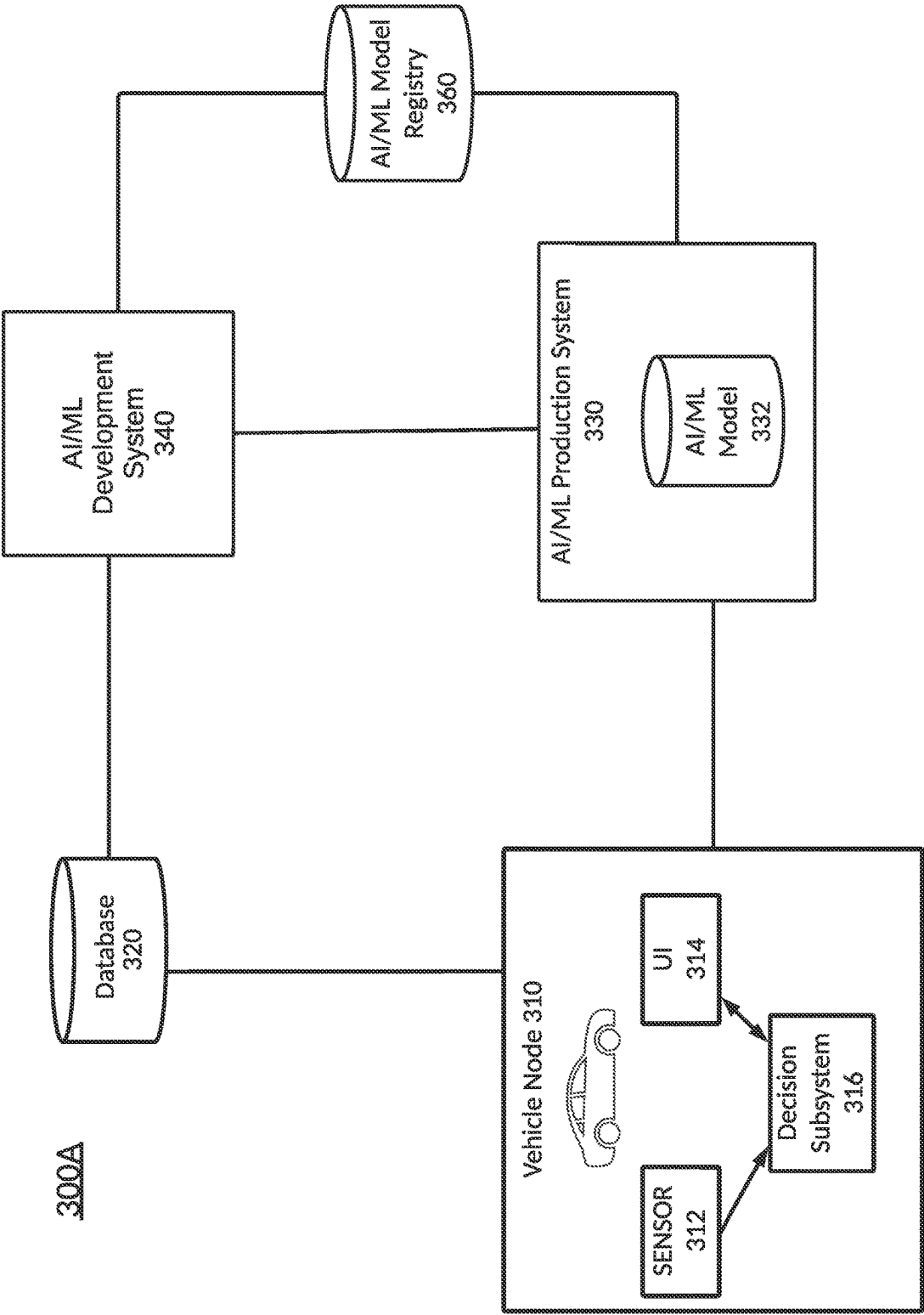


FIG. 3A

TMNA Reference No. IP-A-7092
12/27

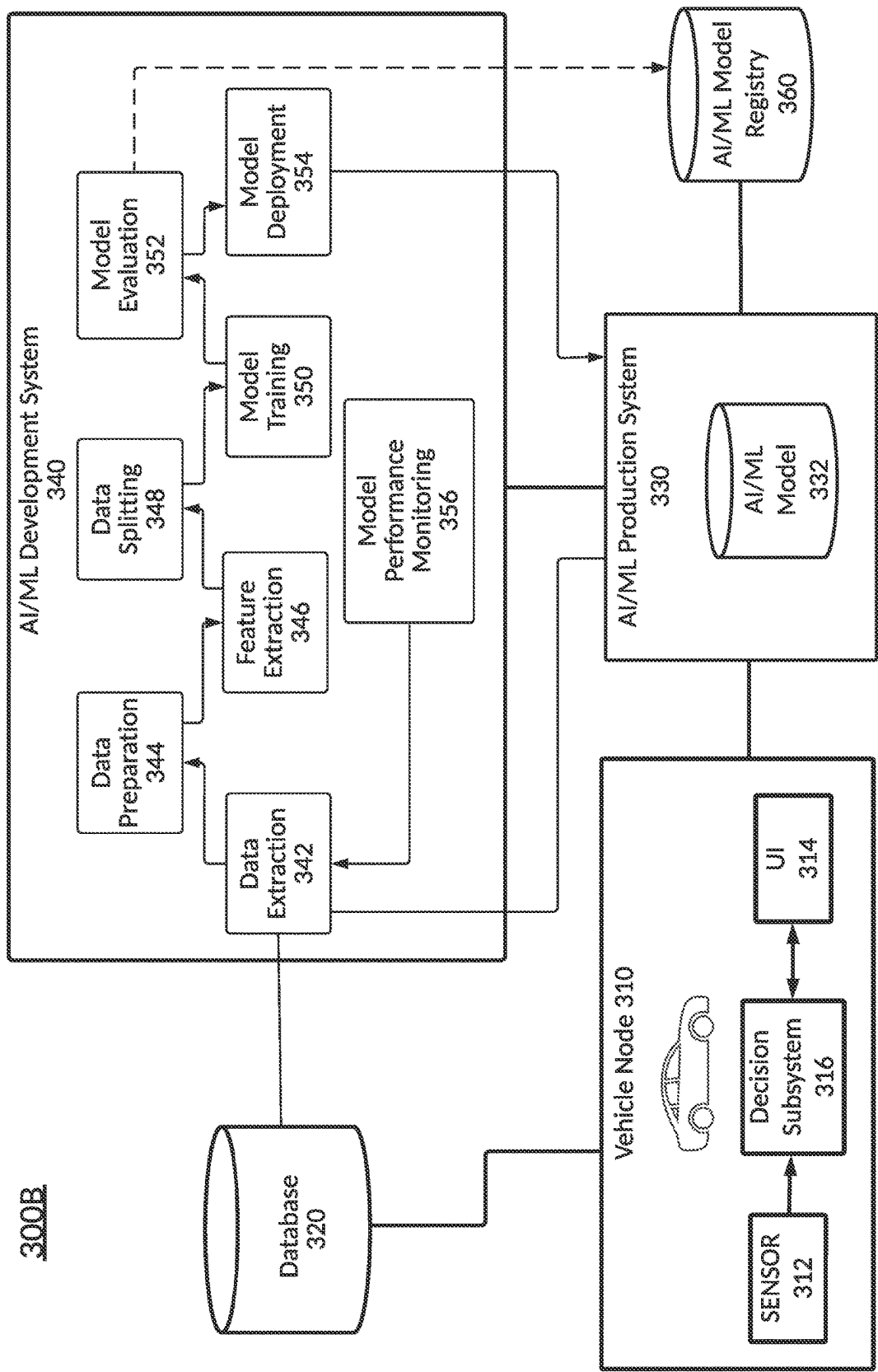


FIG. 3B

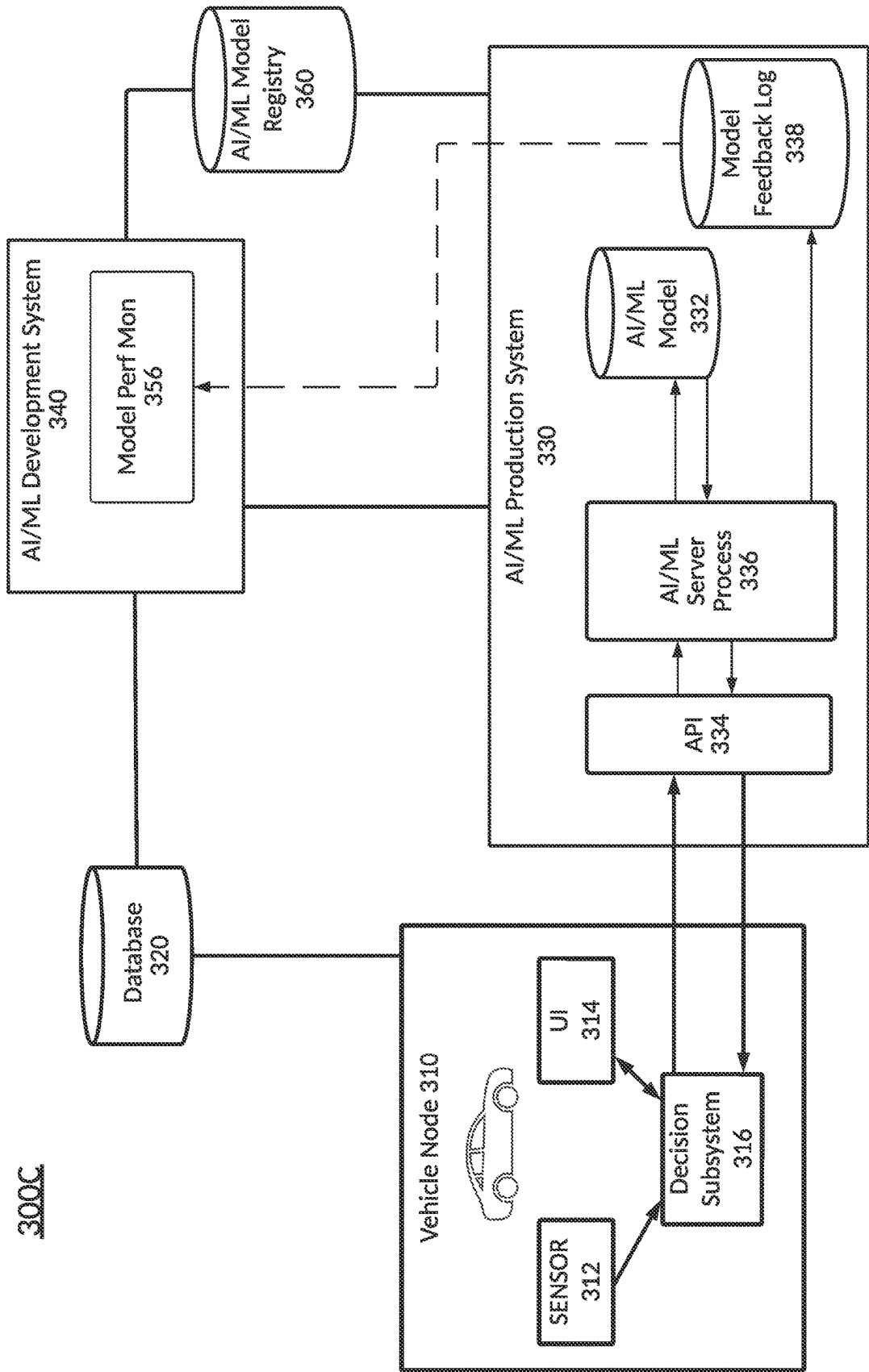


FIG. 3C

300D

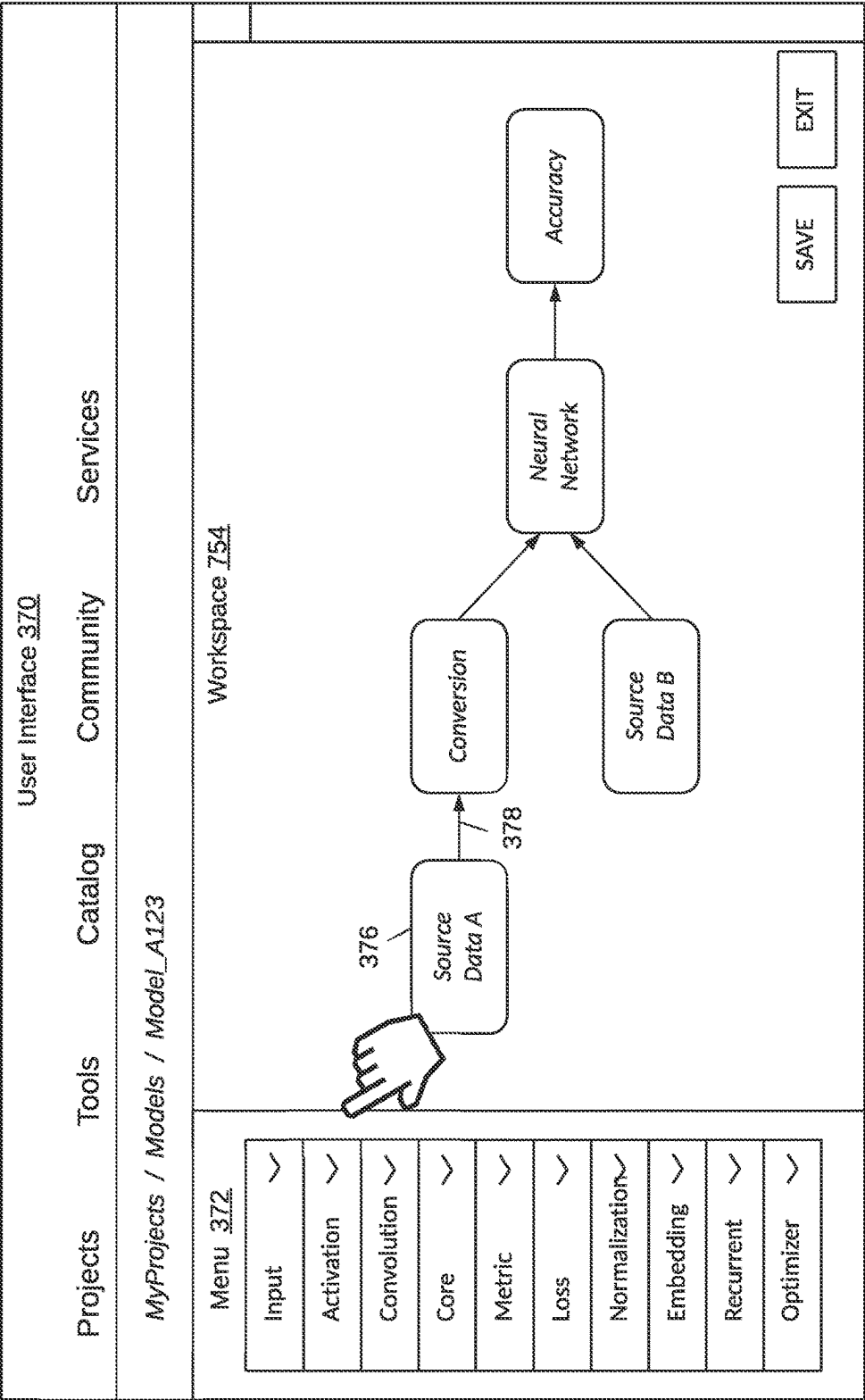


FIG. 3D

300E

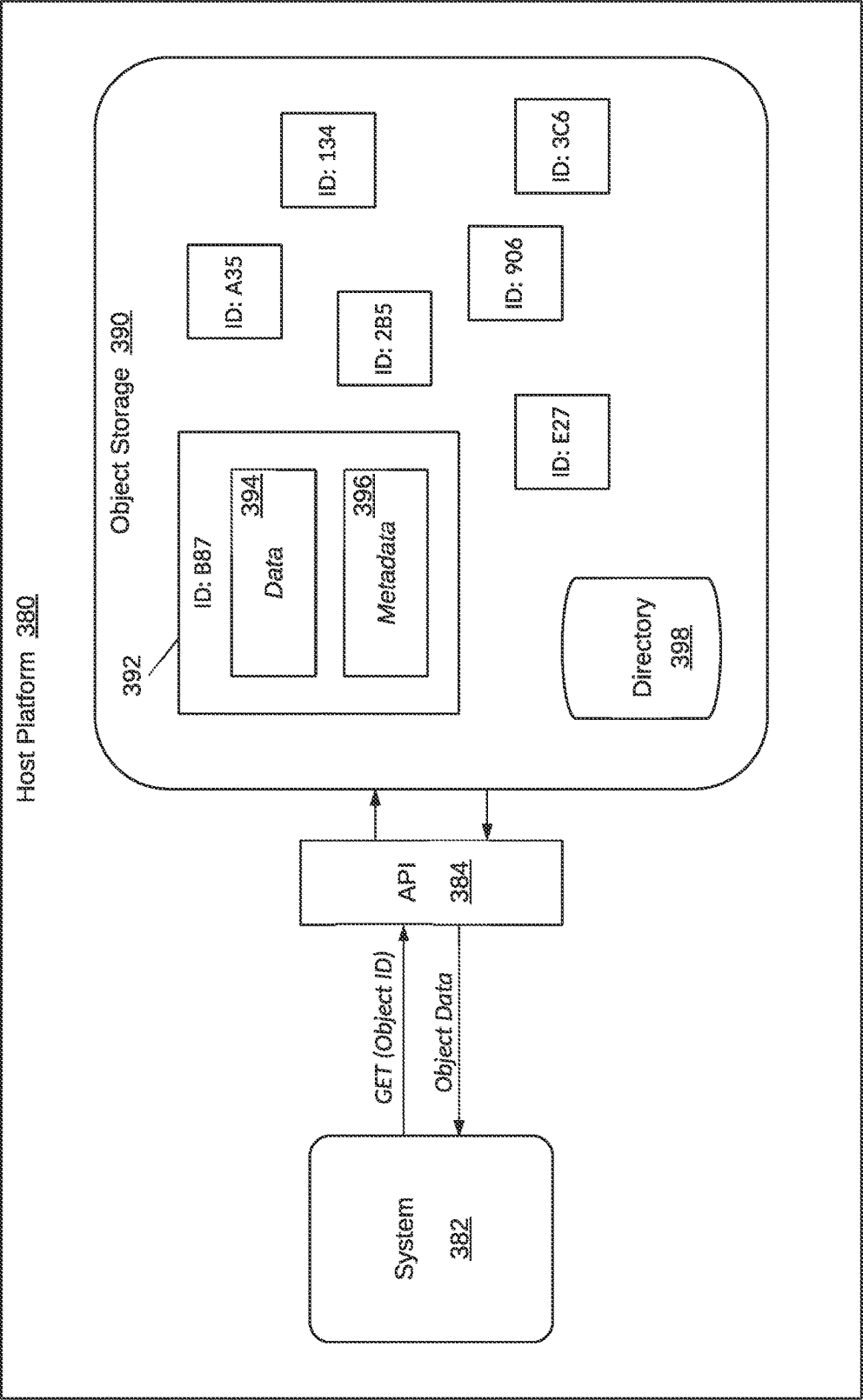


FIG. 3E

400A

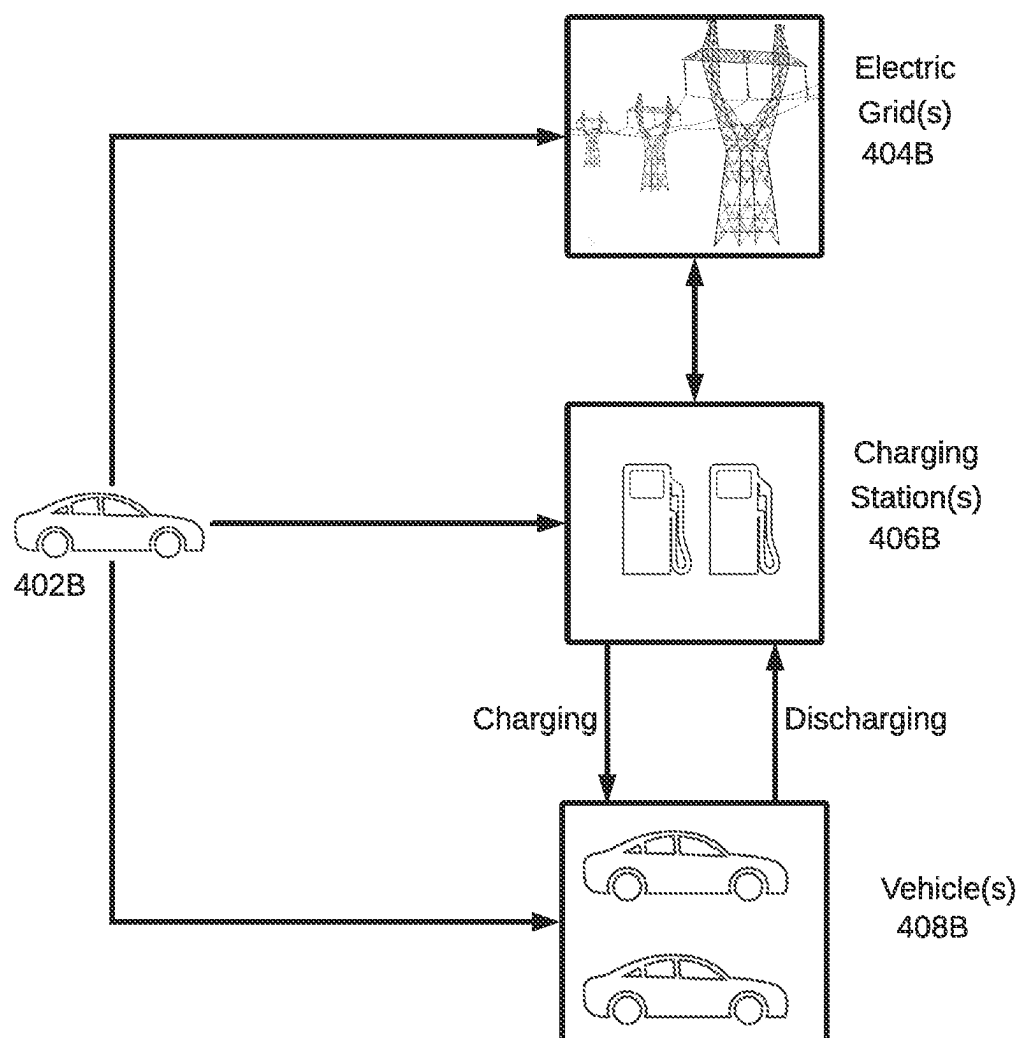


FIG. 4A

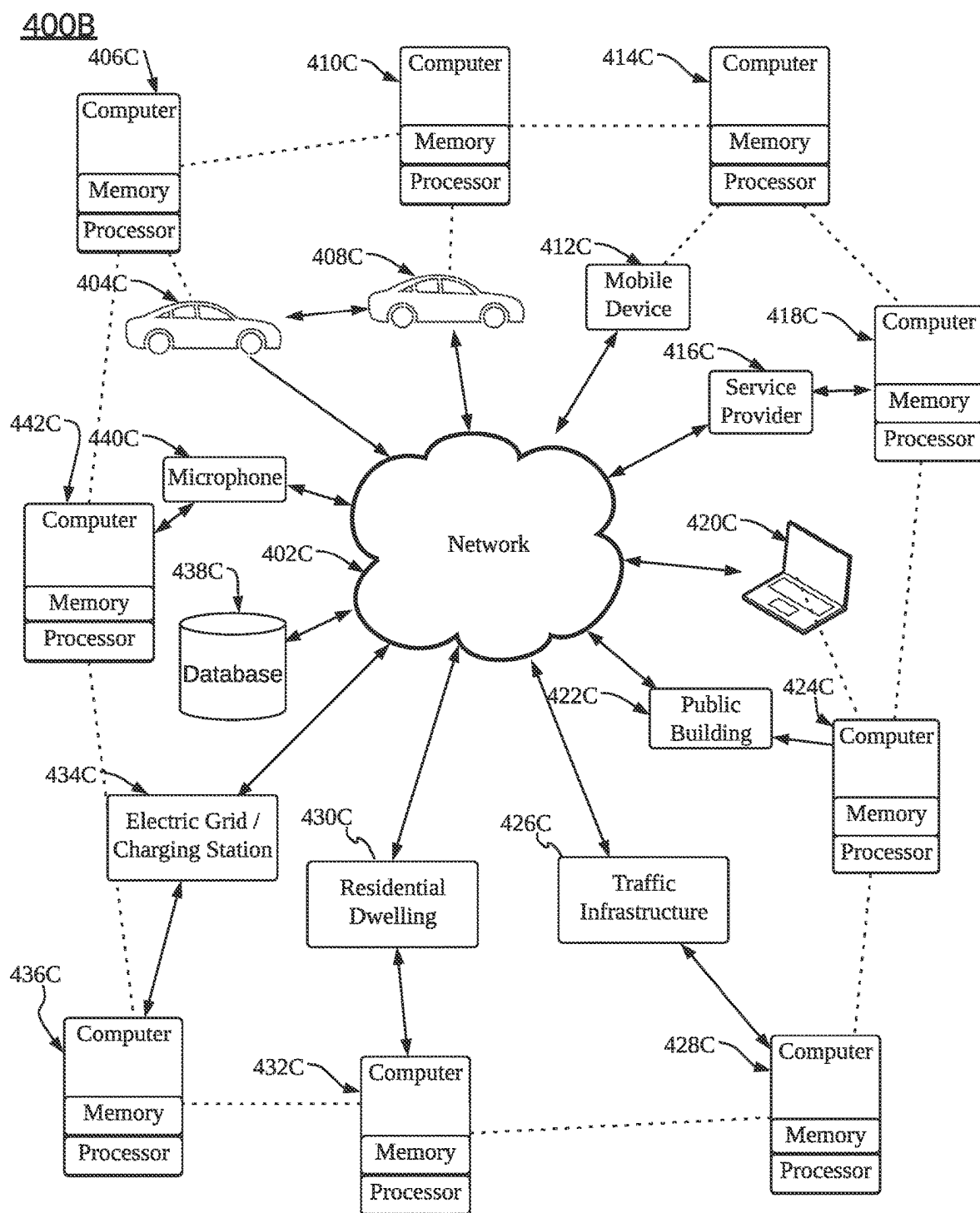


FIG. 4B

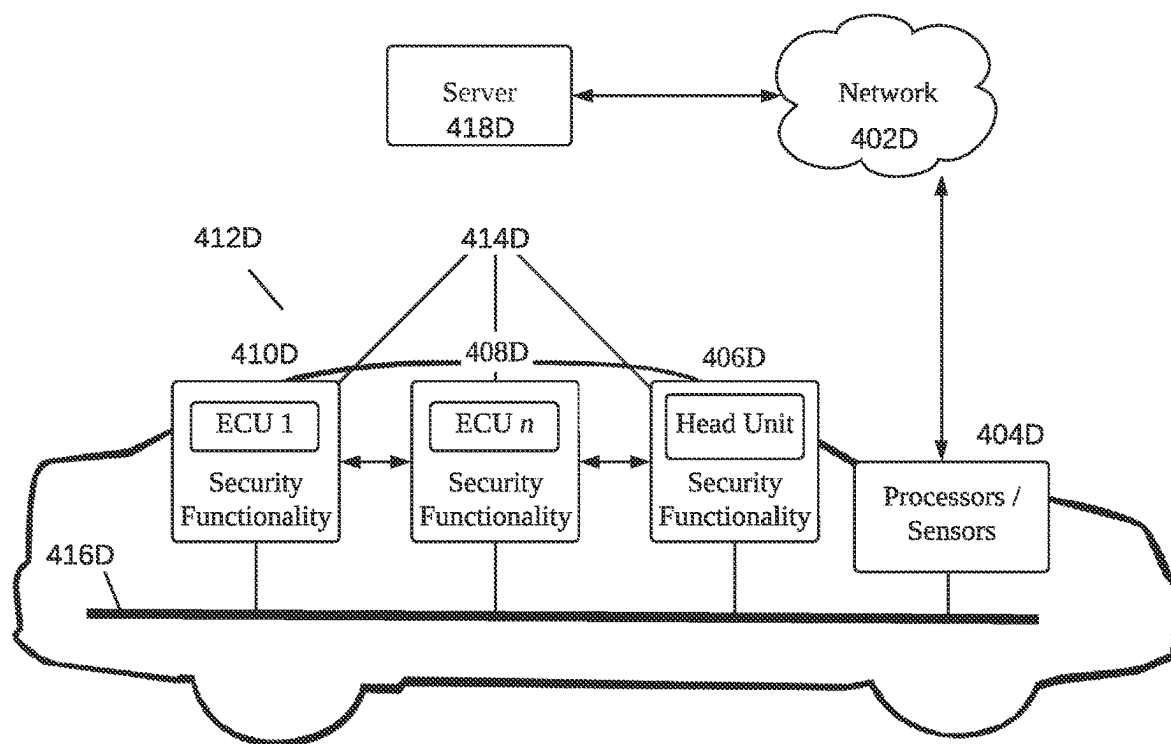


FIG. 4C

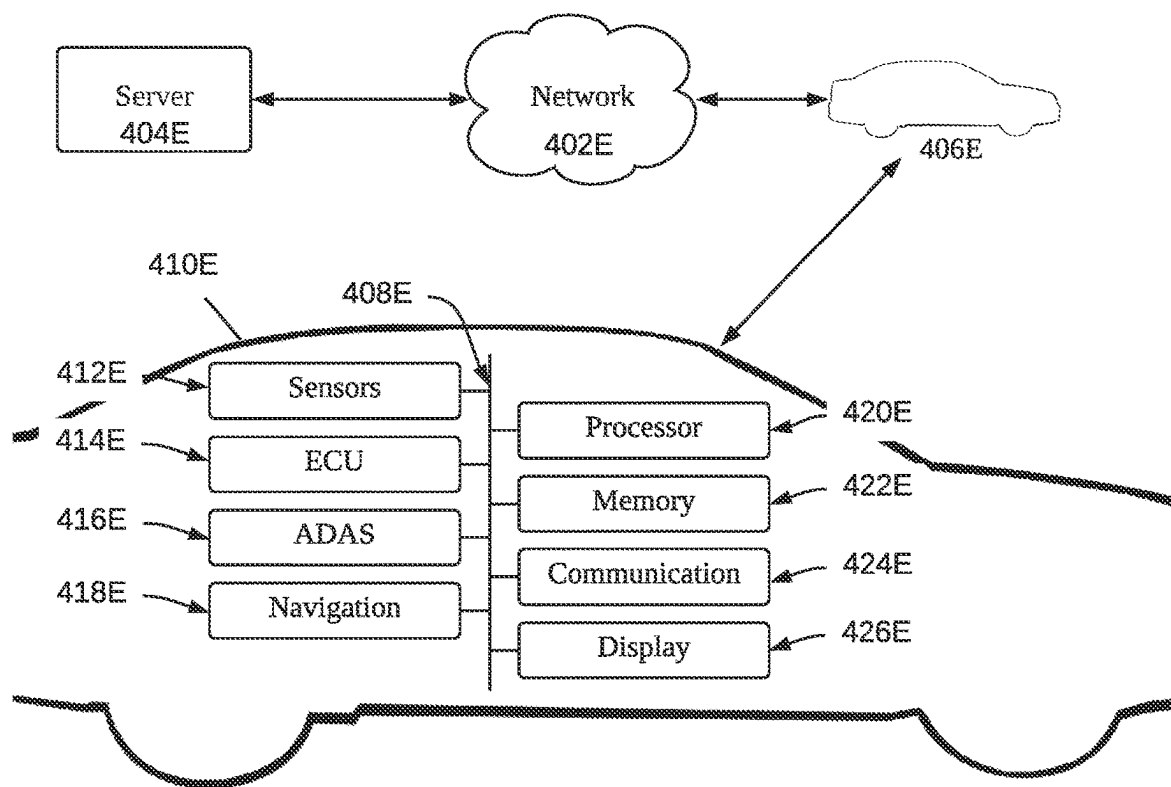


FIG. 4D

400E

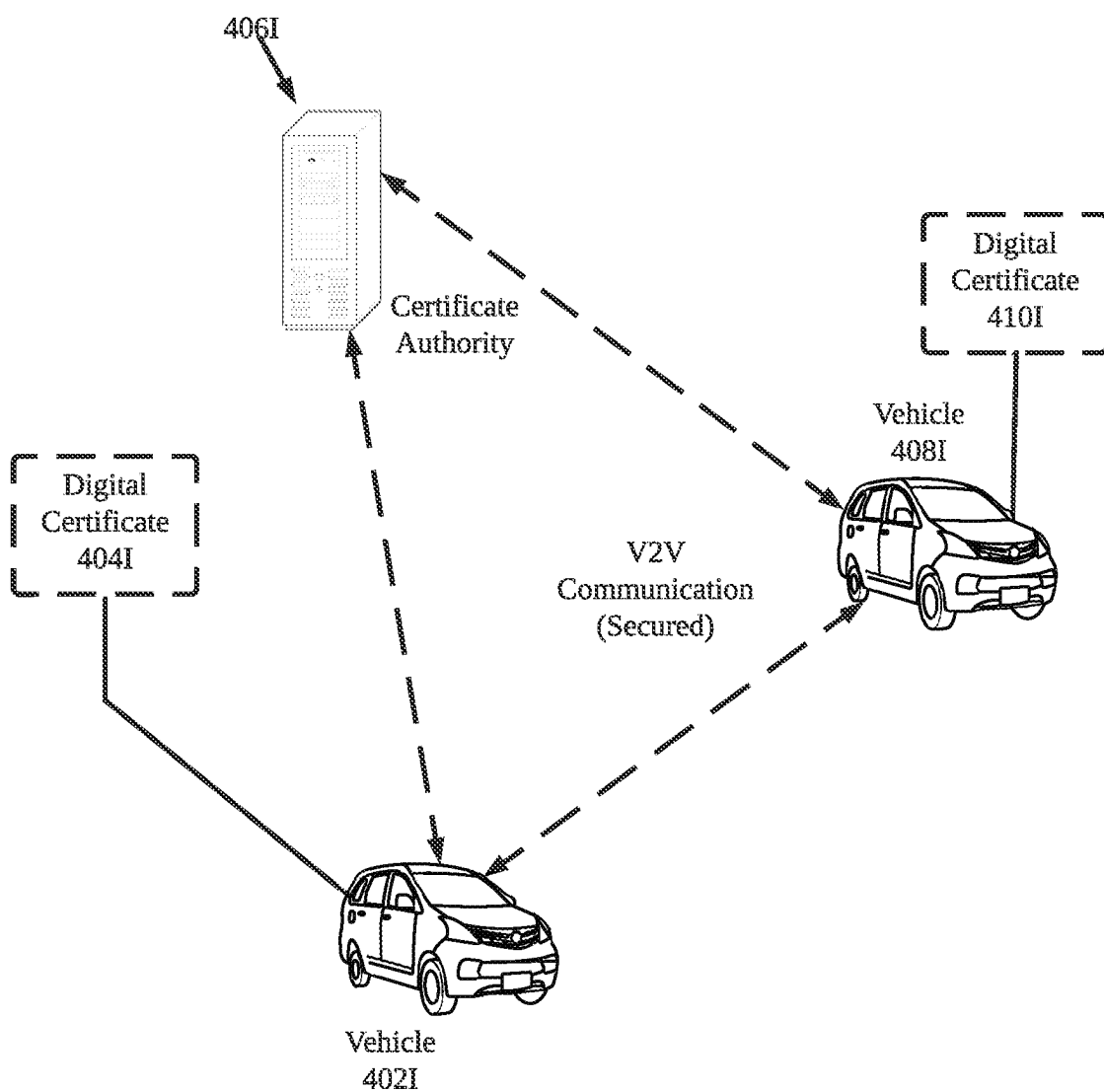


FIG. 4E

500A

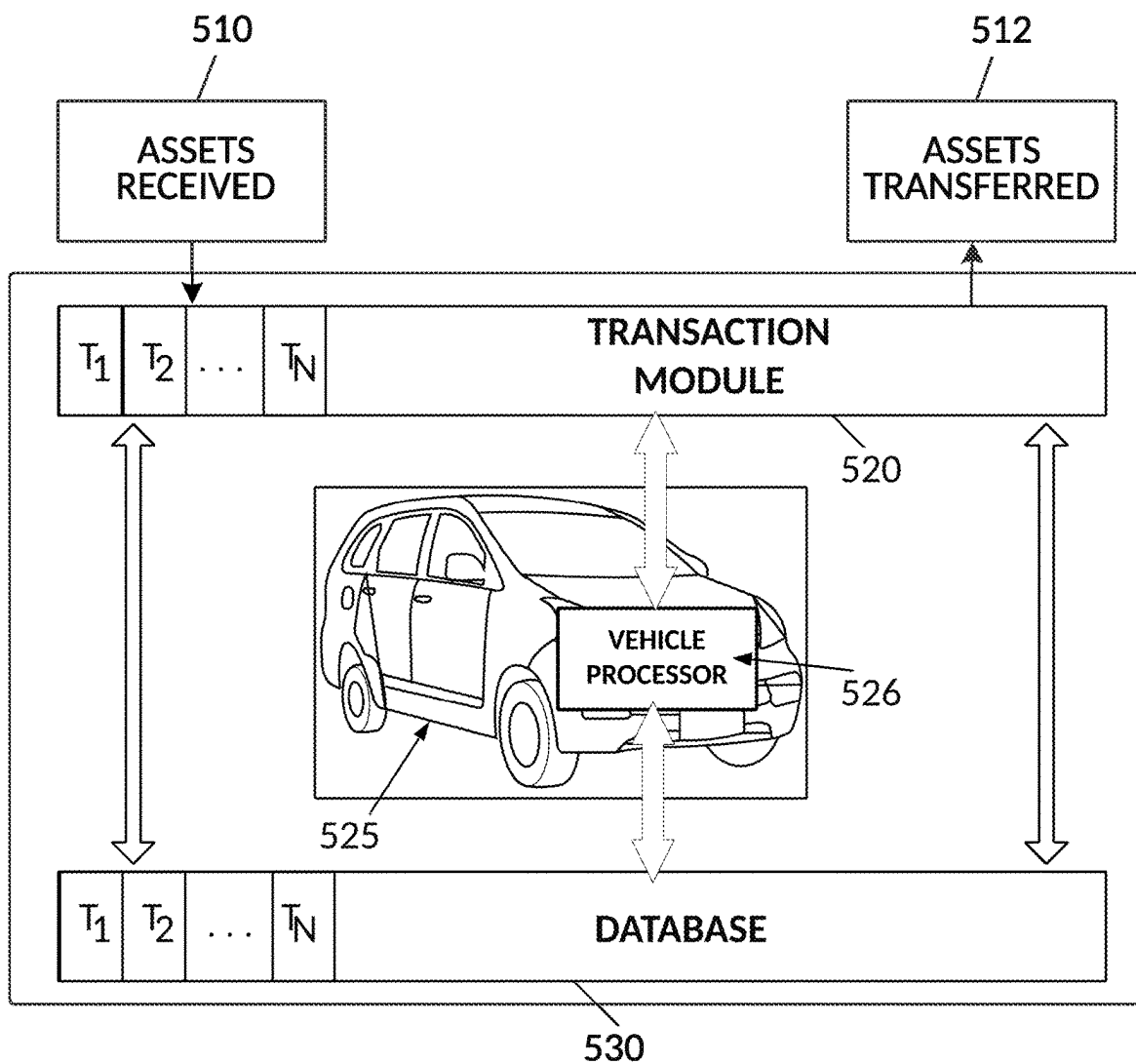


FIG. 5A

500B

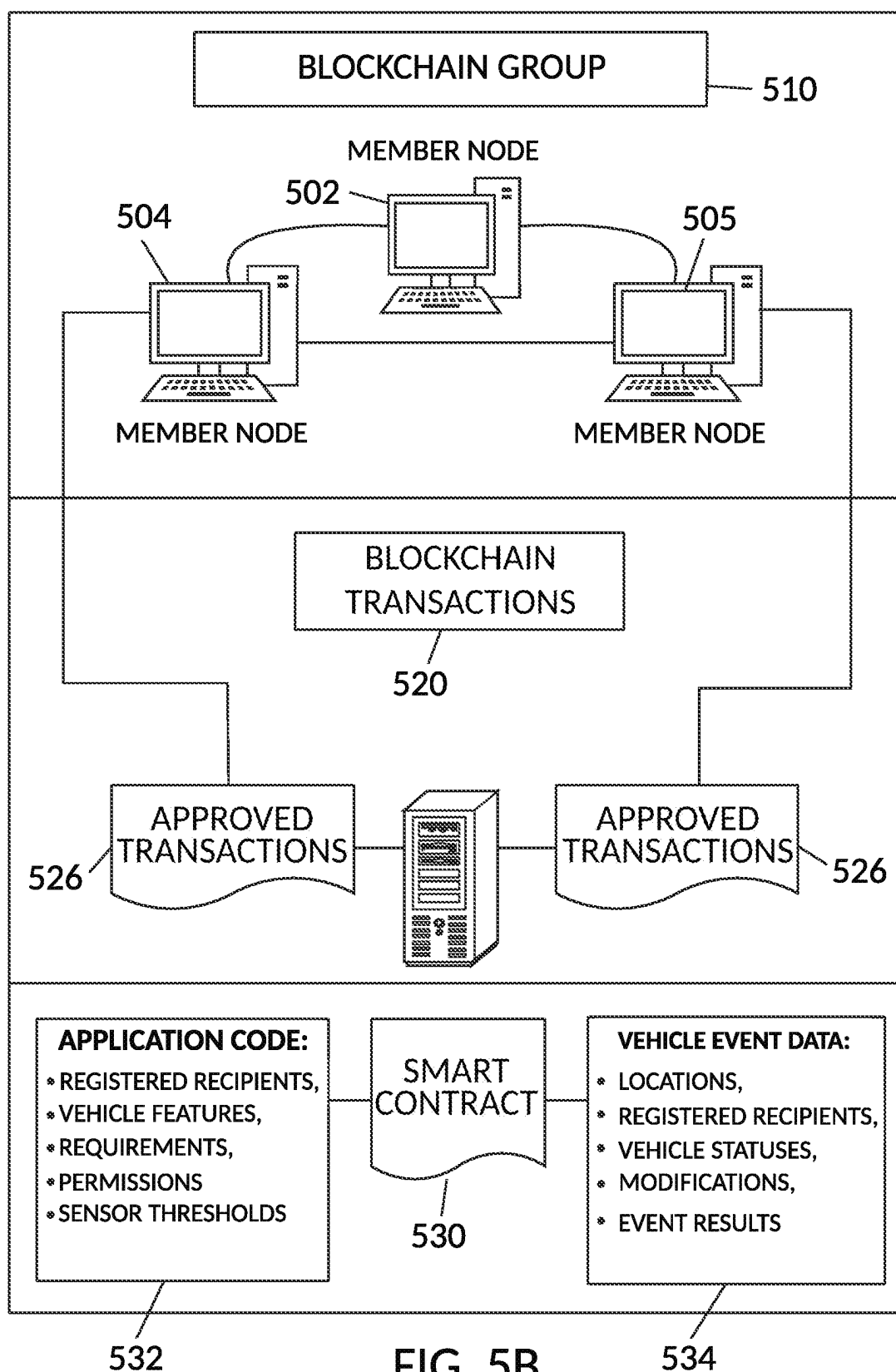


FIG. 5B

500C

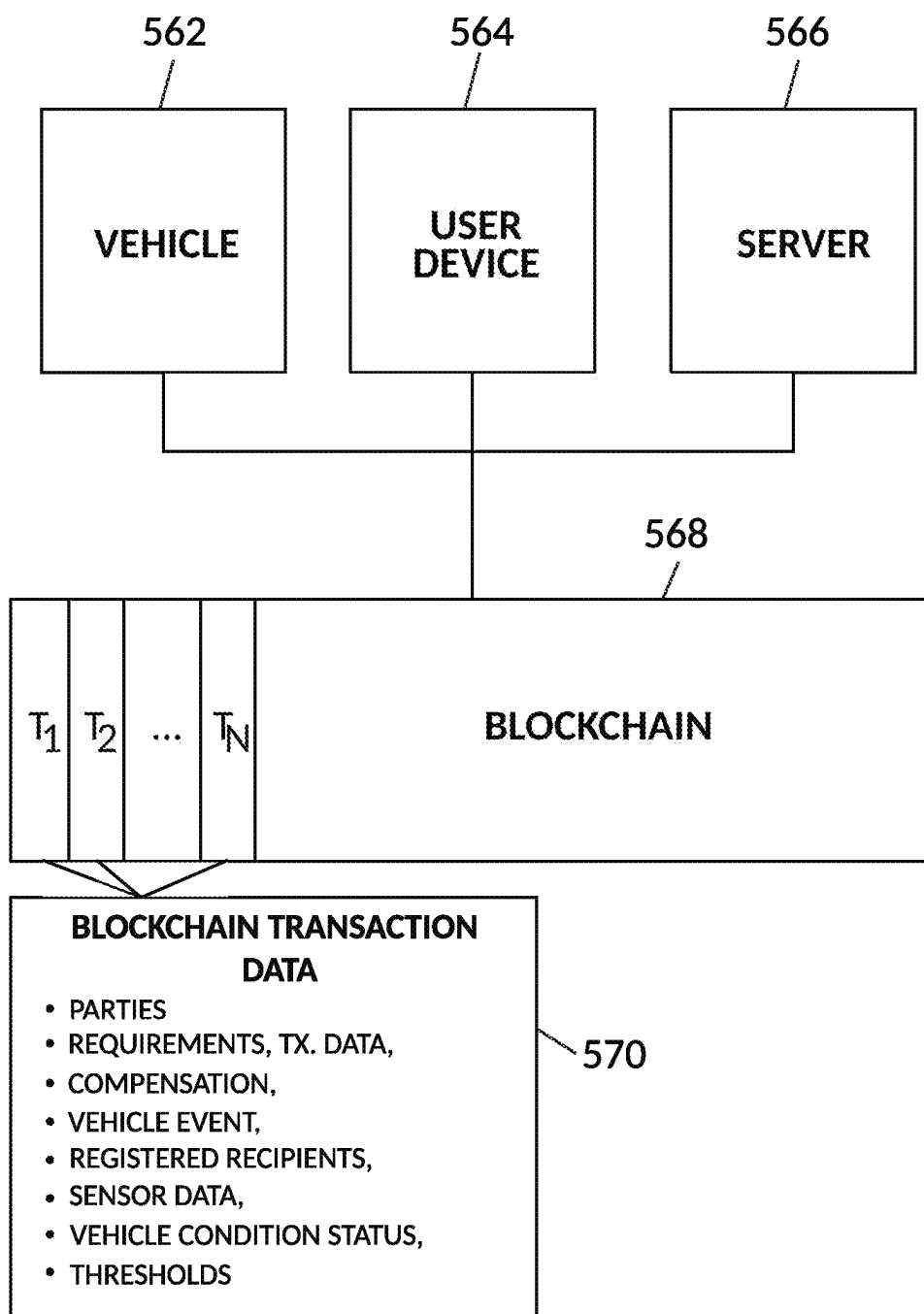
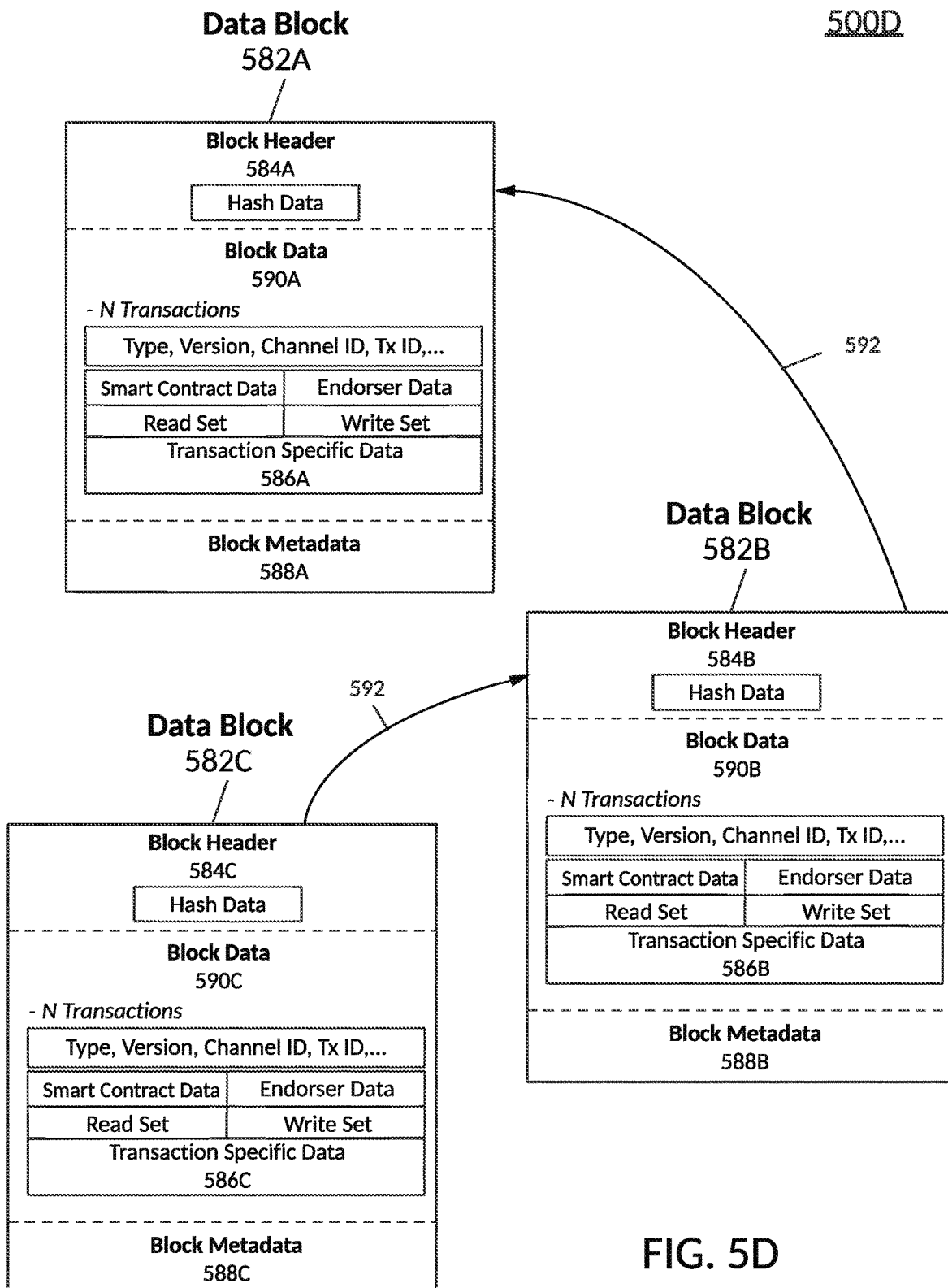


FIG. 5C



500E

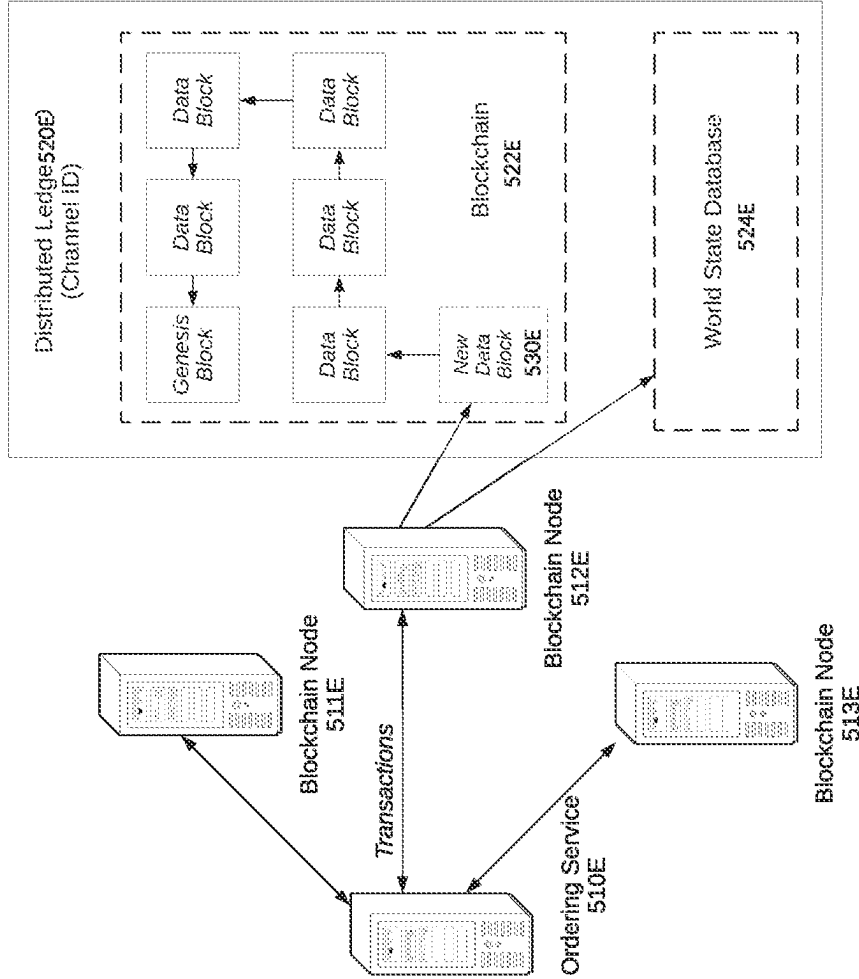


FIG. 5E

500E

New Data Block 530

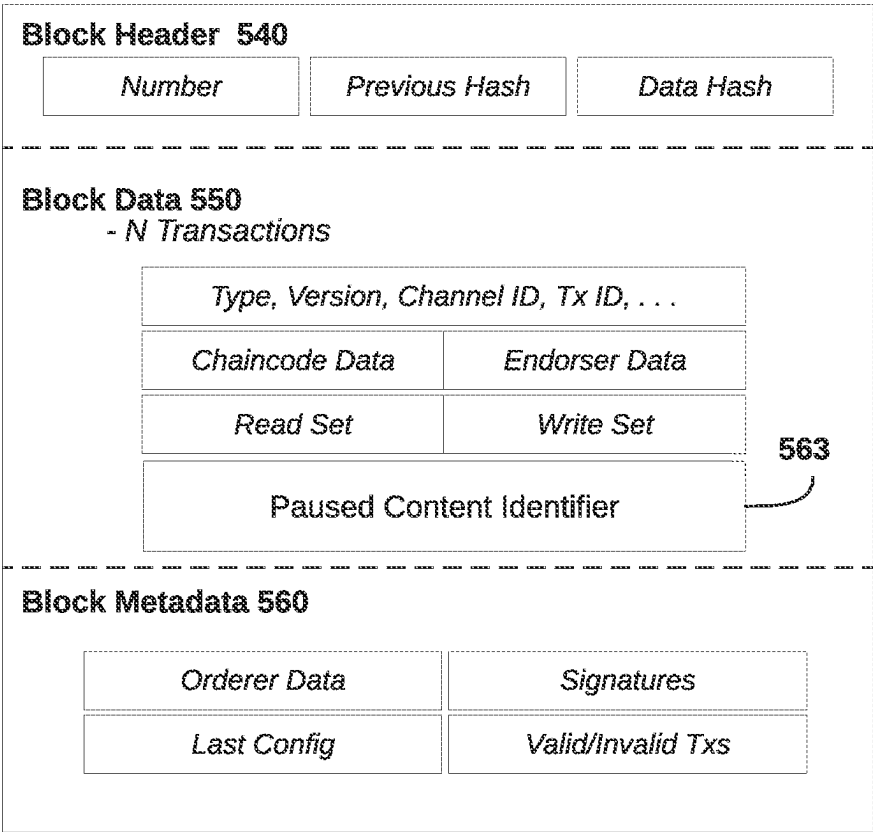


FIG. 5F

600

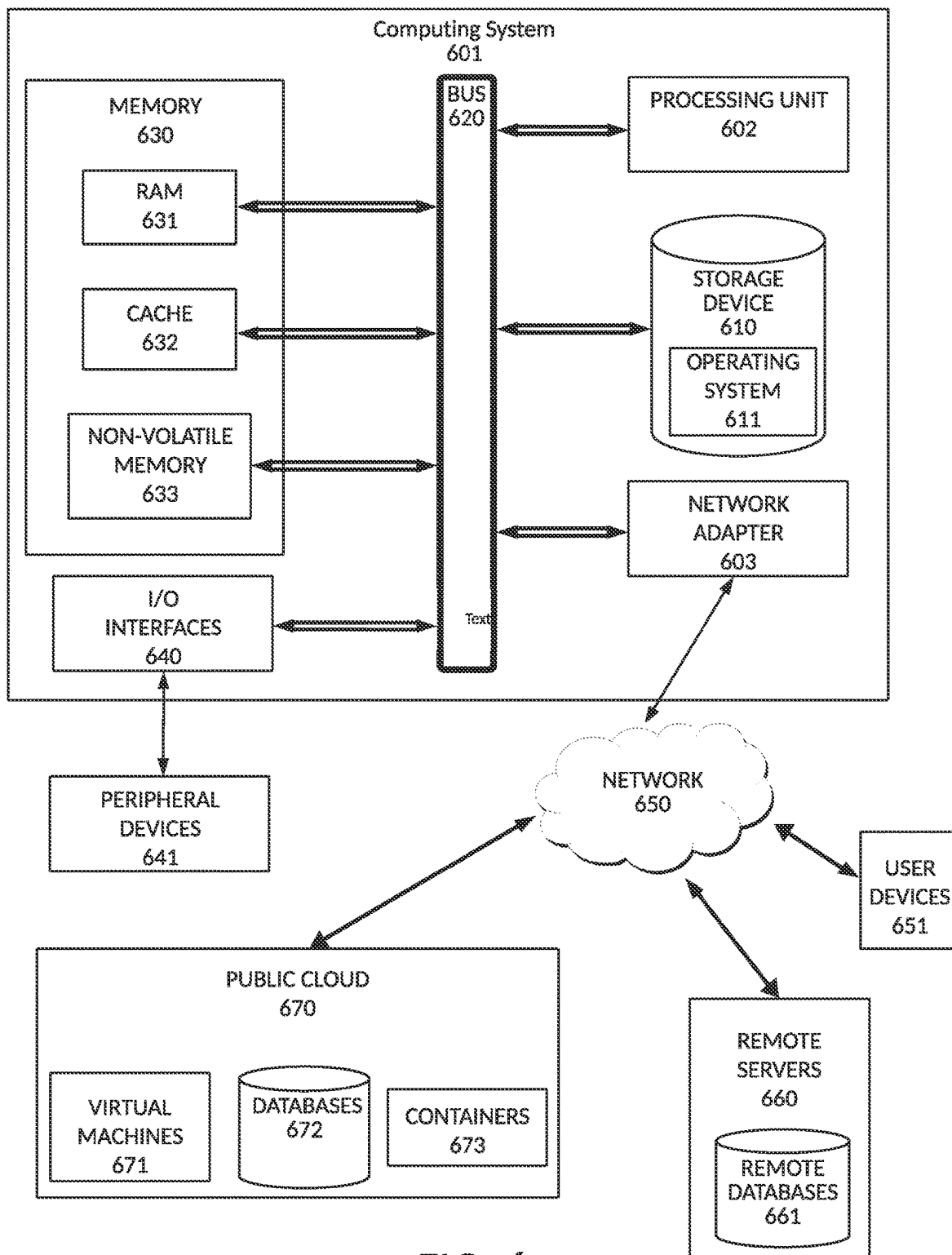


FIG. 6

ENERGY STORAGE CONFIGURATION MANAGEMENT BETWEEN A VEHICLE AND A PROPERTY ASSOCIATED WITH THE VEHICLE

BACKGROUND

[0001] Vehicles or transports, such as cars, motorcycles, trucks, planes, trains, etc., generally provide transportation needs to occupants and/or goods in a variety of ways. Functions related to vehicles may be identified and utilized by various computing devices, such as a smartphone or a computer located on and/or off the vehicle.

SUMMARY

[0002] One example embodiment provides a method that includes one or more of receiving, via a power management apparatus, mobility data comprising movements of a vehicle from one or more systems associated with the vehicle, receiving, via the power management apparatus, energy consumption data from one or more power consuming systems at a real property that is associated with the vehicle, determining a recommended battery for the vehicle and a recommended power generation system for the real property based on the mobility data and the energy consumption data, and displaying an identifier of the recommended battery and an identifier of the recommended power generation system via a user interface of the power management apparatus.

[0003] Another example embodiment provides an apparatus that includes a memory communicably coupled to a processor, wherein the processor may one or more of receive, via a power management apparatus, mobility data comprising movements of a vehicle from one or more systems associated with the vehicle, receive, via the power management apparatus, energy consumption data from one or more power consuming systems at a real property that is associated with the vehicle, determine a recommended battery for the vehicle and a recommended power generation system for the real property based on the mobility data and the energy consumption data, and display an identifier of the recommended battery and an identifier of the recommended power generation system via a user interface of the power management apparatus.

[0004] A further example embodiment provides a computer readable storage medium comprising instructions, that when read by a processor, cause the processor to perform one or more of receiving, via a power management apparatus, mobility data comprising movements of a vehicle from one or more systems associated with the vehicle, receiving, via the power management apparatus, energy consumption data from one or more power consuming systems at a real property that is associated with the vehicle, determining a recommended battery for the vehicle and a recommended power generation system for the real property based on the mobility data and the energy consumption data, and displaying an identifier of the recommended battery and an identifier of the recommended power generation system via a user interface of the power management apparatus.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1A is a diagram illustrating a process of managing power and making recommendations between a vehicle and a property associated with the vehicle according to example embodiments.

[0006] FIG. 1B is a diagram illustrating a process of recommending storage types for the vehicle and the property according to example embodiments.

[0007] FIG. 1C is a diagram illustrating a process of transferring energy between the vehicle and the property according to example embodiments.

[0008] FIG. 1D is a diagram illustrating a process of transferring power to a different property from one or more of the vehicle and the property associated with the vehicle according to example embodiments.

[0009] FIG. 2A illustrates a vehicle network diagram, according to example embodiments.

[0010] FIG. 2B illustrates another vehicle network diagram, according to example embodiments.

[0011] FIG. 2C illustrates yet another vehicle network diagram, according to example embodiments.

[0012] FIG. 2D illustrates a further vehicle network diagram, according to example embodiments.

[0013] FIG. 2E illustrates a flow diagram, according to example embodiments.

[0014] FIG. 2F illustrates another flow diagram, according to example embodiments.

[0015] FIG. 3A illustrates an Artificial Intelligence (AI)/Machine Learning (ML) network diagram for integrating an artificial intelligence (AI) model into any decision point in the example embodiments.

[0016] FIG. 3B illustrates a process for developing an Artificial Intelligence (AI)/Machine Learning (ML) model that supports AI-assisted vehicle or occupant decision points.

[0017] FIG. 3C illustrates a process for utilizing an Artificial Intelligence (AI)/Machine Learning (ML) model that supports AI-assisted vehicle or occupant decision points.

[0018] FIG. 3D illustrates a machine learning network diagram, according to example embodiments.

[0019] FIG. 3E illustrates a machine learning network diagram, according to example embodiments.

[0020] FIG. 4A illustrates a diagram depicting electrification of one or more elements, according to example embodiments.

[0021] FIG. 4B illustrates a diagram depicting interconnections between different elements, according to example embodiments.

[0022] FIG. 4C illustrates a further diagram depicting interconnections between different elements, according to example embodiments.

[0023] FIG. 4D illustrates yet a further diagram depicting interconnections between elements, according to example embodiments.

[0024] FIG. 4E illustrates yet a further diagram depicting an example of vehicles performing secured Vehicle-to-Vehicle (V2V) communications using security certificates, according to example embodiments.

[0025] FIG. 5A illustrates an example vehicle configuration for managing database transactions associated with a vehicle, according to example embodiments.

[0026] FIG. 5B illustrates an example blockchain group, according to example embodiments.

[0027] FIG. 5C illustrates an example interaction between elements and a blockchain, according to example embodiments.

[0028] FIG. 5D illustrates an example data block interaction, according to example embodiments.

[0029] FIG. 5E illustrates a blockchain network diagram, according to example embodiments.

[0030] FIG. 5F illustrates an example new data block, according to example embodiments.

[0031] FIG. 6 illustrates an example system that supports one or more of the example embodiments.

DETAILED DESCRIPTION

[0032] It will be readily understood that the instant components, as generally described and illustrated in the figures herein, may be arranged and designed in a wide variety of different configurations. Thus, the following detailed description of the embodiments of at least one of a method, apparatus, computer readable storage medium and system, as represented in the attached figures, is not intended to limit the scope of the application as claimed but is merely representative of selected embodiments. Multiple embodiments depicted herein are not intended to limit the scope of the solution. The computer-readable storage medium may be a non-transitory computer readable medium or a non-transitory computer readable storage medium.

[0033] Communications between the vehicle(s) and certain entities, such as remote servers, other vehicles and local computing devices (e.g., smartphones, personal computers, vehicle-embedded computers, etc.) may be sent and/or received and processed by one or more ‘components’ which may be hardware, firmware, software, or a combination thereof. The components may be part of any of these entities or computing devices or certain other computing devices. In one example, consensus decisions related to blockchain transactions may be performed by one or more computing devices or components (which may be any element described and/or depicted herein) associated with the vehicle (s) and one or more of the components outside or at a remote location from the vehicle(s).

[0034] The instant features, structures, or characteristics described in this specification may be combined in any suitable manner in one or more embodiments. For example, the usage of the phrases “example embodiments,” “some embodiments,” “a first embodiment”, or other similar language throughout this specification refers to the fact that a particular feature, structure, or characteristic described in connection with the one or more embodiments may be included in one or more other embodiments described or depicted herein. Thus, the one or more embodiments, described or depicted throughout this specification can all refer to the same embodiment. Thus, these embodiments may work in conjunction with any of the other embodiments, may not be functionally separate, and the described features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. Although described in a particular manner, by example only, or more feature(s), element(s), and step(s) described herein may be utilized together and in various combinations, without exclusivity, unless expressly indicated otherwise herein. In the figures, any connection between elements can permit one-way and/or two-way communication, even if the depicted connection is a one-way or two-way connection, such as an arrow.

[0035] In the instant solution, a vehicle may include one or more of cars, trucks, Internal Combustion Engine (ICE) vehicles, battery electric vehicle (BEV), fuel cell vehicles, any vehicle utilizing renewable sources, hybrid vehicles, e-Pallettes, buses, motorcycles, scooters, bicycles, boats,

recreational vehicles, planes, drones, Unmanned Aerial Vehicle (UAV) and any object that may be used to transport people and/or goods from one location to another.

[0036] In addition, while the term “message” may have been used in the description of embodiments, other types of network data, such as, a packet, frame, datagram, etc. may also be used. Furthermore, while certain types of messages and signaling may be depicted in exemplary embodiments they are not limited to a certain type of message and signaling.

[0037] Example embodiments provide methods, systems, components, non-transitory computer readable medium, devices, and/or networks, which provide at least one of a transport (also referred to as a vehicle or car herein), a data collection system, a data monitoring system, a verification system, an authorization system, and a vehicle data distribution system. The vehicle status condition data received in the form of communication messages, such as wireless data network communications and/or wired communication messages, may be processed to identify vehicle status conditions and provide feedback on the condition and/or changes of a vehicle. In one example, a user profile may be applied to a particular vehicle to authorize a current vehicle event, service stops at service stations, to authorize subsequent vehicle rental services, and enable vehicle-to-vehicle communications.

[0038] Within the communication infrastructure, a decentralized database is a distributed storage system which includes multiple nodes that communicate with each other. A blockchain is an example of a decentralized database, which includes an append-only immutable data structure (i.e., a distributed ledger) capable of maintaining records between untrusted parties. The untrusted parties are referred to herein as peers, nodes, or peer nodes. Each peer maintains a copy of the database records, and no single peer can modify the database records without a consensus being reached among the distributed peers. For example, the peers may execute a consensus protocol to validate blockchain storage entries, group the storage entries into blocks, and build a hash chain via the blocks. This process forms the ledger by ordering the storage entries, as is necessary, for consistency. In public or permissionless blockchains, anyone can participate without a specific identity. Public blockchains can involve crypto-currencies and use consensus-based on various protocols such as proof of work (PoW). Conversely, a permissioned blockchain database can secure interactions among a group of entities, which share a common goal, but which do not or cannot fully trust one another, such as businesses that exchange funds, goods, information, and the like. The instant solution can function in a permissioned and/or a permissionless blockchain setting.

[0039] Smart contracts are trusted distributed applications which leverage tamper-proof properties of the shared or distributed ledger (which may be in the form of a blockchain) and an underlying agreement between member nodes, which is referred to as an endorsement or endorsement policy. In general, blockchain entries are “endorsed” before being committed to the blockchain while entries which are not endorsed are disregarded. A typical endorsement policy allows smart contract executable code to specify endorses for an entry in the form of a set of peer nodes that are necessary for endorsement. When a client sends the entry to the peers specified in the endorsement policy, the entry is executed to validate the entry. After validation, the entries

enter an ordering phase in which a consensus protocol produces an ordered sequence of endorsed entries grouped into blocks.

[0040] Nodes are the communication entities of the blockchain system. A “node” may perform a logical function in the sense that multiple nodes of different types can run on the same physical server. Nodes are grouped in trust domains and are associated with logical entities that control them in various ways. Nodes may include different types, such as a client or submitting-client node, which submits an entry-invocation to an endorser (e.g., peer), and broadcasts entry proposals to an ordering service (e.g., ordering node). Another type of node is a peer node, which can receive client submitted entries, commit the entries, and maintain a state and a copy of the ledger of blockchain entries. Peers can also have the role of an endorser. An ordering-service-node or orderer is a node running the communication service for all nodes and which implements a delivery guarantee, such as a broadcast to each of the peer nodes in the system when committing entries and modifying a world state of the blockchain. The world state can constitute the initial blockchain entry, which normally includes control and setup information.

[0041] A ledger is a sequenced, tamper-resistant record of all state transitions of a blockchain. State transitions may result from smart contract executable code invocations (i.e., entries) submitted by participating parties (e.g., client nodes, ordering nodes, endorser nodes, peer nodes, etc.). An entry may result in a set of asset key-value pairs being committed to the ledger as one or more operands, such as creates, updates, deletes, and the like. The ledger includes a blockchain (also referred to as a chain), which stores an immutable, sequenced record in blocks. The ledger also includes a state database, which maintains a current state of the blockchain. There is typically one ledger per channel. Each peer node maintains a copy of the ledger for each channel of which they are a member.

[0042] A chain is an entry log structured as hash-linked blocks, and each block contains a sequence of N entries where N is equal to or greater than one. The block header includes a hash of the blocks’ entries, as well as a hash of the prior block’s header. In this way, all entries on the ledger may be sequenced and cryptographically linked together. Accordingly, it is not possible to tamper with the ledger data without breaking the hash links. A hash of a most recently added blockchain block represents every entry on the chain that has come before it, making it possible to ensure that all peer nodes are in a consistent and trusted state. The chain may be stored on a peer node file system (i.e., local, attached storage, cloud, etc.), efficiently supporting the append-only nature of the blockchain workload.

[0043] The current state of the immutable ledger represents the latest values for all keys that are included in the chain entry log. Since the current state represents the latest key values known to a channel, it is sometimes referred to as a world state. Smart contract executable code invocations execute entries against the current state data of the ledger. To make these smart contract executable code interactions efficient, the latest values of the keys may be stored in a state database. The state database may be simply an indexed view into the chain’s entry log and can therefore be regenerated from the chain at any time. The state database may automatically be recovered (or generated if needed) upon peer node startup and before entries are accepted.

[0044] A blockchain is different from a traditional database in that the blockchain is not a central storage but rather a decentralized, immutable, and secure storage, where nodes must share in changes to records in the storage. Some properties that are inherent in blockchain and which help implement the blockchain include, but are not limited to, an immutable ledger, smart contracts, security, privacy, decentralization, consensus, endorsement, accessibility, and the like.

[0045] Example embodiments provide a service to a particular vehicle and/or a user profile that is applied to the vehicle. For example, a user may be the owner of a vehicle or the operator of a vehicle owned by another party. The vehicle may require service at certain intervals, and the service needs may require authorization before permitting the services to be received. Also, service centers may offer services to vehicles in a nearby area based on the vehicle’s current route plan and a relative level of service requirements (e.g., immediate, severe, intermediate, minor, etc.). The vehicle needs may be monitored via one or more vehicle and/or road sensors or cameras, which report sensed data to a central controller computer device in and/or apart from the vehicle. This data is forwarded to a management server for review and action. A sensor may be located on one or more of the interior of the vehicle, the exterior of the vehicle, on a fixed object apart from the vehicle, and on another vehicle proximate the vehicle. The sensor may also be associated with the vehicle’s speed, the vehicle’s braking, the vehicle’s acceleration, fuel levels, service needs, the gear-shifting of the vehicle, the vehicle’s steering, and the like. A sensor, as described herein, may also be a device, such as a wireless device in and/or proximate to the vehicle. Also, sensor information may be used to identify whether the vehicle is operating safely and whether an occupant has engaged in any unexpected vehicle conditions, such as during a vehicle access and/or utilization period. Vehicle information collected before, during and/or after a vehicle’s operation may be identified and stored in a transaction on a shared/distributed ledger, which may be generated and committed to the immutable ledger as determined by a permission granting consortium, and thus in a “decentralized” manner, such as via a blockchain membership group.

[0046] Each interested party (i.e., owner, user, company, agency, etc.) may want to limit the exposure of private information, and therefore the blockchain and its immutability can be used to manage permissions for each particular user vehicle profile. A smart contract may be used to provide compensation, quantify a user profile score/rating/review, apply vehicle event permissions, determine when service is needed, identify a collision and/or degradation event, identify a safety concern event, identify parties to the event and provide distribution to registered entities seeking access to such vehicle event data. Also, the results may be identified, and the necessary information can be shared among the registered companies and/or individuals based on a consensus approach associated with the blockchain. Such an approach may not be implemented on a traditional centralized database.

[0047] Various driving systems of the instant solution can utilize software, an array of sensors as well as machine learning functionality, light detection and ranging (Lidar) projectors, radar, ultrasonic sensors, etc. to create a map of terrain and road that a vehicle can use for navigation and

other purposes. In some embodiments, GPS, maps, cameras, sensors, and the like can also be used in autonomous vehicles in place of Lidar.

[0048] The instant solution includes, in certain embodiments, authorizing a vehicle for service via an automated and quick authentication scheme. For example, driving up to a charging station or fuel pump may be performed by a vehicle operator or an autonomous vehicle and the authorization to receive charge or fuel may be performed without any delays provided the authorization is received by the service and/or charging station. A vehicle may provide a communication signal that provides an identification of a vehicle that has a currently active profile linked to an account that is authorized to accept a service, which can be later rectified by compensation. Additional measures may be used to provide further authentication, such as another identifier may be sent from the user's device wirelessly to the service center to replace or supplement the first authorization effort between the vehicle and the service center with an additional authorization effort.

[0049] Data shared and received may be stored in a database, which maintains data in one single database (e.g., database server) and generally at one particular location. This location is often a central computer, for example, a desktop central processing unit (CPU), a server CPU, or a mainframe computer. Information stored on a centralized database is typically accessible from multiple different points. A centralized database is easy to manage, maintain, and control, especially for purposes of security because of its single location. Within a centralized database, data redundancy is minimized as a single storing place of all data also implies that a given set of data only has one primary record. A blockchain may be used for storing vehicle-related data and transactions.

[0050] Any of the actions described herein may be performed by one or more processors (such as a microprocessor, a sensor, an Electronic Control Unit (ECU), a head unit, and the like), with or without memory, which may be located on-board the vehicle and/or off-board the vehicle (such as a server, computer, mobile/wireless device, etc.). The one or more processors may communicate with other memory and/or other processors on-board or off-board other vehicles to utilize data being sent by and/or to the vehicle. The one or more processors and the other processors can send data, receive data, and utilize this data to perform one or more of the actions described or depicted herein.

[0051] The example embodiments are directed to a system that may use mobility data from a vehicle and energy consumption data from a property (such as a home, office, etc.) associated with the vehicle, to optimize the sizing of home energy systems. This includes determining the appropriate size and type of on-premises energy storage, such as batteries, and the extent of carbon-free energy generation capabilities, like solar or wind installations. Furthermore, the system may also manage energy stored between the vehicle and the property based on current priority and needs of the vehicle and the property. For example, when a power outage is predicted, the system may transfer energy from the vehicle to the property. As another example, when a vehicle has significant use coming up, the system may transfer energy from the property to the vehicle. The system may also recommend storage types for the vehicle and for the real property based on energy consumption by the vehicle and the property.

[0052] For example, the system least selecting a number of solar panels, etc., for a home, a size of a rechargeable battery (small, medium, large) for a vehicle, and the like. The system predicts future power outages and suggests appropriate solar/wind capabilities and on-premises storage based on the type of car battery a customer has. This helps in ensuring seamless energy flow even during grid failures. The system accounts for changes in the customer's vehicle. For example, newer vehicles might have better battery storage, affecting the need for on-premises storage capacity.

[0053] The system may also analyze a customer's driving patterns based on mobility data from the vehicle and local energy pricing to make recommendations on a size of a home battery and/or an electric vehicle (EV) battery. The system may also consider a potential need for expanding the energy system in the future due to factors like extreme weather events, life changes (e.g., having a child), or transitioning to different vehicle types (like a hybrid or full EV). The system has a modular design to facilitate easy expansion without needing to replace core hardware components and considers the environmental impact of EVs versus internal combustion engine (ICE) vehicles and the potential for reusing components (like batteries) in different applications. This invention provides a customizable and future-proof energy solution that adapts to individual customer needs and changing circumstances, ensuring efficient and sustainable energy use.

[0054] In some embodiments, the system may be integrated with a smart home energy management system that is interconnected to one or more energy storage devices of the property and the charging point, etc. of the vehicle. As another example, the system may receive communications from an application installed on a computer of the vehicle. The communications may include data about energy consumed, tracking behavior, etc. The system may also utilize real-time data from smart home devices (e.g., thermostats, smart appliances, lighting systems, etc.) along with mobility data from the vehicle to dynamically adjust energy storage and distribution between a battery of the vehicle and an energy storage system of the property. Based on user behavior patterns and predicted future energy needs (inferred from both mobility and home appliance usage data), the system allocates energy between home storage and the electric vehicle battery. For example, during times of low home energy use but high solar generation, excess energy could be diverted to charge the electric vehicle. In the event of a predicted power outage, the system adjusts home energy consumption and storage, ensuring essential services (like refrigeration and heating) are maintained. It optimizes the charging and discharging cycles of the electric vehicle to support home energy needs more efficiently during the outage. The system learns user preferences over time (e.g., preferred indoor temperature, typical appliance usage times) and optimizes energy management to enhance comfort and efficiency, all while ensuring the home energy system is sized and used optimally.

[0055] In some embodiments, the system extends its capabilities to a community level by aggregating mobility and energy usage data from multiple households within a community or neighborhood. Leveraging this aggregated data, the system optimizes energy distribution across the community, identifying houses with surplus energy (e.g., those with larger solar installations and lower consumption) and redistributing it to homes with higher energy needs. A peer-to-

peer energy-sharing model is incorporated where households can “trade” energy. For example, a home with excess stored solar energy could provide energy to a neighbor’s electric vehicle or home storage system, enhancing overall community energy resilience and efficiency. The system could be used for community-level energy resilience planning, particularly in areas prone to frequent power outages.

[0056] In some embodiments, energy usage and energy needs during a power outage are determined. The system determines how long an electric vehicle (EV) will be associated with the location in one embodiment. If it is predicted that the EV will be replaced with a new EV or the EV will be unassociated with the environment for a period of time, the system may utilize this information in the recommendation. In some embodiments, the system recommends upgrading the recommended energy storage system configuration after a period of time based on the degraded capacity of the battery in the EV and/or the battery in the energy storage system. For example, after a year of degradation, add 5% to the system and another 10% for later battery degradation.

[0057] In some embodiments, the frequency and duration of usage of the EV and a type of battery type and capacity of the EV are used in the recommendation of the energy storage system. In some embodiments, the energy storage system configuration should include the EV battery(s), and the on-premises solution (electricity-based, gas-powered, and/or renewable). In some embodiments, the usage data of the EV battery will indicate the health of the battery and the future health of the battery. In some embodiments, the period of time may be, for example, one or two years. In some embodiments, due to technical advancements, the price of modifications to the system may be cheaper over time. The system predicts these price modifications.

[0058] FIG. 1A illustrates a process 100A of managing power and making recommendations between a vehicle and a property associated with the vehicle according to example embodiments. Referring to FIG. 1A, a power management system 140 may receive mobility data from a vehicle 110, a mobile device 120 associated with the vehicle 110 such as a mobile device that is “paired” with the vehicle on a regular basis, and the like. The mobility data may include Global Positioning System (GPS) data of trips that the vehicle 110 has taken which have been tracked using a software application, or the like, which is installed within the vehicle 110, the mobile device 120, or the like. The GPS data may include both location data and temporal data (timestamps, etc.) which can be used to identify the amount of time that the vehicle 110 is powered on and the distance traveled by the vehicle 110 when it is on. The mobility data may be recorded in real-time while the vehicle 110 is travelling, and subsequently extracted from the software application on the vehicle 110, the mobile device 120, and the like.

[0059] To extract the mobility data, the power management system 140 may transmit a query to a computer of the vehicle 110 (or the mobile device 120), such as a software application installed on the computer (or the mobile device 120), over a wireless computer network. The query may include an identifier of a period of time which the power management system 140 desires mobility data, for example, a one day period of time, a one week period of time, a one month period of time, and the like. In response, the software application may identify trips that have occurred during the period of time set forth in the query, and return the GPS data,

the timing data, the location data, the occupant data, etc. from the trips that occurred during the period of time to the power management system 140.

[0060] As another example, the power management system 140 may receive mobility data including charging data from a charging point 130 associated with the vehicle 110. The charging data may include an amount of charge provided to the vehicle 110, a source of the power (e.g., renewable, non-renewable, etc.) However, it should be appreciated that the vehicle 110 does not need to be an electric vehicle (EV). Instead, the vehicle 110 may be a gas-powered vehicle, or the like.

[0061] In addition, the power management system 140 may receive energy consumption data from a real property 150 that is associated with the vehicle 110. In this example, the real property 150 may correspond to a home, an office, a school, etc. of an owner or other occupant of the vehicle 110. The power management system 140 may transmit a query to a central controller of the real property, queries to individual appliances, queries to a power meter, and the like, over a wireless computer network. The appliances may include heating ventilation and air conditioning (HVAC) systems, laundry machines, refrigerator, and other appliances. The energy consumption data may also include power data from battery chargers in the home, solar panels (if installed), and the like. The query may include an identifier of a period of time which the power management system 140 desires energy consumption data, for example, a one day period of time, a one week period of time, a one month period of time, and the like.

[0062] The energy consumption data returned from the controller, the power meter, the appliances, etc., may include power consumption data by one or more appliances and other systems within the real property 150. The energy consumption data may include an amount of power consumed over a specific period of time (e.g., wattage consumed over the last day, week, month, etc.). As another example, the energy consumption data may include settings that are currently set on the systems, such as temperatures settings, cycle settings, number of cycles that occur, speed settings, and the like. The energy consumption data may be read from gauges, thermostats, meters, etc. of the appliances within the real property 150.

[0063] The power management system 140 may analyze the mobility data associated with the vehicle 110 and the energy consumption data from the real property 150, and make recommendations on energy configurations for the vehicle 110 and the real property 150. For example, the power management system 140 may analyze the current cost of gas, electricity, etc. and also the amount of energy consumed by the vehicle 110 and/or the real property 150, and make recommendations on a type of rechargeable battery to be used within the vehicle 110, a type of renewable energy source (e.g., solar panel, small wind turbine, etc.) for the real property 150, a number/amount of renewable energy sources for the real property 150, and the like.

[0064] FIG. 1B illustrates a process 100B of recommending storage types for the vehicle and the property according to example embodiments. According to various embodiments, the power management system 140 may include logic installed therein (software) which can determine a type of battery, energy storage, etc. that is recommended for the vehicle 110 and the real property 150 based on the mobility data and the energy consumption data. Here, the logic may

include a list of possible battery types and the amount of energy they store, a list of possible energy sources such as solar panels, wind turbines, carbon-based energy sources (e.g., propane, heating oil, etc.) and the amounts of energy that they can create. The power management system 140 may analyze the mobility data of the vehicle 110 and recommend a type of rechargeable battery for the vehicle, or a new type of vehicle, such as an electric vehicle, etc. As another example, the power management system 140 may analyze the energy consumption data and recommend different energy generation/storage configuration for power for the real property 150.

[0065] In this case, both the vehicle battery and the energy storage configuration for the real property 150 may both be determined in consideration of the vehicle 110 and the real property 150. Here, the vehicle 110 and the real property 150 may share energy if necessary, via the power management system 140. Accordingly, the power management system 140 may generate a smart configuration that is based on the shared needs of the vehicle 110 and the real property 150 including the times at which the vehicle 110 is expected to consume power and the time when the real property 150 is expected to consume more power. For example, the power management system 140 may recommend a type of rechargeable battery for the vehicle 110, a size of the battery (e.g., Watts, etc.), and the like. Meanwhile, the power management system 140 may recommend on-premises energy storage for the real property 150 including, but not limited to, solar panel installations, wind turbine installations, carbon-based energy sources, carbon-free energy generation sources, and the like.

[0066] Furthermore, the type of vehicle 110 may also be a factor in the recommended rechargeable battery for the vehicle 110. For example, newer vehicles might have better battery storage while older vehicles do not. In addition, the power management system 140 may also recommend a new vehicle to be purchased to replace the vehicle 110 based on the energy consuming needs of the owner of the vehicle 110, for example, based on the mobility data from the vehicle, and the like. Furthermore, the power management system 140 may make recommendations on how to “expand” existing energy storage systems already located on-premises at the real property 150.

[0067] For example, FIG. 1B illustrates a user interface 142 of the power management system 140 which may be included within a smart home energy management system, or the like. Here, the user interface 142 includes an identifier 144 of a recommendation to upgrade a battery of the vehicle 110, and an identifier 146 of a recommendation to expand on the renewable energy sources at the real property 150.

[0068] FIG. 1C illustrates a process 100C of transferring energy between the vehicle 110 and the real property 150 according to example embodiments, and FIG. 1D illustrates a process 100D of transferring power to a second real property 160 from one or more of the vehicle 110 and the real property 150 associated with the vehicle 110 according to example embodiments. According to various embodiments, the power management system 140 may provide a bi-directional charging cable or other means by which energy/power can be transferred from a battery within the vehicle 110 to an electricity distribution panel (also referred to herein as an energy storage system 152) of the real property 150. As an example, the vehicle 110 may be plugged into the charging point 130, such as a home

charging station or the like. The charging point 130 may be electrically connected to the power management system 140 (or the electricity distribution panel, etc.) of the real property 150 via a cable or other means.

[0069] To charge the energy storage system 152 of the real property 150, the power management system 140 may send a control signal to the charging point 130 to draw power from the battery of the vehicle 110 and transfer the power to the energy storage system 152 of the real property 150, via the cable, etc. Likewise, to draw power from the real property 150, the power management system 140 may send a control signal to the energy storage system 152 of the real property 150 to transfer power to the charging point 130 via the cable, etc. The decision to transfer power may be based on mobility data, energy consumption data, energy prices, weather forecasts, etc. which are obtained by the power management system 140 and analyzed to determine priority/power needs between the vehicle 110 and the real property 150.

[0070] Referring now to FIG. 1D, the power management system 140 may provide a bi-directional charging cable or other means by which energy/power can be transferred from a battery within the vehicle 110 and/or the energy storage system 152 of the real property 150, to an energy storage system 162 of a second real property 160. Here, the second real property 160 may be a property that is located within a predetermined geographical area with respect to the real property 150, such as an adjacent neighboring house, a home located on the same block, a home located in the same town, or the like. Here, the vehicle 110 may be plugged into the charging point 130, such as a home charging station or the like.

[0071] To charge the energy storage system 162 of the second real property 160, the power management system 140 may send a control signal to the charging point 130 and/or the energy storage system 152 of the real property 150, to draw power from the battery of the vehicle 110 and/or the energy storage system 152 of the real property 150. Here, the charging point 130 and/or the energy storage system 152 may transfer power to the energy storage system 162 of the second real property 160, via another cable, etc. As another example, the power management system 140 may draw power from the energy storage system 162 of the second real property 160, based on a control signal sent to the energy storage system 162 of the second real property 160 to transfer power to the charging point 130 and/or the energy storage system 152 of the real property 150, via the cable(s), etc. The decision to transfer power may be based on mobility data, energy consumption data, energy prices, weather forecasts, etc. which are obtained by the power management system 140 and analyzed to determine priority/power needs between the vehicle 110, the real property 150 and the second real property 160.

[0072] Although not shown in FIG. 1D, the power management system 140 may be connected to one or more external systems such as an external host server/platform of a weather-related website, an energy-related website, or the like. Here, the power management system 140 may receive a data feed from the one or more external systems and detect upcoming events such as forecasted weather events, power emergencies, etc. In this case, the power management system 140 may take steps to proactively safeguard the power that is currently stored within the battery of the vehicle 110 and the energy storage system 152 of the real property. For

example, the power management system **140** may trigger a transfer of power from the battery of the vehicle **110** to the energy storage system **152** of the real property when the power management system **140** predicts a potential power outage, large storm, etc.

[0073] Flow diagrams depicted herein, such as FIG. 2C, FIG. 2D, FIG. 2E, and FIG. 2F, are separate examples but may be the same or different embodiments. Any of the operations in one flow diagram may be adopted and shared with another flow diagram. No example operation is intended to limit the subject matter of any embodiment or corresponding claim.

[0074] It is important to note that all the flow diagrams and corresponding processes derived from FIG. 2C, FIG. 2D, FIG. 2E, and FIG. 2F may be part of a same process or may share sub-processes with one another thus making the diagrams combinable into a single preferred embodiment that does not require any one specific operation but which performs certain operations from one example process and from one or more additional processes. All the example processes are related to the same physical system and can be used separately or interchangeably.

[0075] The instant solution can be used in conjunction with one or more types of vehicles: battery electric vehicles, hybrid vehicles, fuel cell vehicles, internal combustion engine vehicles and/or vehicles utilizing renewable sources.

[0076] FIG. 2A illustrates a vehicle network diagram **200**, according to example embodiments. The network comprises elements including a vehicle **202** including a processor **204**, as well as a vehicle **202'** including a processor **204'**. The vehicles **202**, **202'** communicate with one another via the processors **204**, **204'**, as well as other elements (not shown) including transceivers, transmitters, receivers, storage, sensors, and other elements capable of providing communication. The communication between the vehicles **202**, and **202'** can occur directly, via a private and/or a public network (not shown), or via other vehicles and elements comprising one or more of a processor, memory, and software. Although depicted as single vehicles and processors, a plurality of vehicles and processors may be present. One or more of the applications, features, steps, solutions, etc., described and/or depicted herein may be utilized and/or provided by the instant elements.

[0077] FIG. 2B illustrates another vehicle network diagram **210**, according to example embodiments. The network comprises elements including a vehicle **202** including a processor **204**, as well as a vehicle **202'** including a processor **204'**. The vehicles **202**, **202'** communicate with one another via the processors **204**, **204'**, as well as other elements (not shown), including transceivers, transmitters, receivers, storage, sensors, and other elements capable of providing communication. The communication between the vehicles **202**, and **202'** can occur directly, via a private and/or a public network (not shown), or via other vehicles and elements comprising one or more of a processor, memory, and software. The processors **204**, **204'** can further communicate with one or more elements **230** including sensor **212**, wired device **214**, wireless device **216**, database **218**, mobile phone **220**, vehicle **222**, computer **224**, input/output (I/O) device **226**, and voice application **228**. The processors **204**, **204'** can further communicate with elements comprising one or more of a processor, memory, and software.

[0078] Although depicted as single vehicles, processors and elements, a plurality of vehicles, processors and ele-

ments may be present. Information or communication can occur to and/or from any of the processors **204**, **204'** and elements **230**. For example, the mobile phone **220** may provide information to the processor **204**, which may initiate the vehicle **202** to take an action, may further provide the information or additional information to the processor **204'**, which may initiate the vehicle **202'** to take an action, may further provide the information or additional information to the mobile phone **220**, the vehicle **222**, and/or the computer **224**. One or more of the applications, features, steps, solutions, etc., described and/or depicted herein may be utilized and/or provided by the instant elements.

[0079] FIG. 2C illustrates yet another vehicle network diagram **240**, according to example embodiments. The network comprises elements including a vehicle **202**, a processor **204**, and a non-transitory computer readable medium **242C**. The processor **204** is communicably coupled to the computer readable medium **242C** and elements **230** (which were depicted in FIG. 2B). The vehicle **202** may be a vehicle, server, or any device with a processor and memory.

[0080] The processor **204** performs one or more of receiving, via a power management apparatus, mobility data comprising movements of a vehicle from one or more systems associated with the vehicle in **244C**, receiving, via the power management apparatus, energy consumption data from one or more power consuming systems at a real property that is associated with the vehicle in **246C**, determining a recommended battery for the vehicle and a recommended power generation system for the real property based on the mobility data and the energy consumption data in **248C**, and displaying an identifier of the recommended battery and an identifier of the recommended power generation system via a user interface of the power management apparatus. FIG. 2D illustrates a further vehicle network diagram **250**, according to example embodiments. The network comprises elements including a vehicle **202** a processor **204**, and a non-transitory computer readable medium **242D**. The processor **204** is communicably coupled to the computer readable medium **242D** and elements **230** (which were depicted in FIG. 2B). The vehicle **202** may be a vehicle, server or any device with a processor and memory.

[0081] The processor **204** performs one or more of dynamically adjusting an amount of energy stored within a battery of the vehicle and an amount of energy stored within an energy storage device of the real property via the power management apparatus based on the mobility data and the energy consumption data in **244D**, determining that a priority of energy storage for the real property is greater than a priority of energy storage for the vehicle based on one or more of the mobility data and the energy consumption data, and in response, transferring energy stored within a battery of the vehicle to an energy storage device of the real property in **245D**, determining that a priority of energy storage for the vehicle is greater than a priority of energy storage for the real property based on one or more of the mobility data and the energy consumption data, and in response, transferring energy stored within a battery of an energy storage device of a real property to the battery of the vehicle in **246D**, receiving Global Positioning System (GPS) data and temporal data of tracked vehicle activity from one or more of a mobile device associated with the vehicle and a navigation system installed in the vehicle, and receiving energy consumption data from one or more electrical appliances installed within the real property in **247D**, receiving second

energy consumption data from a second energy storage device of a second real property that is located within a predetermined distance of the real property, and transferring energy from at least one of energy stored within a battery of the vehicle, and energy stored within an energy storage device of the real property to the second energy storage device of the second real property in **248D**, and receiving forecasted weather data from an external data source, predicting a power outage will occur based on the forecasted weather data, and in response, transferring energy stored within a battery of the vehicle to an energy storage device of the real property in **249D**.

[0082] While this example describes in detail only one vehicle **202**, multiple such nodes may be connected to the blockchain. It should be understood that the vehicle **202** may include additional components and that some of the components described herein may be removed and/or modified without departing from a scope of the instant application. The vehicle **202** may have a computing device or a server computer, or the like, and may include a processor **204**, which may be a semiconductor-based microprocessor, a central processing unit (CPU), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), and/or another hardware device. Although a single processor **204** is depicted, it should be understood that the vehicle **202** may include multiple processors, multiple cores, or the like without departing from the scope of the instant application. The vehicle **202** may be a vehicle, server or any device with a processor and memory.

[0083] The processor **204** performs one or more of receiving a confirmation of an event from one or more elements described or depicted herein, wherein the confirmation comprises a blockchain consensus between peers represented by any of the elements and executing a smart contract to record the confirmation on the blockchain consensus. Consensus is formed between one or more of any element **230** and/or any element described or depicted herein, including a vehicle, a server, a wireless device, etc. In another example, the vehicle **202** can be one or more of any element **230** and/or any element described or depicted herein, including a server, a wireless device, etc.

[0084] The processors and/or computer readable medium may fully or partially reside in the interior or exterior of the vehicles. The steps or features stored in the computer readable medium may be fully or partially performed by any of the processors and/or elements in any order. Additionally, one or more steps or features may be added, omitted, combined, performed at a later time, etc.

[0085] FIG. 2E illustrates a flow diagram **260**, according to example embodiments. Referring to FIG. 2E, the instant solution includes one or more of receiving, via a power management apparatus, mobility data comprising movements of a vehicle from one or more systems associated with the vehicle in **244E**, receiving, via the power management apparatus, energy consumption data from one or more power consuming systems at a real property that is associated with the vehicle in **246E**, determining a recommended battery for the vehicle and a recommended power generation system for the real property based on the mobility data and the energy consumption data in **248E**, and displaying an identifier of the recommended battery and an identifier of the recommended power generation system via a user interface of the power management apparatus in **250E**.

[0086] FIG. 2F illustrates another flow diagram **270**, according to example embodiments. Referring to FIG. 2F, the instant solution includes one or more of dynamically adjusting an amount of energy stored within a battery of the vehicle and an amount of energy stored within an energy storage device of the real property via the power management apparatus based on the mobility data and the energy consumption data in **244F**, determining that a priority of energy storage for the real property is greater than a priority of energy storage for the vehicle based on one or more of the mobility data and the energy consumption data, and in response, transferring energy stored within a battery of the vehicle to an energy storage device of the real property in **245F**, determining that a priority of energy storage for the vehicle is greater than a priority of energy storage for the real property based on one or more of the mobility data and the energy consumption data, and in response, transferring energy stored within a battery of an energy storage device of a real property to the battery of the vehicle in **246F**, receiving Global Positioning System (GPS) data and temporal data of tracked vehicle activity from one or more of a mobile device associated with the vehicle and a navigation system installed in the vehicle, and receiving energy consumption data from one or more electrical appliances installed within the real property in **247F**, receiving second energy consumption data from a second energy storage device of a second real property that is located within a predetermined distance of the real property, and transferring energy from at least one of energy stored within a battery of the vehicle, and energy stored within an energy storage device of the real property to the second energy storage device of the second real property in **248F**, and receiving forecasted weather data from an external data source, predicting a power outage will occur based on the forecasted weather data, and in response, transferring energy stored within a battery of the vehicle to an energy storage device of the real property in **249F**.

[0087] Technological advancements typically build upon the fundamentals of predecessor technologies; such is the case with Artificial Intelligence (AI) models. An AI classification system describes the stages of AI progression. The first classification is known as “Reactive Machines,” followed by present-day AI classification “Limited Memory Machines” (also known as “Artificial Narrow Intelligence”), then progressing to “Theory of Mind” (also known as “Artificial General Intelligence”), and reaching the AI classification “Self-Aware” (also known as “Artificial Superintelligence”). Present-day Limited Memory Machines are a growing group of AI models built upon the foundation of its predecessor, Reactive Machines. Reactive Machines emulate human responses to stimuli; however, they are limited in their capabilities as they cannot typically learn from prior experience. Once the AI model’s learning abilities emerged, its classification was promoted to Limited Memory Machines. In this present-day classification, AI models learn from large volumes of data, detect patterns, solve problems, generate and predict data, and the like, while inheriting all of the capabilities of Reactive Machines. Examples of AI models classified as Limited Memory Machines include, but are not limited to, Chatbots, Virtual Assistants, Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), Generative AI (GenAI) models, and any future AI models that are yet to be developed possessing characteristics of Limited Memory Machines. Generative AI models combine Limited Memory Machine technologies,

incorporating ML and DL, forming the foundational building blocks of future AI models. For example, Theory of Mind is the next progression of AI that may be able to perceive, connect, and react by generating appropriate reactions in response to an entity with which the AI model is interacting; all of these capabilities rely on the fundamentals of Generative AI. Furthermore, in an evolution into the Self-Aware classification, AI models will be able to understand and evoke emotions in the entities they interact with, as well as possessing their own emotions, beliefs, and needs, all of which rely on Generative AI fundamentals of learning from experiences to generate and draw conclusions about itself and its surroundings. Generative AI models are integral and core to future artificial intelligence models. As described herein, Generative AI refers to present-day Generative AI models and future AI models.

[0088] FIG. 3A illustrates an AI/ML network diagram 300A that supports AI-assisted vehicle or occupant decision points. Other branches of AI, such as, but not limited to, computer vision, fuzzy logic, expert systems, neural networks/deep learning, generative AI, and natural language processing, may all be employed in developing the AI model shown in these embodiments. Further, the AI model included in these embodiments is not limited to particular AI algorithms. Any algorithm or combination of algorithms related to supervised, unsupervised, and reinforcement learning algorithms may be employed.

[0089] In one embodiment, Generative AI (GenAI) may be used by the instant solution in the transformation of data. Vehicles are equipped with diverse sensors, cameras, radars, and LIDARs, which collect a vast array of data, such as images, speed readings, GPS data, and acceleration metrics. However, raw data, once acquired, undergoes preprocessing that may involve normalization, anonymization, missing value imputation, or noise reduction to allow the data to be further used effectively.

[0090] The GenAI executes data augmentation following the preprocessing of the data. Due to the limitation of datasets in capturing the vast complexity of real-world vehicle scenarios, augmentation tools are employed to expand the dataset. This might involve image-specific transformations like rotations, translations, or brightness adjustments. For non-image data, techniques like jittering can be used to introduce synthetic noise, simulating a broader set of conditions.

[0091] In the instant solution, data generation is then performed on the data. Tools like Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) are trained on existing datasets to generate new, plausible data samples. For example, GANs might be tasked with crafting images showcasing vehicles in uncharted conditions or from unique perspectives. As another example, the synthesis of sensor data may be performed to model and create synthetic readings for such scenarios, enabling thorough system testing without actual physical encounters. A critical step in the use of GenAI, given the safety-critical nature of vehicles, is validation. This validation might include the output data being compared with real-world datasets or using specialized tools like a GAN discriminator to gauge the realism of the crafted samples.

[0092] Vehicle node 310 may include a plurality of sensors 312 that may include but are not limited to, light sensors, weight sensors, cameras, lidar, and radar. In some embodiments, these sensors 312 send data to a database 320 that

stores data about the vehicle and occupants of the vehicle. In some embodiments, these sensors 312 send data to one or more decision subsystems 316 in vehicle node 310 to assist in decision-making.

[0093] Vehicle node 310 may include one or more user interfaces (UIs) 314, such as a steering wheel, navigation controls, audio/video controls, temperature controls, etc. In some embodiments, these UIs 314 send data to a database 320 that stores event data about the UIs 314 that includes but is not limited to selection, state, and display data. In some embodiments, these UIs 314 send data to one or more decision subsystems 316 in vehicle node 310 to assist decision-making.

[0094] Vehicle node 310 may include one or more decision subsystems 316 that drive a decision-making process around, but are not limited to, vehicle control, temperature control, charging control, etc. In some embodiments, the decision subsystems 316 gather data from one or more sensors 312 to aid in the decision-making process. In some embodiments, a decision subsystem 316 may gather data from one or more UIs 314 to aid in the decision-making process. In some embodiments, a decision subsystem 316 may provide feedback to a UI 314.

[0095] An AI/ML production system 330 may be used by a decision subsystem 316 in a vehicle node 310 to assist in its decision-making process. The AI/ML production system 330 includes one or more AI/ML models 332 that are executed to retrieve the needed data, such as, but not limited to, a prediction, a categorization, a UI prompt, etc. In some embodiments, an AI/ML production system 330 is hosted on a server. In some embodiments, the AI/ML production system 330 is cloud-hosted. In some embodiments, the AI/ML production system 330 is deployed in a distributed multi-node architecture. In some embodiments, the AI production system resides in vehicle node 310.

[0096] An AI/ML development system 340 creates one or more AI/ML models 332. In some embodiments, the AI/ML development system 340 utilizes data in the database 320 to develop and train one or more AI models 332. In some embodiments, the AI/ML development system 340 utilizes feedback data from one or more AI/ML production systems 330 for new model development and/or existing model re-training. In an embodiment, the AI/ML development system 340 resides and executes on a server. In another embodiment the AI/ML development system 340 is cloud hosted. In a further embodiment, the AI/ML development system 340 utilizes a distributed data pipeline/analytics engine.

[0097] Once an AI/ML model 332 has been trained and validated in the AI/ML development system 340, it may be stored in an AI/ML model registry 360 for retrieval by either the AI/ML development system 340 or by one or more AI/ML production systems 330. The AI/ML model registry 360 resides in a dedicated server in one embodiment. In some embodiments, the AI/ML model registry 360 is cloud-hosted. The AI/ML model registry 360 is a distributed database in other embodiments. In further embodiments, the AI/ML model registry 360 resides in the AI/ML production system 330.

[0098] FIG. 3B illustrates a process 300B for developing one or more AI/ML models that support AI-assisted vehicle or occupant decision points. An AI/ML development system 340 executes steps to develop an AI/ML model 332 that begins with data extraction 342, in which data is loaded and

ingested from one or more data sources. In some embodiments, vehicle and user data is extracted from a database 320. In some embodiments, model feedback data is extracted from one or more AI/ML production systems 330. [0099] Once the required data has been extracted 342, it must be prepared 344 for model training. In some embodiments, this step involves statistical testing of the data to see how well it reflects real-world events, its distribution, the variety of data in the dataset, etc. In some embodiments, the results of this statistical testing may lead to one or more data transformations being employed to normalize one or more values in the dataset. In some embodiments, this step includes cleaning data deemed to be noisy. A noisy dataset includes values that do not contribute to the training, such as but are not limited to, null and long string values. Data preparation 344 may be a manual process or an automated process using one or more of the elements, functions described or depicted herein.

[0100] Features of the data are identified and extracted 346. In some embodiments, a feature of the data is internal to the prepared data from step 344. In other embodiments, a feature of the data requires a piece of prepared data from step 344 to be enriched by data from another data source to be useful in developing an AI/ML model 332. In some embodiments, identifying features is a manual process or an automated process using one or more of the elements, functions described or depicted herein. Once the features have been identified, the values of the features are collected into a dataset that will be used to develop the AI/ML model 332.

[0101] The dataset output from feature extraction step 346 is split 348 into a training and validation data set. The training data set is used to train the AI/ML model 332, and the validation data set is used to evaluate the performance of the AI/ML model 332 on unseen data.

[0102] The AI/ML model 332 is trained and tuned 350 using the training data set from the data splitting step 348. In this step, the training data set is fed into an AI/ML algorithm and an initial set of algorithm parameters. The performance of the AI/ML model 332 is then tested within the AI/ML development system 340 utilizing the validation data set from step 348. These steps may be repeated with adjustments to one or more algorithm parameters until the model's performance is acceptable based on various goals and/or results.

[0103] The AI/ML model 332 is evaluated 352 in a staging environment (not shown) that resembles the ultimate AI/ML production system 330. This evaluation uses a validation dataset to ensure the performance in an AI/ML production system 330 matches or exceeds expectations. In some embodiments, the validation dataset from step 348 is used. In other embodiments, one or more unseen validation datasets are used. In some embodiments, the staging environment is part of the AI/ML development system 340. In other embodiments, the staging environment is managed separately from the AI/ML development system 340. Once the AI/ML model 332 has been validated, it is stored in an AI/ML model registry 360, which can be retrieved for deployment and future updates. As before, in some embodiments, the model evaluation step 352 is a manual process or an automated process using one or more of the elements, functions described or depicted herein.

[0104] Once an AI/ML model 332 has been validated and published to an AI/ML model registry 360, it may be

deployed 354 to one or more AI/ML production systems 330. In some embodiments, the performance of deployed AI/ML models 332 is monitored 356 by the AI/ML development system 340. In some embodiments, AI/ML model 332 feedback data is provided by the AI/ML production system 330 to enable model performance monitoring 356. In some embodiments, the AI/ML development system 340 periodically requests feedback data for model performance monitoring 356. In some embodiments, model performance monitoring includes one or more triggers that result in the AI/ML model 332 being updated by repeating steps 342-354 with updated data from one or more data sources.

[0105] FIG. 3C illustrates a process 300C for utilizing an AI/ML model that supports AI-assisted vehicle or occupant decision points. As stated previously, the AI model utilization process depicted herein reflects ML, which is a particular branch of AI, but the instant solution is not limited to ML and is not limited to any AI algorithm or combination of algorithms.

[0106] Referring to FIG. 3C, an AI/ML production system 330 may be used by a decision subsystem 316 in vehicle node 310 to assist in its decision-making process. The AI/ML production system 330 provides an application programming interface (API) 334, executed by an AI/ML server process 336 through which requests can be made. In some embodiments, a request may include an AI/ML model 332 identifier to be executed. In some embodiments, the AI/ML model 332 to be executed is implicit based on the type of request. In some embodiments, a data payload (e.g., to be input to the model during execution) is included in the request. In some embodiments, the data payload includes sensor 312 data from vehicle node 310. In some embodiments, the data payload includes UI 314 data from vehicle node 310. In some embodiments, the data payload includes data from other vehicle node 310 subsystems (not shown), including but not limited to, occupant data subsystems. In an embodiment, one or more elements or nodes 320, 330, 340, or 360 may be located in the vehicle 310.

[0107] Upon receiving the API 334 request, the AI/ML server process 336 may need to transform the data payload or portions of the data payload to be valid feature values into an AI/ML model 332. Data transformation may include but is not limited to combining data values, normalizing data values, and enriching the incoming data with data from other data sources. Once any required data transformation occurs, the AI/ML server process 336 executes the appropriate AI/ML model 332 using the transformed input data. Upon receiving the execution result, the AI/ML server process 336 responds to the API caller, which is a decision subsystem 316 of vehicle node 310. In some embodiments, the response may result in an update to a UI 314 in vehicle node 310. In some embodiments, the response includes a request identifier that can be used later by the decision subsystem 316 to provide feedback on the AI/ML model 332 performance. Further, in some embodiments, immediate performance feedback may be recorded into a model feedback log 338 by the AI/ML server process 336. In some embodiments, execution model failure is a reason for immediate feedback.

[0108] In some embodiments, the API 334 includes an interface to provide AI/ML model 332 feedback after an AI/ML model 332 execution response has been processed. This mechanism may be used to evaluate the performance of the AI/ML model 332 by enabling the API caller to provide

feedback on the accuracy of the model results. For example, if the AI/ML model 332 provided an estimated time of arrival of 20 minutes, but the actual travel time was 24 minutes, that may be indicated. In some embodiments, the feedback interface includes the identifier of the initial request so that it can be used to associate the feedback with the request. Upon receiving a call into the feedback interface of API 334, the AI/ML server process 336 records the feedback in the model feedback log 338. In some embodiments, the data in this model feedback log 338 is provided to model performance monitoring 356 in the AI/ML development system 340. This log data is streamed to the AI/ML development system 340 in one embodiment. In some embodiments, the log data is provided upon request.

[0109] A number of the steps/features that may utilize the AI/ML process described herein include one or more of: receiving, via a power management apparatus, mobility data comprising movements of a vehicle from one or more systems associated with the vehicle, receiving, via the power management apparatus, energy consumption data from one or more power consuming systems at a real property that is associated with the vehicle, determining a recommended battery for the vehicle and a recommended power generation system for the real property based on the mobility data and the energy consumption data, and displaying an identifier of the recommended battery and an identifier of the recommended power generation system via a user interface of the power management apparatus, dynamically adjusting an amount of energy stored within a battery of the vehicle and an amount of energy stored within an energy storage device of the real property via the power management apparatus based on the mobility data and the energy consumption data, determining that a priority of energy storage for the real property is greater than a priority of energy storage for the vehicle based on one or more of the mobility data and the energy consumption data, and in response, transferring energy stored within a battery of the vehicle to an energy storage device of the real property, determining that a priority of energy storage for the vehicle is greater than a priority of energy storage for the real property based on one or more of the mobility data and the energy consumption data, and in response, transferring energy stored within a battery of an energy storage device of a real property to the battery of the vehicle, receiving Global Positioning System (GPS) data and temporal data of tracked vehicle activity from one or more of a mobile device associated with the vehicle and a navigation system installed in the vehicle, and receiving energy consumption data from one or more electrical appliances installed within the real property, receiving second energy consumption data from a second energy storage device of a second real property that is located within a predetermined distance of the real property, and transferring energy from at least one of energy stored within a battery of the vehicle, and energy stored within an energy storage device of the real property to the second energy storage device of the second real property, and receiving forecasted weather data from an external data source, predicting a power outage will occur based on the forecasted weather data, and in response, transferring energy stored within a battery of the vehicle to an energy storage device of the real property.

[0110] Data associated with any of these steps/features, as well as any other features or functionality described or depicted herein, the AI/ML production system 330, as well

as one or more of the other elements depicted in FIG. 3C may be used to process this data in a pre-transformation and/or post-transformation process. Data related to this process can be used by the vehicle node 310. In one embodiment, data related to this process may be used with a charging station/charging point, a server, a wireless device, and/or any of the processors described or depicted herein.

[0111] FIG. 3D illustrates a process 300D of designing a new machine learning model via a user interface 370 of the system according to example embodiments. As an example, a model may be output as part of the AI/ML Development System 340. Referring to FIG. 3D, a user can use an input mechanism from a menu 372 of a user interface 370 to add pieces/components to a model being developed within a workspace 374 of the user interface 370.

[0112] The menu 372 includes a plurality of graphical user interface (GUI) menu options which can be selected to reveal additional components that can be added to the model design shown in the workspace 374. The GUI menu includes options for adding elements to the workspace, such as features which may include neural networks, machine learning models, AI models, data sources, conversion processes (e.g., vectorization, encoding, etc.), analytics, etc. The user can continue to add features to the model and connect them using edges or other means to create a flow within the workspace 374. For example, the user may add a node 376 to a flow of a new model within the workspace 374. For example, the user may connect the node 376 to another node in the diagram via an edge 378, creating a dependency within the diagram. When the user is done, the user can save the model for subsequent training/testing.

[0113] In another example, the name of the object can be identified from a web page or a user interface 370 where the object is visible within a browser or the workspace 374 on the user device. A pop-up within the browser or the workspace 374 can be overlaid where the object is visible, which includes an option to navigate to the identified web page corresponding to the alternative object via a rule set.

[0114] FIG. 3E illustrates a process 300E of accessing an object 392 from an object storage 390 of the host platform 380 according to example embodiments. For example, the object storage 390 may store data that is used by the AI models and machine learning (ML) models, training data, expected outputs for testing, training results, and the like. The object storage 390 may also store any other kind of data. Each object may include a unique identifier, a data section 394, and a metadata section 396, which provide a descriptive context associated with the data, including data that can later be extracted for purposes of machine learning. The unique identifier may uniquely identify an object with respect to all other objects in the object storage 390. The data section 394 may include unstructured data such as web pages, digital content, images, audio, text, and the like.

[0115] Instead of breaking files into blocks stored on disks in a file system, the object storage 390 handles objects as discrete units of data stored in a structurally flat data environment. Here, the object storage may not use folders, directories, or complex hierarchies. Instead, each object may be a simple, self-contained repository that includes the data, the metadata, and the unique identifier that a client application can use to locate and access it. In this case, the metadata is more descriptive than a file-based approach. The

metadata can be customized with additional context that can later be extracted and leveraged for other purposes, such as data analytics.

[0116] The objects that are stored in the object storage **390** may be accessed via an API **384**. The API **384** may be a Hypertext Transfer Protocol (HTTP)-based RESTful API (also known as a RESTful Web service). The API **384** can be used by the client application to query an object's metadata to locate the desired object (data) via the Internet from anywhere on any device. The API **384** may use HTTP commands such as "PUT" or "POST" to upload an object, "GET" to retrieve an object, "DELETE" to remove an object, and the like.

[0117] The object storage **390** may provide a directory **398** that uses the metadata of the objects to locate appropriate data files. The directory **398** may contain descriptive information about each object stored in the object storage **390**, such as a name, a unique identifier, a creation timestamp, a collection name, etc. To query the object within the object storage **390**, the client application may submit a command, such as an HTTP command, with an identifier of the object **392**, a payload, etc. The object storage **390** can store the actions and results described herein, including associating two or more lists of ranked assets with one another based on variables used by the two or more lists of ranked assets that have a correlation above a predetermined threshold.

[0118] FIG. 4A illustrates a diagram **400A** depicting the electrification of one or more elements. In one example, a vehicle **402B** may provide power stored in its batteries to one or more elements, including other vehicle(s) **408B**, charging station(s) **406B**, and electric grid(s) **404B**. The electric grid(s) **404B** is/are coupled to one or more of the charging stations **406B**, which may be coupled to one or more of the vehicles **408B**. This configuration allows the distribution of electricity/power received from the vehicle **402B**. The vehicle **402B** may also interact with the other vehicle(s) **408B**, such as via V2V technology, communication over cellular, Wi-Fi, and the like. The vehicle **402B** may also interact wirelessly and/or wired with other vehicles **408B**, the charging station(s) **406B** and/or with the electric grid(s) **404B**. In one example, the vehicle **402B** is routed (or routes itself) in a safe and efficient manner to the electric grid(s) **404B**, the charging station(s) **406B**, or the other vehicle(s) **408B**. Using one or more embodiments of the instant solution, the vehicle **402B** can provide energy to one or more of the elements depicted herein in various advantageous ways as described and/or depicted herein. Further, the safety and efficiency of the vehicle may be increased, and the environment may be positively affected as described and/or depicted herein.

[0119] The term 'energy', 'electricity', 'power', and the like may be used to denote any form of energy received, stored, used, shared, and/or lost by the vehicle(s). The energy may be referred to in conjunction with a voltage source and/or a current supply of charge provided from an entity to the vehicle(s) during a charge/use operation. Energy may also be in the form of fossil fuels (for example, for use with a hybrid vehicle) or via alternative power sources, including but not limited to lithium-based, nickel-based, hydrogen fuel cells, atomic/nuclear energy, fusion-based energy sources, and energy generated during an energy sharing and/or usage operation for increasing or decreasing one or more vehicles energy levels at a given time.

[0120] In one example, the charging station **406B** manages the amount of energy transferred from the vehicle **402B** such that there is sufficient charge remaining in the vehicle **402B** to arrive at a destination. In one example, a wireless connection is used to wirelessly direct an amount of energy transfer between vehicles **408B**, wherein the vehicles may both be in motion. In one embodiment, wireless charging may occur via a fixed charger and batteries of the vehicle in alignment with one another (such as a charging mat in a garage or parking space). In one example, an idle vehicle, such as a vehicle **402B** (which may be autonomous) is directed to provide an amount of energy to a charging station **406B** and return to the original location (for example, its original location or a different destination). In one example, a mobile energy storage unit (not shown) is used to collect surplus energy from at least one other vehicle **408B** and transfer the stored surplus energy at a charging station **406B**. In one example, factors determine an amount of energy to transfer to a charging station **406B**, such as distance, time, as well as traffic conditions, road conditions, environmental/weather conditions, the vehicle's condition (weight, etc.), an occupant(s) schedule while utilizing the vehicle, a prospective occupant(s) schedule waiting for the vehicle, etc. In one example, the vehicle(s) **408B**, the charging station(s) **406B** and/or the electric grid(s) **404B** can provide energy to the vehicle **402B**.

[0121] In one embodiment, a location such as a building, a residence, or the like (not depicted), communicably coupled to one or more of the electric grid **404B**, the vehicle **402B**, and/or the charging station(s) **406B**. The rate of electric flow to one or more of the location, the vehicle **402B**, the other vehicle(s) **408B** is modified, depending on external conditions, such as weather. For example, when the external temperature is extremely hot or extremely cold, raising the chance for an outage of electricity, the flow of electricity to a connected vehicle **402B/408B** is slowed to help minimize the chance for an outage.

[0122] In one embodiment, vehicles **402B** and **408B** may be utilized as bidirectional vehicles. Bidirectional vehicles are those that may serve as mobile microgrids that can assist in the supplying of electrical power to the grid **404B** and/or reduce the power consumption when the grid is stressed. Bidirectional vehicles incorporate bidirectional charging, which in addition to receiving a charge to the vehicle, the vehicle can transfer energy from the vehicle to the grid **404B**, otherwise referred to as "V2G". In bidirectional charging, the electricity flows both ways; to the vehicle and from the vehicle. When a vehicle is charged, alternating current (AC) electricity from the grid **404B** is converted to direct current (DC). This may be performed by one or more of the vehicle's own converter or a converter on the charging station **406B**. The energy stored in the vehicle's batteries may be sent in an opposite direction back to the grid. The energy is converted from DC to AC through a converter usually located in the charging station **406B**, otherwise referred to as a bidirectional charger. Further, the instant solution as described and depicted with respect to FIG. 4B can be utilized in this and other networks and/or systems.

[0123] FIG. 4B is a diagram showing interconnections between different elements **400B**. The instant solution may be stored and/or executed entirely or partially on and/or by one or more computing devices **414C**, **418C**, **424C**, **428C**, **432C**, **436C**, **406C**, **442C** and **410C** associated with various entities, all communicably coupled and in communication

with a network 402C. A database 438C is communicably coupled to the network and allows for the storage and retrieval of data. In one example, the database is an immutable ledger. One or more of the various entities may be a vehicle 404C, one or more service provider 416C, one or more public buildings 422C, one or more traffic infrastructure 426C, one or more residential dwellings 430C, an electric grid/charging station 434C, a microphone 440C, and/or another vehicle 408C. Other entities and/or devices, such as one or more private users using a smartphone 412C, a laptop 420C, an augmented reality (AR) device, a virtual reality (VR) device, and/or any wearable device may also interwork with the instant solution. The smartphone 412C, laptop 420C, the microphone 440C, and other devices may be connected to one or more of the connected computing devices 414C, 418C, 424C, 428C, 432C, 436C, 406C, 442C, and 410C. The one or more public buildings 422C may include various agencies. The one or more public buildings 422C may utilize a computing device 424C. The one or more service provider 416C may include a dealership, a tow truck service, a collision center, or other repair shop. The one or more service provider 416C may utilize a computing apparatus 418C. These various computer devices may be directly and/or communicably coupled to one another, such as via wired networks, wireless networks, blockchain networks, and the like. The microphone 440C may be utilized as a virtual assistant, in one example. In one example, the one or more traffic infrastructure 426C may include one or more traffic signals, one or more sensors including one or more cameras, vehicle speed sensors or traffic sensors, and/or other traffic infrastructure. The one or more traffic infrastructure 426C may utilize a computing device 428C.

[0124] In one embodiment, anytime an electrical charge is given or received to/from a charging station and/or an electrical grid, the entities that allow that to occur are one or more of a vehicle, a charging station, a server, and a network communicably coupled to the vehicle, the charging station, and the electrical grid.

[0125] In one example, a vehicle 408C/404C can transport a person, an object, a permanently or temporarily affixed apparatus, and the like. In one example, the vehicle 408C may communicate with vehicle 404C via V2V communication through the computers associated with each vehicle 406C and 410C and may be referred to as a car, vehicle, automobile, and the like. The vehicle 404C/408C may be a self-propelled wheeled conveyance, such as a car, a sports utility vehicle, a truck, a bus, a van, or other motor or battery-driven or fuel cell-driven vehicle. For example, vehicle 404C/408C may be an electric vehicle, a hybrid vehicle, a hydrogen fuel cell vehicle, a plug-in hybrid vehicle, or any other type of vehicle with a fuel cell stack, a motor, and/or a generator. Other examples of vehicles include bicycles, scooters, trains, planes, boats, and any other form of conveyance that is capable of transportation. The vehicle 404C/408C may be semi-autonomous or autonomous. For example, vehicle 404C/408C may be self-maneuvering and navigate without human input. An autonomous vehicle may have and use one or more sensors and/or a navigation unit to drive autonomously. All of the data described or depicted herein can be stored, analyzed, processed and/or forwarded by one or more of the elements in FIG. 4B.

[0126] FIG. 4C is another block diagram showing interconnections between different elements in one example

400C. A vehicle 412D is presented and includes ECUs 410D, 408D, and a Head Unit (otherwise known as an Infotainment System) 406D. An ECU is an embedded system in automotive electronics controlling one or more of the electrical systems or subsystems in a vehicle. ECUs may include but are not limited to the management of a vehicle's engine, brake system, gearbox system, door locks, dashboard, airbag system, infotainment system, electronic differential, and active suspension. ECUs are connected to the vehicle's Controller Area Network (CAN) bus 416D. The ECUs may also communicate with a vehicle computer 404D via the CAN bus 416D. The vehicle's processors/sensors (such as the vehicle computer) 404D can communicate with external elements, such as a server 418D via a network 402D (such as the Internet). Each ECU 410D, 408D, and Head Unit 406D may contain its own security policy. The security policy defines permissible processes that can be executed in the proper context. In one example, the security policy may be partially or entirely provided in the vehicle computer 404D.

[0127] ECUs 410D, 408D, and Head Unit 406D may each include a custom security functionality element 414D defining authorized processes and contexts within which those processes are permitted to run. Context-based authorization to determine validity if a process can be executed allows ECUs to maintain secure operation and prevent unauthorized access from elements such as the vehicle's CAN Bus. When an ECU encounters a process that is unauthorized, that ECU can block the process from operating. Automotive ECUs can use different contexts to determine whether a process is operating within its permitted bounds, such as proximity contexts, nearby objects, distance to approaching objects, speed, and trajectory relative to other moving objects, and operational contexts such as an indication of whether the vehicle is moving or parked, the vehicle's current speed, the transmission state, user-related contexts such as devices connected to the transport via wireless protocols, use of the infotainment, cruise control, parking assist, driving assist, location-based contexts, and/or other contexts.

[0128] Referring to FIG. 4D, an operating environment 400D for a connected vehicle, is illustrated according to some embodiments. As depicted, the vehicle 410E includes a CAN bus 408E connecting elements 412E-426E of the vehicle. Other elements may be connected to the CAN bus and are not depicted herein. The depicted elements connected to the CAN bus include a sensor set 412E, Electronic Control Units 414E, autonomous features or Advanced Driver Assistance Systems (ADAS) 416E, and the navigation system 418E. In some embodiments, the vehicle 410E includes a processor 420E, a memory 422E, a communication unit 424E, and an electronic display 426E.

[0129] The processor 420E includes an arithmetic logic unit, a microprocessor, a general-purpose controller, and/or a similar processor array to perform computations and provide electronic display signals to a display unit 426E. The processor 420E processes data signals and may include various computing architectures, including a complex instruction set computer (CISC) architecture, a reduced instruction set computer (RISC) architecture, or an architecture implementing a combination of instruction sets. The vehicle 410E may include one or more processors 420E. Other processors, operating systems, sensors, displays, and

physical configurations that are communicably coupled to one another (not depicted) may be used with the instant solution.

[0130] Memory 422E is a non-transitory memory storing instructions or data that may be accessed and executed by the processor 420E. The instructions and/or data may include code to perform the techniques described herein. The memory 422E may be a dynamic random-access memory (DRAM) device, a static random-access memory (SRAM) device, flash memory, or another memory device. In some embodiments, the memory 422E also may include non-volatile memory or a similar permanent storage device and media, which may include a hard disk drive, a floppy disk drive, a compact disc read only memory (CD-ROM) device, a digital versatile disk read only memory (DVD-ROM) device, a digital versatile disk random access memory (DVD-RAM) device, a digital versatile disk rewritable (DVD-RW) device, a flash memory device, or some other mass storage device for storing information on a permanent basis. A portion of the memory 422E may be reserved for use as a buffer or virtual random-access memory (virtual RAM). The vehicle 410E may include one or more memories 422E without deviating from the current solution.

[0131] The memory 422E of the vehicle 410E may store one or more of the following types of data: navigation route data 418E, and autonomous features data 416E. In some embodiments, the memory 422E stores data that may be necessary for the navigation application 418E to provide the functions.

[0132] The navigation system 418E may describe at least one navigation route including a start point and an endpoint. In some embodiments, the navigation system 418E of the vehicle 410E receives a request from a user for navigation routes wherein the request includes a starting point and an ending point. The navigation system 418E may query a real-time data server 404E (via a network 402E), such as a server that provides driving directions, for navigation route data corresponding to navigation routes, including the start point and the endpoint. The real-time data server 404E transmits the navigation route data to the vehicle 410E via a wireless network 402E, and the communication system 424E stores the navigation data 418E in the memory 422E of the vehicle 410E.

[0133] The ECU 414E controls the operation of many of the systems of the vehicle 410E, including the ADAS systems 416E. The ECU 414E may, responsive to instructions received from the navigation system 418E, deactivate any unsafe and/or unselected autonomous features for the duration of a journey controlled by the ADAS systems 416E. In this way, the navigation system 418E may control whether ADAS systems 416E are activated or enabled so that they may be activated for a given navigation route.

[0134] The sensor set 412E may include any sensors in the vehicle 410E generating sensor data. For example, the sensor set 412E may include short-range sensors and long-range sensors. In some embodiments, the sensor set 412E of the vehicle 410E may include one or more of the following vehicle sensors: a camera, a Light Detection and Ranging (Lidar) sensor, an ultrasonic sensor, an automobile engine sensor, a radar sensor, a laser altimeter, a manifold absolute pressure sensor, an infrared detector, a motion detector, a thermostat, a sound detector, a carbon monoxide sensor, a carbon dioxide sensor, an oxygen sensor, a mass airflow

sensor, an engine coolant temperature sensor, a throttle position sensor, a crankshaft position sensor, a valve timer, an air-fuel ratio meter, a blind spot meter, a curb feeler, a defect detector, a Hall effect sensor, a parking sensor, a radar gun, a speedometer, a speed sensor, a tire-pressure monitoring sensor, a torque sensor, a transmission fluid temperature sensor, a turbine speed sensor (TSS), a variable reluctance sensor, a vehicle speed sensor (VSS), a water sensor, a wheel speed sensor, a global positioning system (GPS) sensor, a mapping functionality, and any other type of automotive sensor. The navigation system 418E may store the sensor data in the memory 422E.

[0135] The communication unit 424E transmits and receives data to and from the network 402E or to another communication channel. In some embodiments, the communication unit 424E may include a dedicated short-range communication (DSRC) transceiver, a DSRC receiver, and other hardware or software necessary to make the vehicle 410E a DSRC-equipped device.

[0136] The vehicle 410E may interact with other vehicles 406E via V2V technology. V2V communication includes sensing radar information corresponding to relative distances to external objects, receiving GPS information of the vehicles, setting areas where the other vehicles 406E are located based on the sensed radar information, calculating probabilities that the GPS information of the object vehicles will be located at the set areas, and identifying vehicles and/or objects corresponding to the radar information and the GPS information of the object vehicles based on the calculated probabilities, in one example.

[0137] For a vehicle to be adequately secured, the vehicle must be protected from unauthorized physical access as well as unauthorized remote access (e.g., cyber-threats). To prevent unauthorized physical access, a vehicle is equipped with a secure access system such as a keyless entry in one example. Meanwhile, security protocols are added to a vehicle's computers and computer networks to facilitate secure remote communications to and from the vehicle in one example.

[0138] ECUs are nodes within a vehicle that control tasks such as activating the windshield wipers to tasks such as an anti-lock brake system. ECUs are often connected to one another through the vehicle's central network, which may be referred to as a controller area network (CAN). State-of-the-art features such as autonomous driving are strongly reliant on implementing new, complex ECUs such as ADAS, sensors, and the like. While these new technologies have helped improve the safety and driving experience of a vehicle, they have also increased the number of externally-communicating units inside of the vehicle, making them more vulnerable to attack. Below are some examples of protecting the vehicle from physical intrusion and remote intrusion.

[0139] In one embodiment, a CAN includes a CAN bus with a high and low terminal and a plurality of ECUs, which are connected to the CAN bus via wired connections. The CAN bus is designed to allow microcontrollers and devices to communicate with each other in an application without a host computer. The CAN bus implements a message-based protocol (i.e., ISO 11898 standards) that allows ECUs to send commands to one another at a root level. Meanwhile, the ECUs represent controllers for controlling electrical systems or subsystems within the vehicle. Examples of the

electrical systems include power steering, anti-lock brakes, air-conditioning, tire pressure monitoring, cruise control, and many other features.

[0140] In this example, the ECU includes a transceiver and a microcontroller. The transceiver may be used to transmit and receive messages to and from the CAN bus. For example, the transceiver may convert the data from the microcontroller into a format of the CAN bus and also convert data from the CAN bus into a format for the microcontroller. Meanwhile, the microcontroller interprets the messages and also decides what messages to send using ECU software installed therein in one example.

[0141] To protect the CAN from cyber threats, various security protocols may be implemented. For example, sub-networks (e.g., sub-networks A and B, etc.) may be used to divide the CAN into smaller sub-CANs and limit an attacker's capabilities to access the vehicle remotely. In one embodiment, a firewall (or gateway, etc.) may be added to block messages from crossing the CAN bus across sub-networks. If an attacker gains access to one sub-network, the attacker will not have access to the entire network. To make sub-networks even more secure, the most critical ECUs are not placed on the same sub-network, in one example.

[0142] In addition to protecting a vehicle's internal network, vehicles may also be protected when communicating with external networks such as the Internet. One of the benefits of having a vehicle connection to a data source such as the Internet is that information from the vehicle can be sent through a network to remote locations for analysis. Examples of vehicle information include GPS, onboard diagnostics, tire pressure, and the like. These communication systems are often referred to as telematics because they involve the combination of telecommunications and informatics. Further, the instant solution as described and depicted can be utilized in this and other networks and/or systems, including those that are described and depicted herein.

[0143] FIG. 4E illustrates an example 400E of vehicles 402I and 408I performing secured V2V communications using security certificates, according to example embodiments. Referring to FIG. 4E, the vehicles 402I and 408I may communicate via V2V communications over a short-range network, a cellular network, or the like. Before sending messages, the vehicles 402I and 408I may sign the messages using a respective public key certificate. For example, the vehicle 402I may sign a V2V message using a public key certificate 404I. Likewise, the vehicle 408I may sign a V2V message using a public key certificate 410I. The public key certificates 404I and 410I are associated with the vehicles 402I and 408I, respectively, in one example.

[0144] Upon receiving the communications from each other, the vehicles may verify the signatures with a certificate authority 406I or the like. For example, the vehicle 408I may verify with the certificate authority 406I that the public key certificate 404I used by vehicle 402I to sign a V2V communication is authentic. If the vehicle 408I successfully verifies the public key certificate 404I, the vehicle knows that the data is from a legitimate source. Likewise, the vehicle 402I may verify with the certificate authority 406I that the public key certificate 410I used by the vehicle 408I to sign a V2V communication is authentic. Further, the instant solution as described and depicted with respect to

FIG. 4E can be utilized in this and other networks and/or systems including those that are described and depicted herein.

[0145] In some embodiments, a computer may include a security processor. In particular, the security processor may perform authorization, authentication, cryptography (e.g., encryption), and the like, for data transmissions that are sent between ECUs and other devices on a CAN bus of a vehicle, and also data messages that are transmitted between different vehicles. The security processor may include an authorization module, an authentication module, and a cryptography module. The security processor may be implemented within the vehicle's computer and may communicate with other vehicle elements, for example, the ECUs/CAN network, wired and wireless devices such as wireless network interfaces, input ports, and the like. The security processor may ensure that data frames (e.g., CAN frames, etc.) that are transmitted internally within a vehicle (e.g., via the ECUs/CAN network) are secure. Likewise, the security processor can ensure that messages transmitted between different vehicles and devices attached or connected via a wire to the vehicle's computer are also secured.

[0146] For example, the authorization module may store passwords, usernames, PIN codes, biometric scans, and the like for different vehicle users. The authorization module may determine whether a user (or technician) has permission to access certain settings such as a vehicle's computer. In some embodiments, the authorization module may communicate with a network interface to download any necessary authorization information from an external server. When a user desires to make changes to the vehicle settings or modify technical details of the vehicle via a console or GUI within the vehicle or via an attached/connected device, the authorization module may require the user to verify themselves in some way before such settings are changed. For example, the authorization module may require a username, a password, a PIN code, a biometric scan, a predefined line drawing or gesture, and the like. In response, the authorization module may determine whether the user has the necessary permissions (access, etc.) being requested.

[0147] The authentication module may be used to authenticate internal communications between ECUs on the CAN network of the vehicle. As an example, the authentication module may provide information for authenticating communications between the ECUs. As an example, the authentication module may transmit a bit signature algorithm to the ECUs of the CAN network. The ECUs may use the bit signature algorithm to insert authentication bits into the CAN fields of the CAN frame. All ECUs on the CAN network typically receive each CAN frame. The bit signature algorithm may dynamically change the position, amount, etc., of authentication bits each time a new CAN frame is generated by one of the ECUs. The authentication module may also provide a list of ECUs that are exempt (safe list) and that do not need to use the authentication bits. The authentication module may communicate with a remote server to retrieve updates to the bit signature algorithm and the like.

[0148] The encryption module may store asymmetric key pairs to be used by the vehicle to communicate with other external user devices and vehicles. For example, the encryption module may provide a private key to be used by the vehicle to encrypt/decrypt communications, while the corresponding public key may be provided to other user devices

and vehicles to enable the other devices to decrypt/encrypt the communications. The encryption module may communicate with a remote server to receive new keys, updates to keys, keys of new vehicles, users, etc., and the like. The encryption module may also transmit any updates to a local private/public key pair to the remote server.

[0149] FIG. 5A illustrates an example vehicle configuration 500A for managing database transactions associated with a vehicle, according to example embodiments. Referring to FIG. 5A, as a particular vehicle 525 is engaged in transactions (e.g., vehicle service, dealer transactions, delivery/pickup, transportation services, etc.), the vehicle may receive assets 510 and/or expel/transfer assets 512 according to a transaction(s). A vehicle processor 526 resides in the vehicle 525 and communication exists between the vehicle processor 526, a database 530, and the transaction module 520. The transaction module 520 may record information, such as assets, parties, credits, service descriptions, date, time, location, results, notifications, unexpected events, etc. Those transactions in the transaction module 520 may be replicated into a database 530. The database 530 can be one of a SQL database, a relational database management system (RDBMS), a relational database, a non-relational database, a blockchain, a distributed ledger, and may be on board the vehicle, may be off-board the vehicle, may be accessed directly and/or through a network, or be accessible to the vehicle.

[0150] In one embodiment, a vehicle may engage with another vehicle to perform various actions such as to share, transfer, acquire service calls, etc. when the vehicle has reached a status where the services need to be shared with another vehicle. For example, the vehicle may be due for a battery charge and/or may have an issue with a tire and may be in route to pick up a package for delivery. A vehicle processor resides in the vehicle and communication exists between the vehicle processor, a first database, and a transaction module. The vehicle may notify another vehicle, which is in its network and which operates on its blockchain member service. A vehicle processor resides in another vehicle and communication exists between the vehicle processor, a second database, the vehicle processor, and a transaction module. The another vehicle may then receive the information via a wireless communication request to perform the package pickup from the vehicle and/or from a server (not shown). The transactions are logged in the transaction modules and of both vehicles. The credits are transferred from the vehicle to the other vehicle and the record of the transferred service is logged in the first database, assuming that the blockchains are different from one another, or are logged in the same blockchain used by all members. The first database can be one of a SQL database, an RDBMS, a relational database, a non-relational database, a blockchain, a distributed ledger, and may be on board the vehicle, may be off-board the vehicle, may be accessible directly and/or through a network.

[0151] FIG. 5B illustrates a blockchain architecture configuration 500B, according to example embodiments. Referring to FIG. 5B, the blockchain architecture 500B may include certain blockchain elements, for example, a group of blockchain member nodes 502-505 as part of a blockchain group 510. In one example embodiment, a permissioned blockchain is not accessible to all parties but only to those members with permissioned access to the blockchain data. The blockchain nodes participate in a number of activities,

such as blockchain entry addition and validation process (consensus). One or more of the blockchain nodes may endorse entries based on an endorsement policy and may provide an ordering service for all blockchain nodes. A blockchain node may initiate a blockchain action (such as an authentication) and seek to write to a blockchain immutable ledger stored in the blockchain, a copy of which may also be stored on the underpinning physical infrastructure.

[0152] The blockchain transactions 520 are stored in memory of computers as the transactions are received and approved by the consensus model dictated by the members' nodes. Approved transactions 526 are stored in current blocks of the blockchain and committed to the blockchain via a committal procedure, which includes performing a hash of the data contents of the transactions in a current block and referencing a previous hash of a previous block. Within the blockchain, one or more smart contracts 530 may exist that define the terms of transaction agreements and actions included in smart contract executable application code 532, such as registered recipients, vehicle features, requirements, permissions, sensor thresholds, etc. The code may be configured to identify whether requesting entities are registered to receive vehicle services, what service features they are entitled/required to receive given their profile statuses and whether to monitor their actions in subsequent events. For example, when a service event occurs and a user is riding in the vehicle, the sensor data monitoring may be triggered, and a certain parameter, such as a vehicle charge level, may be identified as being above/below a particular threshold for a particular period of time, then the result may be a change to a current status, which requires an alert to be sent to the managing party (i.e., vehicle owner, vehicle operator, server, etc.) so the service can be identified and stored for reference. The vehicle sensor data collected may be based on types of sensor data used to collect information about vehicle's status. The sensor data may also be the basis for the vehicle event data 534, such as a location(s) to be traveled, an average speed, a top speed, acceleration rates, whether there were any collisions, was the expected route taken, what is the next destination, whether safety measures are in place, whether the vehicle has enough charge/fuel, etc. All such information may be the basis of smart contract terms 530, which are then stored in a blockchain. For example, sensor thresholds stored in the smart contract can be used as the basis for whether a detected service is necessary and when and where the service should be performed.

[0153] In one embodiment, a blockchain logic example includes a blockchain application interface as an API or plug-in application that links to the computing device and execution platform for a particular transaction. The blockchain configuration may include one or more applications, which are linked to application programming interfaces (APIs) to access and execute stored program/application code (e.g., smart contract executable code, smart contracts, etc.), which can be created according to a customized configuration sought by participants and can maintain their own state, control their own assets, and receive external information. This can be deployed as an entry and installed, via appending to the distributed ledger, on all blockchain nodes.

[0154] The smart contract application code provides a basis for the blockchain transactions by establishing application code, which when executed causes the transaction

terms and conditions to become active. The smart contract, when executed, causes certain approved transactions to be generated, which are then forwarded to the blockchain platform. The platform includes a security/authorization, computing devices, which execute the transaction management and a storage portion as a memory that stores transactions and smart contracts in the blockchain.

[0155] The blockchain platform may include various layers of blockchain data, services (e.g., cryptographic trust services, virtual execution environment, etc.), and underpinning physical computer infrastructure that may be used to receive and store new entries and provide access to auditors, which are seeking to access data entries. The blockchain may expose an interface that provides access to the virtual execution environment necessary to process the program code and engage the physical infrastructure. Cryptographic trust services may be used to verify entries such as asset exchange entries and keep information private.

[0156] The blockchain architecture configuration of FIGS. 5A and 5B may process and execute program/application code via one or more interfaces exposed, and services provided, by the blockchain platform. As a non-limiting example, smart contracts may be created to execute reminders, updates, and/or other notifications subject to the changes, updates, etc. The smart contracts can themselves be used to identify rules associated with authorization and access requirements and usage of the ledger. For example, the information may include a new entry, which may be processed by one or more processing entities (e.g., processors, virtual machines, etc.) included in the blockchain layer. The result may include a decision to reject or approve the new entry based on the criteria defined in the smart contract and/or a consensus of the peers. The physical infrastructure may be utilized to retrieve any of the data or information described herein.

[0157] Within smart contract executable code, a smart contract may be created via a high-level application and programming language, and then written to a block in the blockchain. The smart contract may include executable code that is registered, stored, and/or replicated with a blockchain (e.g., distributed network of blockchain peers). An entry is an execution of the smart contract code, which can be performed in response to conditions associated with the smart contract being satisfied. The executing of the smart contract may trigger a trusted modification(s) to a state of a digital blockchain ledger. The modification(s) to the blockchain ledger caused by the smart contract execution may be automatically replicated throughout the distributed network of blockchain peers through one or more consensus protocols.

[0158] The smart contract may write data to the blockchain in the format of key-value pairs. Furthermore, the smart contract code can read the values stored in a blockchain and use them in application operations. The smart contract code can write the output of various logic operations into the blockchain. The code may be used to create a temporary data structure in a virtual machine or other computing platform. Data written to the blockchain can be public and/or can be encrypted and maintained as private. The temporary data that is used/generated by the smart contract is held in memory by the supplied execution environment, then deleted once the data needed for the blockchain is identified.

[0159] A smart contract executable code may include the code interpretation of a smart contract, with additional features. As described herein, the smart contract executable code may be program code deployed on a computing network, where it is executed and validated by chain validators together during a consensus process. The smart contract executable code receives a hash and retrieves from the blockchain a hash associated with the data template created by use of a previously stored feature extractor. If the hashes of the hash identifier and the hash created from the stored identifier template data match, then the smart contract executable code sends an authorization key to the requested service. The smart contract executable code may write to the blockchain data associated with the cryptographic details.

[0160] FIG. 5C illustrates a blockchain configuration for storing blockchain transaction data, according to example embodiments. Referring to FIG. 5C, the example configuration 500C provides for the vehicle 562, the user device 564 and a server 566 sharing information with a distributed ledger (i.e., blockchain) 568. The server may represent a service provider entity inquiring with a vehicle service provider to share user profile rating information in the event that a known and established user profile is attempting to rent a vehicle with an established rated profile. The server 566 may be receiving and processing data related to a vehicle's service requirements. As the service events occur, such as the vehicle sensor data indicates a need for fuel/charge, a maintenance service, etc., a smart contract may be used to invoke rules, thresholds, sensor information gathering, etc., which may be used to invoke the vehicle service event. The blockchain transaction data 570 is saved for each transaction, such as the access event, the subsequent updates to a vehicle's service status, event updates, etc. The transactions may include the parties, the requirements (e.g., 18 years of age, service eligible candidate, valid driver's license, etc.), compensation levels, the distance traveled during the event, the registered recipients permitted to access the event and host a vehicle service, rights/permissions, sensor data retrieved during the vehicle event operation to log details of the next service event and identify a vehicle's condition status, and thresholds used to make determinations about whether the service event was completed and whether the vehicle's condition status has changed.

[0161] FIG. 5D illustrates blockchain blocks that can be added to a distributed ledger, according to example embodiments, and contents of block structures 582A to 582N. Referring to FIG. 5D, clients (not shown) may submit entries to blockchain nodes to enact activity on the blockchain. As an example, clients may be applications that act on behalf of a requester, such as a device, person, or entity to propose entries for the blockchain. The plurality of blockchain peers (e.g., blockchain nodes) may maintain a state of the blockchain network and a copy of the distributed ledger. Different types of blockchain nodes/peers may be present in the blockchain network including endorsing peers, which simulate and endorse entries proposed by clients and committing peers which verify endorsements, validate entries, and commit entries to the distributed ledger. In this example, the blockchain nodes may perform the role of endorser node, committer node, or both.

[0162] The instant system includes a blockchain that stores immutable, sequenced records in blocks, and a state database (current world state) maintaining a current state of

the blockchain. One distributed ledger may exist per channel and each peer maintains its own copy of the distributed ledger for each channel of which they are a member. The instant blockchain is an entry log, structured as hash-linked blocks where each block contains a sequence of N entries. Blocks may include various components such as those shown in FIG. 5D. The linking of the blocks may be generated by adding a hash of a prior block's header within a block header of a current block. In this way, all entries on the blockchain are sequenced and cryptographically linked together preventing tampering with blockchain data without breaking the hash links. Furthermore, because of the links, the latest block in the blockchain represents every entry that has come before it. The instant blockchain may be stored on a peer file system (local or attached storage), which supports an append-only blockchain workload.

[0163] The current state of the blockchain and the distributed ledger may be stored in the state database. Here, the current state data represents the latest values for all keys ever included in the chain entry log of the blockchain. Smart contract executable code invocations execute entries against the current state in the state database. To make these smart contract executable code interactions extremely efficient, the latest values of all keys are stored in the state database. The state database may include an indexed view into the entry log of the blockchain, it can therefore be regenerated from the chain at any time. The state database may automatically get recovered (or generated if needed) upon peer startup, before entries are accepted.

[0164] Endorsing nodes receive entries from clients and endorse the entry based on simulated results. Endorsing nodes hold smart contracts, which simulate the entry proposals. When an endorsing node endorses an entry, the endorsing nodes creates an entry endorsement, which is a signed response from the endorsing node to the client application indicating the endorsement of the simulated entry. The method of endorsing an entry depends on an endorsement policy that may be specified within smart contract executable code. An example of an endorsement policy is "the majority of endorsing peers must endorse the entry." Different channels may have different endorsement policies. Endorsed entries are forwarded by the client application to an ordering service.

[0165] The ordering service accepts endorsed entries, orders them into a block, and delivers the blocks to the committing peers. For example, the ordering service may initiate a new block when a threshold of entries has been reached, a timer times out, or another condition. In this example, blockchain node is a committing peer that has received a data block 582A for storage on the blockchain. The ordering service may be made up of a cluster of orderers. The ordering service does not process entries, smart contracts, or maintain the shared ledger. Rather, the ordering service may accept the endorsed entries and specifies the order in which those entries are committed to the distributed ledger. The architecture of the blockchain network may be designed such that the specific implementation of 'ordering' becomes a pluggable component.

[0166] Entries are written to the distributed ledger in a consistent order. The order of entries is established to ensure that the updates to the state database are valid when they are committed to the network. Unlike a cryptocurrency blockchain system where ordering occurs through the solving of a cryptographic puzzle, or mining, in this example the

parties of the distributed ledger may choose the ordering mechanism that best suits that network.

[0167] Referring to FIG. 5D, a block 582A (also referred to as a data block) that is stored on the blockchain and/or the distributed ledger may include multiple data segments such as a block header 584A to 584n, transaction-specific data 586A to 586n, and block metadata 588A to 588n. It should be appreciated that the various depicted blocks and their contents, such as block 582A and its contents are merely for purposes of an example and are not meant to limit the scope of the example embodiments. In some cases, both the block header 584A and the block metadata 588A may be smaller than the transaction-specific data 586A, which stores entry data; however, this is not a requirement. The block 582A may store transactional information of N entries (e.g., 100, 500, 1000, 2000, 3000, etc.) within the block data 590A to 590n. The block 582A may also include a link to a previous block (e.g., on the blockchain) within the block header 584A. In particular, the block header 584A may include a hash of a previous block's header. The block header 584A may also include a unique block number, a hash of the block data 590A of the current block 582A, and the like. The block number of the block 582A may be unique and assigned in an incremental/sequential order starting from zero. The first block in the blockchain may be referred to as a genesis block, which includes information about the blockchain, its members, the data stored therein, etc.

[0168] The block data 590A may store entry information of each entry that is recorded within the block. For example, the entry data may include one or more of a type of the entry, a version, a timestamp, a channel ID of the distributed ledger, an entry ID, an epoch, a payload visibility, a smart contract executable code path (deploy tx), a smart contract executable code name, a smart contract executable code version, input (smart contract executable code and functions), a client (creator) identify such as a public key and certificate, a signature of the client, identities of endorsers, endorser signatures, a proposal hash, smart contract executable code events, response status, namespace, a read set (list of key and version read by the entry, etc.), a write set (list of key and value, etc.), a start key, an end key, a list of keys, a Merkel tree query summary, and the like. The entry data may be stored for each of the N entries.

[0169] In some embodiments, the block data 590A may also store transaction-specific data 586A, which adds additional information to the hash-linked chain of blocks in the blockchain. Accordingly, the data 586A can be stored in an immutable log of blocks on the distributed ledger. Some of the benefits of storing such data 586A are reflected in the various embodiments disclosed and depicted herein. The block metadata 588A may store multiple fields of metadata (e.g., as a byte array, etc.). Metadata fields may include signature on block creation, a reference to a last configuration block, an entry filter identifying valid and invalid entries within the block, last offset persisted of an ordering service that ordered the block, and the like. The signature, the last configuration block, and the orderer metadata may be added by the ordering service. Meanwhile, a committer of the block (such as a blockchain node) may add validity/invalidity information based on an endorsement policy, verification of read/write sets, and the like. The entry filter may include a byte array of a size equal to the number of entries in the block data and a validation code identifying whether an entry was valid/invalid.

[0170] The other blocks **582B** to **582n** in the blockchain also have headers, files, and values. However, unlike the first block **582A**, each of the headers **584A** to **584n** in the other blocks includes the hash value of an immediately preceding block. The hash value of the immediately preceding block may be just the hash of the header of the previous block or may be the hash value of the entire previous block. By including the hash value of a preceding block in each of the remaining blocks, a trace can be performed from the Nth block back to the genesis block (and the associated original file) on a block-by-block basis, as indicated by arrows **592**, to establish an auditable and immutable chain-of-custody.

[0171] FIG. **5E** illustrates a process **500E** of a new block being added to a distributed ledger **520E**, according to example embodiments, and FIG. **5D** illustrates the contents of FIG. **5E**'s new data block structure **530E** for blockchain, according to example embodiments. Referring to FIG. **5E**, clients (not shown) may submit transactions to blockchain nodes **511E**, **512E**, and/or **513E**. Clients may be instructions received from any source to enact activity on the blockchain **522E**. As an example, clients may be applications that act on behalf of a requester, such as a device, person, or entity to propose transactions for the blockchain. The plurality of blockchain peers (e.g., blockchain nodes **511E**, **512E**, and **513E**) may maintain a state of the blockchain network and a copy of the distributed ledger **520E**. Different types of blockchain nodes/peers may be present in the blockchain network including endorsing peers which simulate and endorse transactions proposed by clients and committing peers which verify endorsements, validate transactions, and commit transactions to the distributed ledger **520E**. In this example, the blockchain nodes **511E**, **512E**, and **513E** may perform the role of endorser node, committer node, or both.

[0172] The distributed ledger **520E** includes a blockchain which stores immutable, sequenced records in blocks, and a state database **524E** (current world state) maintaining a current state of the blockchain **522E**. One distributed ledger **520E** may exist per channel and each peer maintains its own copy of the distributed ledger **520E** for each channel of which they are a member. The blockchain **522E** is a transaction log, structured as hash-linked blocks where each block contains a sequence of N transactions. The linking of the blocks (shown by arrows in FIG. **5E**) may be generated by adding a hash of a prior block's header within a block header of a current block. In this way, all transactions on the blockchain **522E** are sequenced and cryptographically linked together preventing tampering with blockchain data without breaking the hash links. Furthermore, because of the links, the latest block in the blockchain **522E** represents every transaction that has come before it. The blockchain **522E** may be stored on a peer file system (local or attached storage), which supports an append-only blockchain workload.

[0173] The current state of the blockchain **522E** and the distributed ledger **520E** may be stored in the state database **524E**. Here, the current state data represents the latest values for all keys ever included in the chain transaction log of the blockchain **522E**. Chaincode invocations execute transactions against the current state in the state database **524E**. To make these chaincode interactions extremely efficient, the latest values of all keys are stored in the state database **524E**. The state database **524E** may include an indexed view into the transaction log of the blockchain **522E**, and it can therefore be regenerated from the chain at any time. The

state database **524E** may automatically get recovered (or generated if needed) upon peer startup, before transactions are accepted.

[0174] Endorsing nodes receive transactions from clients and endorse the transaction based on simulated results. Endorsing nodes hold smart contracts which simulate the transaction proposals. When an endorsing node endorses a transaction, the endorsing node creates a transaction endorsement which is a signed response from the endorsing node to the client application indicating the endorsement of the simulated transaction. The method of endorsing a transaction depends on an endorsement policy which may be specified within chaincode. An example of an endorsement policy is "the majority of endorsing peers must endorse the transaction." Different channels may have different endorsement policies. Endorsed transactions are forwarded by the client application to the ordering service **510E**.

[0175] The ordering service **510E** accepts endorsed transactions, orders them into a block, and delivers the blocks to the committing peers. For example, the ordering service **510E** may initiate a new block when a threshold of transactions has been reached, a timer times out, or another condition. In the example of FIG. **5E**, blockchain node **512E** is a committing peer that has received a new data block **530E** for storage on blockchain **522E**. The first block in the blockchain may be referred to as a genesis block which includes information about the blockchain, its members, the data stored therein, etc.

[0176] The ordering service **510E** may be made up of a cluster of orderers. The ordering service **510E** does not process transactions, smart contracts, or maintain the shared ledger. Rather, the ordering service **510E** may accept the endorsed transactions and specifies the order in which those transactions are committed to the distributed ledger **522E**. The architecture of the blockchain network may be designed such that the specific implementation of 'ordering' becomes a pluggable component.

[0177] Transactions are written to the distributed ledger **520E** in a consistent order. The order of transactions is established to ensure that the updates to the state database **524E** are valid when they are committed to the network. Unlike a cryptocurrency blockchain system where ordering occurs through the solving of a cryptographic puzzle, or mining, in this example the parties of the distributed ledger **520E** may choose the ordering mechanism that best suits the network.

[0178] When the ordering service **510E** initializes a new data block **530E**, the new data block **530E** may be broadcast to committing peers (e.g., blockchain nodes **511E**, **512E**, and **513E**). In response, each committing peer validates the transaction within the new data block **530E** by checking to make sure that the read set and the write set still match the current world state in the state database **524E**. Specifically, the committing peer can determine whether the read data that existed when the endorsers simulated the transaction is identical to the current world state in the state database **524E**. When the committing peer validates the transaction, the transaction is written to the blockchain **522E** on the distributed ledger **520E**, and the state database **524E** is updated with the write data from the read-write set. If a transaction fails, that is, if the committing peer finds that the read-write set does not match the current world state in the state database **524E**, the transaction ordered into a block will

still be included in that block, but it will be marked as invalid, and the state database 524E will not be updated.

[0179] Referring to FIG. 5F 500F, a new data block 530 (also referred to as a data block) that is stored on the blockchain 522E of the distributed ledger 520E may include multiple data segments such as a block header 540, block data 550, and block metadata 560. It should be appreciated that the various depicted blocks and their contents, such as new data block 530 and its contents shown in FIG. 5F, are merely examples and are not meant to limit the scope of the example embodiments. The new data block 530 may store transactional information of N transaction(s) (e.g., 1, 10, 100, 500, 1000, 2000, 3000, etc.) within the block data 550. The new data block 530 may also include a link to a previous block (e.g., on the blockchain 522E in FIG. 5E) within the block header 540. In particular, the block header 540 may include a hash of a previous block's header. The block header 540 may also include a unique block number, a hash of the block data 550 of the new data block 530, and the like. The block number of the new data block 530 may be unique and assigned in various orders, such as an incremental/sequential order starting from zero.

[0180] The block data 550 may store transactional information of each transaction that is recorded within the new data block 530. For example, the transaction data may include one or more of a type of the transaction, a version, a timestamp, a channel ID of the distributed ledger 520E (shown in FIG. 5E), a transaction ID, an epoch, a payload visibility, a chaincode path (deploy tx), a chaincode name, a chaincode version, input (chaincode and functions), a client (creator) identify such as a public key and certificate, a signature of the client, identities of endorsers, endorser signatures, a proposal hash, chaincode events, response status, namespace, a read set (list of key and version read by the transaction, etc.), a write set (list of key and value, etc.), a start key, an end key, a list of keys, a Merkel tree query summary, and the like. The transaction data may be stored for each of the N transactions.

[0181] In one embodiment of the instant solution, the block data 564 may include data comprising one or more of receiving a request from a display device associated with a vehicle, wherein the request comprises a request to view content on the display device, determining content consuming attributes of a user of the display device based on content that has been consumed by the display device, determining that a different user of a different vehicle has similar content consuming attributes as the user based on content that has been consumed by a display device associated with the different vehicle, and outputting a content-sharing session to the display device associated with the vehicle and the display device associate with the different vehicle.

[0182] Although in FIG. 5F the blockchain data 563 is depicted in the block data 550 but may also be located in the block header 540 or the block metadata 560. In some embodiments, the an identifier of the paused content, a timestamp associated with the content at the time of the pause, etc. may be written to the blockchain data 563 and committed to a blockchain ledger.

[0183] The block metadata 560 may store multiple fields of metadata (e.g., as a byte array, etc.). Metadata fields may include signature on block creation, a reference to a last configuration block, a transaction filter identifying valid and invalid transactions within the block, last offset persisted of an ordering service that ordered the block, and the like. The

signature, the last configuration block, and the orderer metadata may be added by the ordering service 510E in FIG. 5E. Meanwhile, a committer of the block (such as blockchain node 512E in FIG. 5E) may add validity/invalidity information based on an endorsement policy, verification of read/write sets, and the like. The transaction filter may include a byte array of a size equal to the number of transactions in the block data and a validation code identifying whether a transaction was valid/invalid.

[0184] The above embodiments may be implemented in hardware, in a computer program executed by a processor, in firmware, or in a combination of the above. A computer program may be embodied on a computer readable medium, such as a storage medium. For example, a computer program may reside in random access memory ("RAM"), flash memory, read-only memory ("ROM"), erasable programmable read-only memory ("EPROM"), electrically erasable programmable read-only memory ("EEPROM"), registers, hard disk, a removable disk, a compact disk read-only memory ("CD-ROM"), or any other form of storage medium known in the art.

[0185] An exemplary storage medium may be coupled to the processor such that the processor may read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an application-specific integrated circuit ("ASIC"). In the alternative, the processor and the storage medium may reside as discrete components. For example, FIG. 6 illustrates an example computer system architecture 600, which may represent or be integrated in any of the above-described components, etc.

[0186] FIG. 6 illustrates a computing environment according to example embodiments. FIG. 6 is not intended to suggest any limitation as to the scope of use or functionality of embodiments of the application described herein. Regardless, the computing environment 600 can be implemented to perform any of the functionalities described herein. In computer environment 600, computer system 601 is operational within numerous other general-purpose or special-purpose computing system environments or configurations.

[0187] Computer system 601 may take the form of a desktop computer, laptop computer, tablet computer, smartphone, smartwatch or other wearable computer, server computer system, thin client, thick client, network PC, mini-computer system, mainframe computer, quantum computer, and distributed cloud computing environment that include any of the described systems or devices, and the like or any other form of computer or mobile device now known or to be developed in the future that is capable of running a program, accessing a network 650 or querying a database. Depending upon the technology, the performance of a computer-implemented method may be distributed among multiple computers and between multiple locations. However, in this presentation of the computing environment 600, a detailed discussion is focused on a single computer, specifically computer system 601, to keep the presentation as simple as possible.

[0188] Computer system 601 may be located in a cloud, even though it is not shown in a cloud in FIG. 6. On the other hand, computer system 601 is not required to be in a cloud except to any extent as may be affirmatively indicated. Computer system 601 may be described in the general context of computer system-executable instructions, such as

program modules, executed by a computer system **601**. Generally, program modules may include routines, programs, objects, components, logic, data structures, and so on that perform tasks or implement certain abstract data types. As shown in FIG. 6, computer system **601** in computing environment **600** is shown in the form of a general-purpose computing device. The components of computer system **601** may include but are not limited to, one or more processors or processing units **602**, a system memory **630**, and a bus **620** that couples various system components, including system memory **630** to processor **602**.

[0189] Processing unit **602** includes one or more computer processors of any type now known or to be developed. The processing unit **602** may contain circuitry distributed over multiple integrated circuit chips. The processing unit **602** may also implement multiple processor threads and multiple processor cores. Cache **632** is a memory that may be in the processor chip package(s) or located “off-chip,” as depicted in FIG. 6. Cache **632** is typically used for data or code that the threads or cores running on the processing unit **602** should be available for rapid access. In some computing environments, processing unit **602** may be designed to work with qubits and perform quantum computing.

[0190] Network adapter **603** enables the computer system **601** to connect and communicate with one or more networks **650**, such as a local area network (LAN), a wide area network (WAN), and/or a public network (e.g., the Internet). It bridges the computer’s internal bus **620** and the external network, exchanging data efficiently and reliably. The network adapter **603** may include hardware, such as modems or Wi-Fi signal transceivers, and software for packetizing and/or de-packetizing data for communication network transmission. Network adapter **603** supports various communication protocols to ensure compatibility with network standards. For Ethernet connections, it adheres to protocols such as IEEE 802.3, while for wireless communications, it might support IEEE 802.11 standards, Bluetooth, near-field communication (NFC), or other network wireless radio standards.

[0191] Computer system **601** may include a removable/non-removable, volatile/non-volatile computer storage device **610**. By way of example only, storage device **610** can be a non-removable, non-volatile magnetic media (not shown and typically called a “hard drive”). One or more data interfaces can connect it to the bus **620**. In embodiments where computer system **601** is required to have a large amount of storage (for example, where computer system **601** locally stores and manages a large database), then this storage may be provided by peripheral storage devices **610** designed for storing very large amounts of data, such as a storage area network (SAN) that is shared by multiple, geographically distributed computers.

[0192] The operating system **611** is software that manages computer system **601** hardware resources and provides common services for computer programs. Operating system **611** may take several forms, such as various known proprietary operating systems or open-source Portable Operating System Interface type operating systems that employ a kernel.

[0193] The Bus **620** represents one or more of several types of bus structures, including a memory bus or memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using various bus architectures. By way of example, and not limitation, such architectures

include Industry Standard Architecture (ISA) buses, Micro Channel Architecture (MCA) buses, Enhanced ISA (EISA) buses, Video Electronics Standards Association (VESA) local buses, and Peripheral Component Interconnects (PCI) bus. The bus **620** is the signal conduction paths that allow the various components of computer system **601** to communicate with each other.

[0194] Memory **630** is any volatile memory now known or to be developed in the future. Examples include dynamic random-access memory (RAM **631**) or static type RAM **631**. Typically, the volatile memory is characterized by random access, but this is not required unless affirmatively indicated. In computer system **601**, memory **630** is in a single package and is internal to computer system **601**, but alternatively or additionally, the volatile memory may be distributed over multiple packages and/or located externally with respect to computer system **601**. By way of example only, memory **630** can be provided for reading from and writing to a non-removable, non-volatile magnetic media (shown as storage device **610**, and typically called a “hard drive”). Memory **630** may include at least one program product having a set (e.g., at least one) of program modules configured to carry out various functions. A typical computer system **601** may include cache **632**, a specialized volatile memory generally faster than RAM **631** and generally located closer to the processing unit **602**. Cache **632** stores frequently accessed data and instructions accessed by the processing unit **602** to speed up processing time. The computer system **601** may include non-volatile memory **633** in ROM, PROM, EEPROM, and flash memory. Non-volatile memory **633** often contains programming instructions for starting the computer, including the BIOS and information required to start the operating system **611**.

[0195] Computer system **601** may also communicate with one or more peripheral devices **641** via an I/O interface **640**. Such devices may include a keyboard, a pointing device, a display, etc.; one or more devices that enable a user to interact with computer system **601**; and/or any devices (e.g., network card, modem, etc.) that enable computer system **601** to communicate with one or more other computing devices. Such communication can occur via input/output (I/O) interfaces **640**. As depicted, IO interface **640** communicates with the other components of computer system **601** via bus **620**.

[0196] Network **650** is any computer network that can receive and/or transmit data. Network **650** can include a WAN, LAN, private cloud, or public Internet, capable of communicating computer data over non-local distances by any technology that is now known or to be developed in the future. Any connection depicted can be wired and/or wireless and may traverse other components that are not shown. In some embodiments, a network **650** may be replaced and/or supplemented by LANs designed to communicate data between devices located in a local area, such as a Wi-Fi network. The network **650** typically includes computer hardware such as copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers, and edge servers. Computer system **601** connects to network **650** via network adapter **603** and bus **620**.

[0197] User devices **651** are any computer systems used and controlled by an end user in connection with computer system **601**. For example, in a hypothetical case where computer system **601** is designed to provide a recommen-

dation to an end user, this recommendation may typically be communicated from network adapter 603 of computer system 601 through network 650 to a user device 651, allowing user device 651 to display, or otherwise present, the recommendation to an end user. User devices can be a wide array of devices, including PCs, laptops, tablet, hand-held, mobile phones, etc.

[0198] Remote Servers 660 are any computers that serve at least some data and/or functionality over a network 650, for example, WAN, a virtual private network (VPN), a private cloud, or via the Internet to computer system 601. These networks 650 may communicate with a LAN to reach users. The user interface may include a web browser or an application that facilitates communication between the user and remote data. Such applications have been called “thin” desktops or “thin clients.” Thin clients typically incorporate software programs to emulate desktop sessions, such as Microsoft RDP (Remote Desktop Protocol) or Citrix ICA (Independent Computing Architecture). Mobile applications can also be used. Remote servers 660 can also host remote databases 661, with the database located on one remote server 660 or distributed across multiple remote servers 660. Remote databases 661 are accessible from database client applications installed locally on the remote server 660, other remote servers 660, user devices 651, or computer system 601 across a network 650.

[0199] A Public Cloud 670 is an on-demand availability of computer system resources, including data storage and computing power, without direct active management by the user. Public clouds 670 are often distributed, with data centers in multiple locations for availability and performance. Computing resources on public clouds (670) are shared across multiple tenants through virtual computing environments comprising virtual machines 671, databases 672, containers 673, and other resources. A Container 673 is an isolated, lightweight software for running an application on the host operating system 611. Containers 673 are built on top of the host operating system’s kernel and contain only apps and some lightweight operating system APIs and services. In contrast, virtual machine 671 is a software layer that includes a complete operating system 611 and kernel. Virtual machines 671 are built on top of a hypervisor emulation layer designed to abstract a host computer’s hardware from the operating software environment. Public clouds 670 generally offer hosted databases 672 abstracting high-level database management activities. It should be further understood that one or more of the elements described or depicted in FIG. 6 can perform one or more of the actions, functionalities, or features described or depicted herein.

[0200] Although an exemplary embodiment of at least one of a system, method, and non-transitory computer readable medium has been illustrated in the accompanied drawings and described in the foregoing detailed description, it will be understood that the application is not limited to the embodiments disclosed, but is capable of numerous rearrangements, modifications, and substitutions as set forth and defined by the following claims. For example, the capabilities of the system of the various figures can be performed by one or more of the modules or components described herein or in a distributed architecture and may include a transmitter, receiver, or pair of both. For example, all or part of the functionality performed by the individual modules, may be performed by one or more of these modules. Further, the functionality described herein may be performed at various

times and in relation to various events, internal or external to the modules or components. Also, the information sent between various modules can be sent between the modules via at least one of: a data network, the Internet, a voice network, an Internet Protocol network, a wireless device, a wired device and/or via plurality of protocols. Also, the messages sent or received by any of the modules may be sent or received directly and/or via one or more of the other modules.

[0201] One skilled in the art will appreciate that a “system” may be embodied as a personal computer, a server, a console, a personal digital assistant (PDA), a cell phone, a tablet computing device, a smartphone or any other suitable computing device, or combination of devices. Presenting the above-described functions as being performed by a “system” is not intended to limit the scope of the present application in any way but is intended to provide one example of many embodiments. Indeed, methods, systems and apparatuses disclosed herein may be implemented in localized and distributed forms consistent with computing technology.

[0202] It should be noted that some of the system features described in this specification have been presented as modules to more particularly emphasize their implementation independence. For example, a module may be implemented as a hardware circuit comprising custom very-large-scale integration (VLSI) circuits or gate arrays, off-the-shelf semiconductors such as logic chips, transistors, or other discrete components. A module may also be implemented in programmable hardware devices such as field-programmable gate arrays, programmable array logic, programmable logic devices, graphics processing units, or the like.

[0203] A module may also be at least partially implemented in software for execution by various types of processors. An identified unit of executable code may, for instance, comprise one or more physical or logical blocks of computer instructions that may, for instance, be organized as an object, procedure, or function. Nevertheless, the executables of an identified module need not be physically located together but may comprise disparate instructions stored in different locations that, when joined logically together, comprise the module and achieve the stated purpose for the module. Further, modules may be stored on a computer-readable medium, which may be, for instance, a hard disk drive, flash device, random access memory (RAM), tape, or any other such medium used to store data.

[0204] Indeed, a module of executable code may be a single instruction, or many instructions, and may even be distributed over several different code segments, among different programs, and across several memory devices. Similarly, operational data may be identified and illustrated herein within modules and may be embodied in any suitable form and organized within any suitable type of data structure. The operational data may be collected as a single data set or may be distributed over different locations, including over different storage devices, and may exist, at least partially, merely as electronic signals on a system or network.

[0205] It will be readily understood that the components of the application, as generally described and illustrated in the figures herein, may be arranged and designed in a wide variety of different configurations. Thus, the detailed description of the embodiments is not intended to limit the scope of the application as claimed but is merely representative of selected embodiments of the application.

[0206] One having ordinary skill in the art will readily understand that the above may be practiced with steps in a different order and/or with hardware elements in configurations that are different from those which are disclosed. Therefore, although the application has been described based upon these preferred embodiments, it would be apparent to those of skill in the art that certain modifications, variations, and alternative constructions would be apparent. [0207] While preferred embodiments of the present application have been described, it is to be understood that the embodiments described are illustrative only and the scope of the application is to be defined solely by the appended claims when considered with a full range of equivalents and modifications (e.g., protocols, hardware devices, software platforms etc.) thereto.

What is claimed is:

1. A method comprising:
 - receiving, via a power management apparatus, mobility data comprising movements of a vehicle from one or more systems associated with the vehicle;
 - receiving, via the power management apparatus, energy consumption data from one or more power consuming systems at a real property that is associated with the vehicle;
 - determining a recommended battery for the vehicle and a recommended power generation system for the real property based on the mobility data and the energy consumption data; and
 - displaying an identifier of the recommended battery and an identifier of the recommended power generation system via a user interface of the power management apparatus.
2. The method of claim 1, wherein the method further comprises dynamically adjusting an amount of energy stored within a battery of the vehicle and an amount of energy stored within an energy storage device of the real property via the power management apparatus based on the mobility data and the energy consumption data.
3. The method of claim 1, wherein the method further comprises determining that a priority of energy storage for the real property is greater than a priority of energy storage for the vehicle based on one or more of the mobility data and the energy consumption data, and in response, transferring energy stored within a battery of the vehicle to an energy storage device of the real property.
4. The method of claim 1, wherein the determining comprises determining that a priority of energy storage for the vehicle is greater than a priority of energy storage for the real property based on one or more of the mobility data and the energy consumption data, and in response, transferring energy stored within a battery of an energy storage device of the real property to the battery of the vehicle.
5. The method of claim 1, wherein the receiving the mobility data comprises receiving Global Positioning System (GPS) data and temporal data of tracked vehicle activity from one or more of a mobile device associated with the vehicle and a navigation system installed in the vehicle, and the receiving the energy consumption data comprises receiving energy consumption data from one or more electrical appliances installed within the real property.
6. The method of claim 1, wherein the method further comprises receiving second energy consumption data from a second energy storage device of a second real property that is located within a predetermined distance of the real prop-

erty, and transferring energy from at least one of energy stored within a battery of the vehicle, and energy stored within an energy storage device of the real property to the second energy storage device of the second real property.

7. The method of claim 1, wherein the method further comprises receiving forecasted weather data from an external data source, predicting a power outage will occur based on the forecasted weather data, and in response, transferring energy stored within a battery of the vehicle to an energy storage device of the real property.

8. An apparatus comprising:

a memory; and

a processor coupled to the memory, the processor configured to:

receive, via a power management apparatus, mobility data comprising movements of a vehicle from one or more systems associated with the vehicle,

receive, via the power management apparatus, energy consumption data from one or more power consuming systems at a real property that is associated with the vehicle,

determine a recommended battery for the vehicle and a recommended power generation system for the real property based on the mobility data and the energy consumption data, and

display an identifier of the recommended battery and an identifier of the recommended power generation system via a user interface of the power management apparatus.

9. The apparatus of claim 8, wherein the processor is further configured to dynamically adjust an amount of energy stored within a battery of the vehicle and an amount of energy stored within an energy storage device of the real property via the power management apparatus based on the mobility data and the energy consumption data.

10. The apparatus of claim 8, wherein the processor is further configured to determine that a priority of energy storage for the real property is greater than a priority of energy storage for the vehicle based on one or more of the mobility data and the energy consumption data, and in response, transfer energy stored within a battery of the vehicle to an energy storage device of the real property.

11. The apparatus of claim 8, wherein the processor is configured to determine that a priority of energy storage for the vehicle is greater than a priority of energy storage for the real property based on one or more of the mobility data and the energy consumption data, and in response, transfer energy stored within a battery of an energy storage device of the real property to the battery of the vehicle.

12. The apparatus of claim 8, wherein the processor is configured to receive Global Positioning System (GPS) data and temporal data of tracked vehicle activity from one or more of a mobile device associated with the vehicle and a navigation system installed in the vehicle, and receive the energy consumption data from one or more electrical appliances installed within the real property.

13. The apparatus of claim 8, wherein the processor is further configured to receive second energy consumption data from a second energy storage device of a second real property that is located within a predetermined distance of the real property, and transfer energy from at least one of energy stored within a battery of the vehicle, and energy

stored within an energy storage device of the real property to the second energy storage device of the second real property.

14. The apparatus of claim **8**, wherein the processor is further configured to receive forecasted weather data from an external data source, predicting a power outage will occur based on the forecasted weather data, and in response, transfer energy stored within a battery of the vehicle to an energy storage device of the real property.

15. A computer-readable storage medium comprising instructions stored therein which when executed by a processor cause the processor to perform:

receiving, via a power management apparatus, mobility data comprising movements of a vehicle from one or more systems associated with the vehicle;

receiving, via the power management apparatus, energy consumption data from one or more power consuming systems at a real property that is associated with the vehicle;

determining a recommended battery for the vehicle and a recommended power generation system for the real property based on the mobility data and the energy consumption data; and

displaying an identifier of the recommended battery and an identifier of the recommended power generation system via a user interface of the power management apparatus.

16. The computer-readable storage medium of claim **15**, wherein the processor is further configured to perform dynamically adjusting an amount of energy stored within a battery of the vehicle and an amount of energy stored within an energy storage device of the real property via the power management apparatus based on the mobility data and the energy consumption data.

17. The computer-readable storage medium of claim **15**, wherein the processor is further configured to perform determining that a priority of energy storage for the real property is greater than a priority of energy storage for the vehicle based on one or more of the mobility data and the energy consumption data, and in response, transferring energy stored within a battery of the vehicle to an energy storage device of the real property.

18. The computer-readable storage medium of claim **15**, wherein the processor is further configured to perform determining that a priority of energy storage for the vehicle is greater than a priority of energy storage for the real property based on one or more of the mobility data and the energy consumption data, and in response, transferring energy stored within a battery of an energy storage device of the real property to the battery of the vehicle.

19. The computer-readable storage medium of claim **15**, wherein the receiving the mobility data comprises receiving Global Positioning System (GPS) data and temporal data of tracked vehicle activity from one or more of a mobile device associated with the vehicle and a navigation system installed in the vehicle, and the receiving the energy consumption data comprises receiving energy consumption data from one or more electrical appliances installed within the real property.

20. The computer-readable storage medium of claim **15**, wherein the processor is further configured to perform receiving second energy consumption data from a second energy storage device of a second real property that is located within a predetermined distance of the real property, and transferring energy from at least one of energy stored within a battery of the vehicle and energy stored within an energy storage device of the real property to the second energy storage device of the second real property.

* * * * *